Atlantium
Enlightened Water Technologies

UV Water Treatment
Atlantium Hydro-Optic™ Solutions

User Manual

Guide to the Atlantium Hydro-Optic UV System RZ163 series

Models RZ163-11

RZ163-12

RZ163-13

RZ163-14

HP / UHP Lamps compatibility

With All-In-One Control Module and DPM cleaner, for General Applications

Supported Electrical Voltage: 400 / 440 / 480V

Document No. PF42A0D0E

June 2019

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Publication Catalog No. **PF42A0D0E**, Revision **005**, June 2019

IMPORTANT NOTICE

It is strictly forbidden to alter/change the system hardware in any way.

Any change made without written permission from Atlantium shall void the warranty.

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Validations and certifications completed:

Complies with UL 508
C22.2 NO. 14-13
E113027

Revision History

Revision #	Description	Date
001	AIO version based on previous Control Module versions.	Jun 2018
002	Updated screens, corrected images and text for electrical.	Oct 2018
003	Unified files for General, Aquaculture, and Power Plant applications	Jan 2019
004	Added: Electrical requirements for locations with 440Y/254 VAC 3~ +PE 50/60HZ or 480Y/277 VAC 3~ +PE 50/60HZ	Mar 2019
005	Added: compatibility for Ultra High Power lamps	June 2019

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1 Introduction

Congratulations on purchasing an Atlantium **RZ163 series** Hydro-Optic UV Water Treatment System. This user guide provides you with clear guidelines for preparing the site for installation, correct installation and software configuration procedures, and a good understanding of all system functions for maximum benefit from your Atlantium Unit.

The Atlantium **RZ163 series** systems guarantee water bio-security at optimized energy efficiency, and meet EPA criteria for providing bio-security.

Atlantium systems are designed to provide applications for the following Industries:

- **Aquaculture**
- **Dairy**
- **Municipal**
- **Power Plants**
- **UPW**
- **Marine**
- **Bio-Pharma**
- **Bottled Water**
- **Beverages**
- **Beer**
- **Aquariums**
- **and more**

Applications

- **Water Disinfection** - meets US EPA criteria for providing bio-security
- **Chlorine Replacement** - provides a chemical-free alternative to chlorine for treating source water. It easily inactivates microorganisms including those that are resistant to chlorine and heat.
- **Non-Thermal Pasteurization** - replaces energy-intensive heat pasteurization of water - meets US FDA criteria for pasteurized equivalent water,.
- **Post-Active Carbon** - protects against breakthroughs and reduces bio-fouling downstream. Significantly reduces the frequency of aggressive cleaning, and lengthens the lifespan of RO membranes
- **Dechlorination / Chlorine Destruction -** efficient, clean, non-chemical alternative to activated carbon filter or sodium bio-sulfite. Reduces risk, better protection, lower operating expenses.
- **Ozone Destruction –** efficient, cost-effective solution.
- **Ultra-pure water (UPW)** - efficient, cost-effective solution that complies with the uncommonly stringent specifications for ***high-purity water.***
- And more.

Patented Innovative Technology

- Field proven worldwide, used by world-leading brands
- Based on Hydro-Optic engineering that optimizes UV efficiency and enables uniform dose distribution
- Medium Pressure High Intensity UV effective in cold & warm water, compromises microorganisms' repair mechanisms for long-term bio-security
- Guaranteed required dose for complete microbial inactivation.
- Sustainable operation provides the security and stability that the industry demands.

Real-time Monitoring and Control

- The sophisticated data logging system enables traceability over time of all monitored data.
- It maintains a database and provides detailed analysis and reporting functions that can be used for trend analysis or proof of compliance.

Guaranteed Dose Delivery

- Unique monitoring & control software ensures that required UV dose is being delivered at all times.
- Dedicated UVIS sensors per lamp for real-time monitoring
- Automatically adjusts lamp power based on real-time analyses of critical parameters such as water flow rate, lamp intensity and water transmittance to UV (UVT).
- A user-friendly interface displays the monitored data in real-time including the actual delivered dose as well as the other critical parameters, so you can verify that you are actually getting the dose that you need.

Lower Electrical Consumption

- Optimized light distribution, through Hydraulic and Fiber optic principles applied to the UV system design, enable:
- Optimized UV power and uniform dose distribution.
- Efficient use of power, with lower electrical consumption than low or medium-pressure-based UV systems.

Minimum Down Time

- True in-line system.
- Quick lamp replacement (4 minutes) and better safety.

Integrates with your Process Line

- Field proven installation.
- With no additional software, you can integrate the Atlantium system into your current process using Atlantium's customizable settings.

Regulatory Compliance

- Meets the most stringent regulatory criteria including FDA and EPA.
- Automatically generated reports at the push of ab button for proof of compliance and traceability.
- Provides validated minimum required dose and not average dose.

Efficiency and Easy Maintenance

- For total efficiency and dependable performance, the Atlantium system monitors the lamps for performance, ensuring that a lamp is replaced only when absolutely needed.
- By following the easy preventive maintenance programs you keep the system working at peak performance.
- The **Viewport** allows you to safely peek at what's going on inside.

1.1 Before You Begin

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage.
- It is your responsibility to use the product in an appropriate manner.
- This product is designed for use solely indoors.
- You are responsible if the product is used for other than its designated purpose or in disregard of Atlantium instructions. Atlantium shall assume no responsibility in any way for such use of the product.
- The product is used for its designated purpose if it is used in accordance with its product documentation and within its performance limits.
- Using the product requires technical skills. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the users.
- Applicable local or national safety regulations and rules for the prevention of accidents must be observed at all times.

1.2 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers
.

DANGER!

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

WARNING!

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION!

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

ATTENTION!

Indicates the possibility of incorrect operation which can result in damage to the product.

Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.

Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.

Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving caustic chemicals which, if not avoided, can result in death or serious damage.

	Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.
	Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.
	Indicates information related to installation, safety or use of system.

2 About the RZ163 Series Systems

This Atlantium **RZ163-11**, **RZ163-12**, **RZ163-13**, and **RZ163-14** Hydro Optic Water Disinfection Systems includes a disinfection unit, a Ballast Module/Plug-style configuration, and **Control Module**. The quartz disinfection chamber takes advantage of ***Total Internal Reflection*** and reflects UV light back into the water to uniformly distribute the UV dose. Water flows unimpeded, resulting in low head-loss.

The specially designed short (Arc length 155mm/6.1 inch) lamps deliver a broad spectrum disinfection that irrevocably damages DNA and proteins in microorganisms. Housing Atlantium Quick Connect MPHI lamps within a thick quartz sleeve significantly reduces risk of breakage and its consequences. A UV Intensity Sensor (UVIS) provides continuous measurement of each lamp's output.

In addition, a UVT Analyzer reports and tracks the water quality and a UV-proof Viewport allows you to view the inner workings. An optional Deposit Prevention Mechanism (DPM) cleaning system cleans the surface of the quartz sleeves automatically.

2.1 The RZ163 series Units

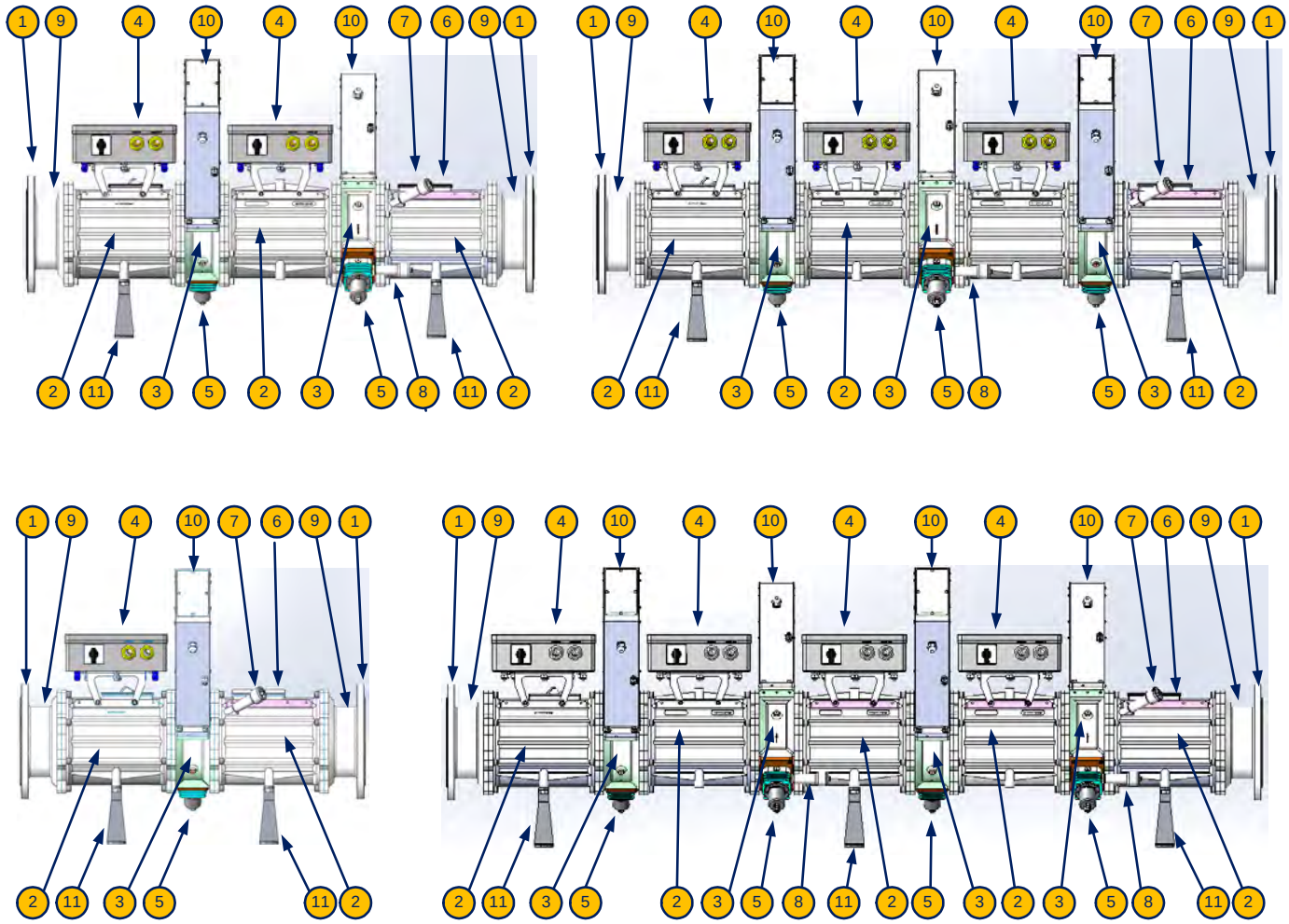


Figure 2-1: RZ163-12, RZ163-13,RZ163-11 and RZ163-14 units with DPM

Table 2-1: Atlantium Unit Key

#	Item	Description
1	Flange or Ferrule/Tri-Clamp interface	The unit connects to the facility piping via a flange interface as per customer order
2	Quartz Chamber	The quartz pipe in which the disinfection occurs
3	Lamp Chamber	The chamber that holds the lamp and optional Cleaning system
4	Connection Box	The box that sits on top of each lamp unit and holds the electrical circuitry

Table 2-1: Atlantium Unit Key

#	Item	Description
5	Lamp Arc length (mm / inch): Lamp total length (mm / inch):	Lamp encased in quartz sleeve assembly Medium Pressure High Intensity (MPHI) 155 / 6.1 322.8 / 12.71
6	Viewport	The UV-proof window to view the Unit working.
7	UVT Analyzer	The sensor that measures the UV transmittance through the water. Together with the UV Intensity sensor and the **Control Module**, it analyzes, tracks and reports the water quality.
8	UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds it to the real time dose calculation.
9	Lifting strap area	The place contains support rings designated for lifting straps to lift and transport the Unit
10	DPM Cleaning system	A sophisticated deposit prevention mechanism is the cleaning system for the quartz sleeve
11	Triangle Support Brackets	Brackets that are used to support and balance a unit anchored to customer supplied supports or surface that hold up the unit at least 750mm / 29.5inch above the floor (see ***RZ163 series Interface Control Drawings*** on page **15**).

2.2 RZ163 series Interface Control Drawings

For the ICDs of the systems in the **RZ163 series**, consult the accompanying ICD files.

2.3 Atlantium System Footprint

The Atlantium system has a small, compact footprint that facilitates in-line installation in any process train.

Table 2-2: **RZ163 Series** Physical Specifications

Item	Detail
Lamps **RZ163-11** **RZ163-12** **RZ163-13** **RZ163-14**	 1 2 3 4
Atlantium Unit Construction Materials: Casing: Quartz Chamber:	 Electro-polished stainless steel 316 High grade fused silica (Quartz)
Standard Pipe Interface Options:	**Flange**
	DIN2576 DN150 PIN16 ANSI B 16.5 6" 150lb
Unit Length (mm / inch) **RZ163-11** **RZ163-12** **RZ163-13** **RZ163-14**	 842 / 33.15 1214.5/ 47.81 1587/ 62.48 1959.5 / 77.15
Inner ∅ Diameter (mm / inch)	161 / 6.34
Outer ∅ Diameter (mm / inch)	279.4 / 11
Width at Widest Point (mm / inch) **RZ163-11** **RZ163-12** **RZ163-13** **RZ163-14**	 564.4/ 22.22 802.1 / 31.58 802.1 / 31.58 802.1 / 31.58
Bolt Circle Diameter	241.3 / 9.5
Bolt Hole Diameter	22.4 / 0.88
Number of Bolts	8
Unit Volume (Liter / Gallon) **RZ163-11** **RZ163-12** **RZ163-13** **RZ163-14**	 17.5 / 4.7 25 / 6.6 35 / 9.2 40 / 10.5

Table 2-2: **RZ163 Series** Physical Specifications

Item	Detail
Weight without water (Kg / lb): With DPM **RZ163-11** **RZ163-12** **RZ163-13** **RZ163-14**	 68 / 150 99.3/ 219 133 / 293 159 / 350
Service Clearance on Sides (mm / inch)	750 / 29.53
Minimum Height above Floor Level (mm / inch)	752 / 29.6
Maximum Working Pressure	10 Bars / 145 psi
Maximum Flow Rate	Application dependent
Distance between Bolt Holes on Supports: **RZ163-11** **RZ163-12** **RZ163-13** **RZ163-14**	 372 / 14.67 745 / 29.33 1117.5 / 44 745 / 29.33 x2
Bracket Nut / Bolt Size (mm / inch)	13 / 0.51

2.4 The Ballast Module

Ballast Module/Plug-style is a water-proof easy plug-n-play box that contains the ballast power supply and related circuitry. Its convenient plugs make it easy to service. Its modular design makes it serviceable and flexible and easier to fit into an existing plant. A Ballast Module is supplied for each lamp in the Atlantium system. Its heat sink frame complies with industry standards and contains one cooling fan on the top. It can be mounted on the wall or other suitable anchoring point. The location of the Ballast Module/Plug-style must be designated according to the maximum distance needed for positioning the **Cable Harness** of the Atlantium Unit. The **Cable Harness** is **5m / 16.4‘** (also available in **10m / 32.8’** length).

For systems with more than one lamp, the Ballast Modules can be positioned one above the other or next to one another. The Ballast Module must have at least 300 mm (11.8 inches) of clearance on its right and left sides for proper ventilation and 300mm (11.8 inches) of clearance on the bottom for cable routing. For more details on mounting, see ***Installing the Ballast Module(s)*** on page **61**.

Table 2-3: Ballast Module Physical Details

Physical	Details
Dimensions (H x W x D) (mm/inch): Ballast Module A (Passive Cooling) Ballast Module B (Active Cooling) Ballast Module C (Power Unit version)	 340 / 13.4 x 241 / 9.5 x 446 / 17.5 313 / 12.3 x 228 / 9.0 x 548 / 21.6 486.7 / 19.16 x 348 / 13.7 x 114.5 / 4.51
Weight (Kg/Lb): Ballast Module A (Passive Cooling version) Ballast Module B (Active Cooling version) Ballast Module C (Power Unit version)	 19 / 42 21.5/ 47.4 6.5 / 14.3
Heat Load per Ballast Module: (For calculating space cooling requirements) Standard Lamps High-Power Lamps Ultra High-Power Lamps	 85W 115W 151W
Protection Category:	IP 56

2.5 Electrical Usage

Utilize a Residual Current Device (RCD) that is Type A with 3000A surge resistance.

Select the appropriate RCD accordingly. Calculating the value, takes into account the power consumption of the system (see the table in the section below) and the mains voltage of your locale.

The tables below detail the Atlantium Unit's electrical usage and power consumption.

Table 2-4: **RZ163 series** Unit Electrical Usage

Electrical	Details
Main power source	400 VAC 3~ + N + PE 50/60Hz 440Y/254 VAC 3~ + PE 50/60Hz 480Y/277 VAC 3~ + PE 50/60Hz
Second power source (only for 440-480 main source)	 100-240 VAC 1ph 50/60Hz
Maximum allowed voltage tolerance	±10% from the nominal voltage

2.5.1 Power Consumption and Local Voltage Requirements

The table below details the power consumption or the Atlantium system according to local voltage.

All circuit breakers are sized according to the tables below and are to be type C.

For Systems with Standard Lamps

Table 2-5: Power Consumption for **RZ163** Systems with Standard Lamps

# of lamps :			1		2		3		4	
Nominal lamp power (kW):			1.7		3.4		5.1		6.8	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No Cleaner	Direct Feeding	400	2.7	3x4A	5.4	3x10A	8.1	3x10A	10.8	3x16A
		440	2.5	3x4A	4.9	3x6A	7.4	3x10A	9.8	3x16A
		480	2.3	3x4A	4.5	3x6A	6.8	3x10A	9.0	3x16A
DPM Cleaner	Direct Feeding	400	3.7	3x6A	6.4	3x10A	9.1	3x10A	11.8	3x16A
	With 48VDC Module	440	3.4	3x6A	5.8	3x10A	8.3	3x10A	10.7	3x16A
		480	3.2	3x6A	5.4	3x10A	7.7	3x10A	9.7	3x16A

For Systems with High-Power Lamps

Table 2-6: Power Consumption for **RZ163** Systems with High-Power Lamps

# of lamps :			1		2		3		4	
Nominal lamp power (kW):			2.3		4.6		6.9		9.2	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No Cleaner	Direct Feeding	400	3.7	3x6A	7.3	3x10A	11.0	3x16A	14.6	3x20A
		440	3.3	3x6A	6.6	3x10A	10.0	3x16A	13.3	3x16A
		480	3.0	3x6A	6.1	3x10A	9.1	3x16A	12.2	3x16A
DPM Cleaner	Direct Feeding	400	4.7	3x6A	8.3	3x10A	12	3x16A	15.6	3x20A

Table 2-6: Power Consumption for **RZ163** Systems with High-Power Lamps

# of lamps :			1		2		3		4	
Nominal lamp power (kW):			2.3		4.6		6.9		9.2	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
DPM Cleaner	With 48VDC Module	440	4.2	3x6A	7.5	3x10A	10.9	3x16A	14.2	3x20A
		480	3.9	3x16A	7	3x10A	10	3x16A	13.1	3x20A

Table 2-7: Power Consumption for **RZ163** Systems with Ultra High-Power Lamps

# of lamps :			1		2		3		4	
Nominal lamp power (kW):			3		3		9		12	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No Cleaner	Direct Feeding	400	4.8	3x10A	9.6	3x20A	14.3	3.20A	19.1	3.25A
		440	4.4	3x6A	8.7	3x16A	13	3.20A	17.3	3.25A
		480	4	3x6A	8.0	3x16A	11.9	3.20A	15.9	3.25A
DPM Cleaner	Direct Feeding	400	5.8	3x10A	10.6	3x20A	15.3	3.20A	20.1	3.25A
	With 48VDC Module	440	5.3	3x10A	9.6	3x20A	13.9	3.20A	18.2	3.25A
		480	4.9	3x10A	8.9	3x20A	12.8	3.20A	16.8	3.25A

2.6 Surge Protection

Every Atlantium system **must** be protected by a device that prevents electrical surges and fluctuations from causing damage. There are three types of surge protection:

- UPS (Uninterruptable Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.
- Surge protection device protects your equipment from variations in electrical current, such as surges and spikes.
- OVPD module (Over Voltage Protection Device) - available from Atlantium. see below.

2.6.1 Surge Protector Devices

Surge protection devices are vital for the protection of industrial power networks. These devices are provided by Atlantium and uses a combination of 3-electrode gas discharge tubes and fast clamping diodes. Typical applications include industrial processing equipment, transmission systems, I/O cards, probes, actuators, and displays.

Table 2-8: Surge Protector Details

Atlantium PN	DATA SHEETS	Module PN	Description
EP0002300	DOC002300	8859660000	SURGE PROTECTOR 400V 3P 40KA TNC CONFIG
EP0002000	DOC002000	4983-DS277-404	SURGE PROTECTOR 440-480 3P 40KA
EP0002600	DOC002600	4983-DS120-402	SURGE PROTECTOR L + N, 115V
EP0002800	DOC002800	1352740000	SURGE PROTECTOR L + N, 230V

2.6.2 The OVPD Module

The Over Voltage Protection Device (OVPD Module) may be required in locations WITH 440/480VAC, where fluctuations in the voltage may occur. If supplied, see the **OVPD Module ICD** accompanying the documentation. Consult your Atlantium representative.

A OVPD Module is not required for Ballast Module version C.

Table 2-9: OVPD Module Requirements Details

System Lamps Total Power	Mains Voltage	
	480VAC	440VAC
<7000W	An OVPD Unit is usually not required	An OVPD Unit is usually not required
<16800W	Requires an OVPD unit	An OVPD Unit is usually not required
≥16800W	Requires an OVPD unit	Requires an OVPD unit

Table 2-10: OVPD Module Details

Physical	Details
Dimensions (W x H x D) (mm/inch):	**400/15.74 x 600/23.62 x 282/11.10**
Weight (Kg/Lb.):	**20/44**
Minimum Clearance for Electrical Cables Routing on each of two sides (top/bottom) (mm/inch): **Total Clearance (mm/inch):**	**300/18.81** **1200/47.24**
Electrical	**Details**

Table 2-10: OVPD Module Details

Physical	Details
Voltage:	277/480VAC*
Current:	Up to 42A or Up to 75A
Frequency:	DC to 60Hz

* The power system must be solidly grounded where the nominal line to ground voltage does not exceed 277V.

The system is not appropriate for use on corner grounded delta, resistance grounded and ungrounded power systems.

If the power system at the site is not appropriate, you can create an appropriate power feed by using a 1:1 isolation transformer (by grounding the middle point of the star configuration at the secondary side of the transformer).

2.7 48VDC Module

In localities with 440/480VAC Electrical Power Distribution Systems

in localities with 440/480VAC **Electrical Power Distribution Systems**, such as, the USA and Mexico, the **48VDC Module** is required for installations **with or without** the optional **Deposit Prevention Mechanism (DPM) Cleaning System**. The **48VDC Module** provides a 48VDC power feed to the Connection Box.

In localities with 400VAC Electrical Power Distribution Systems

In localities with **400VAC Electrical Power Distribution Systems**, such as European countries, China, and the Middle East, the **48VDC Module** is required in installations that include the optional **Deposit Prevention Mechanism (DPM) cleaning system only.**

The **48VDC Module** can be mounted on a wall or suitable anchoring point up to **10m / 32.8'**. from the Atlantium Unit. See the **48VDC Module ICD** accompanying the documentation.

Table 2-11: 48VDC Module Details

Physical	Details
Dimensions (W x D x H) (mm/inch):	300 / 11.8 x 150.7 / 5.9 x 400 / 15.75
Length with open door:	420.9 / 16.6
Minimum Clearance for Electrical Cables Routing:	5 0/ 2

Table 2-11: 48VDC Module Details

Physical	Details
Weight (Kg/Lb.):	10 / 22
Electrical	
Maximum Power Consumption:	Maximum 285W
Voltage: Incoming Outgoing	 200-500VAC 48V
Protection Category:	IP 56.

2.8 The Control Module

The Atlantium system is monitored and controlled from a **Control Module**. It can be mounted on a wall or suitable anchoring point up to 50 meters (164.04 ft.) from the Atlantium Unit. However, it should be located in a close proximity to the Atlantium Unit in a place that it is easy to reach the touch screen and read the monitoring information. See the **AIO Control Module ICD** accompanying the documentation.

Table 2-12: Control Module Details

Physical	Details
Dimensions (W x D x H) (mm/inch):	400/15.75 x 300/11.82 x 144.4/6.12
Weight (Kg/Lb.):	9.8/21.6
Minimum Clearance for Electrical Cables Routing (mm/inch):	50/2
Electrical	
Maximum Power Consumption:	Maximum 40W
Voltage:	1 or 2 phase 120-240VAC
Protection Category:	IP 56.

2.9 Environmental Requirements

The Atlantium system should be situated indoors in an area that is adequately cooled and with sufficient air flow. However, there have been many successful Atlantium installations in less friendly environments. Consult your Atlantium representative to choose the most suitable location.

Table 2-13: **RZ163 series** System Environmental Details

Environmental	Details
Maximum ambient air temperature:	45°C (113°F)
Maximum ambient air temperature in the Ballast Modules environment:	45°C (113°F)
Maximum atmospheric relative humidity:	90%
Water Temperature: Maximum Water Temperature (lamps off) Maximum Water Temperature (lamps on) * Minimum – cold temperature:	 125°C (257°F) ** 60°C (140°F) ** No limit

* For water temperature above 70°C (160°F) consult with Atlantium service engineers

** Any deviation from either of these values require consultation with Atlantium applications department and written approval by Atlantium.

2.10 Regulatory Compliance

Atlantium Systems are designed to comply with the following regulatory standards:

- EMC directive 2014/30/EU
- European Low Voltage Directive (LVD), 2014/35/EU for electrical safety
- Council Directive 98 / 83 / EC of 3 November 1998 for the quality of water intended for human consumption
- ISO 9001:2008 Quality Management Standard – Development, Design, Production and Sales of Water Disinfection Systems
- US Federal Performance Standards 40 CFR 141.720
- Long Term 2 Enhanced Surface Water Treatment Rule of January 5th, 2006
- All components have CE or UL 508 NRTL (MET) certification
- 3rd party validated by HDR/HydroQual, Inc.
- Resolucion exenta N° 2.327 del 31 de diciembre de 2010, en el marco de la Ley General de Pesca y Acuicultura
- GOST Standards Institute, Russian Federation
- National Institute of Public Health, Poland

3 Getting Ready for Installation

Using the guidelines and information in this section, you can prepare the piping and electrical setup for installing the Atlantium Hydro Optic Water Treatment System. If you have any questions, consult with an Atlantium application engineer.

3.1 Planning

Consider where to locate the unit from the perspective of your process. Prepare your installation plan according to the planning points below, as well as the other guidelines in this manual.

3.1.1 Access

- **Premises Access**
  - Take into account that sufficient door space is required to transport the Atlantium Unit into the facility where it is to be placed.
- **Service Access**
  - Take into consideration that sufficient space is required for maintenance personal to access the Atlantium Unit for periodic component replacement.

3.1.2 Hydraulics

- **Piping**
  - Align the upstream and downstream piping with the designated location for the Atlantium Unit.
  - Prepare the flanges/ferrules and piping as specified according to ***The Piping Infrastructure*** on page **34**. To avoid a step in the connection, make sure that the internal diameters where the Atlantium Unit and your piping match.
  - **Do not** use sealing material that contains graphite when sealing the flanges/ferrules and gaskets as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- **Mechanics**
  - Prepare the supports needed for the Atlantium Unit, noting the minimum level above the floor is at least 750mm / 29.5 inch between the floor and the bottom of the provided triangle support brackets.
  - Check that the piping on the inlet and outlet sides has adequate support structures so that there are no mechanical stresses that may endanger the Unit.
  - Check the upstream and downstream piping for vibrations. Make sure that the piping to be attached to the Atlantium Unit is anchored and reinforced to protect the Unit against vibration. A high vibration level can cause damage to any hydraulic system, including the Atlantium unit. High vibration can be caused by an unbalanced pump or by sudden valve close.
- **Hydrodynamics**
  - Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. Make sure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check your facility system procedures, including:
    - When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
    - Sudden valve closures can cause water hammer. Make sure that valves close gradually, enough to avoid this problem.
    - Ensure that all pumps that impact the Atlantium system employ soft start-up procedures.

- The Atlantium Unit must be totally filled with water. Assure that no air can be trapped in the unit and that air can be released. Verify through the Unit's viewport after installation that there are no bubbles.

3.1.3 Flow Data

- Evaluate your flow trends to determine whether you require a flow meter or flow switch. If the flow varies, a **flow meter** is required to detect and report on the flow coming into the unit. Make sure that there are to be **no** branches on or off the pipes between the flow meter and the Unit inlet. If the flow rate is static, a **flow switch** can be used instead.

3.1.4 Application

- Verify that the dose-related parameters, such as the UVT and the flow, are the same as those defined in the Atlantium application scope of supply.
- Determine what a low flow situation means to your process and whether you need to shut down the Unit (low flow indicates upstream problems) or recirculate (low flow normal) when the flow goes below the normal range.

3.2 Typical Installation Set-Up

Use the diagram below as a basis for your installation drawings. Be sure to include service access as you fit the Atlantium system into your facility (i.e., nearby pipes, electrical fixtures, support beams, etc.)

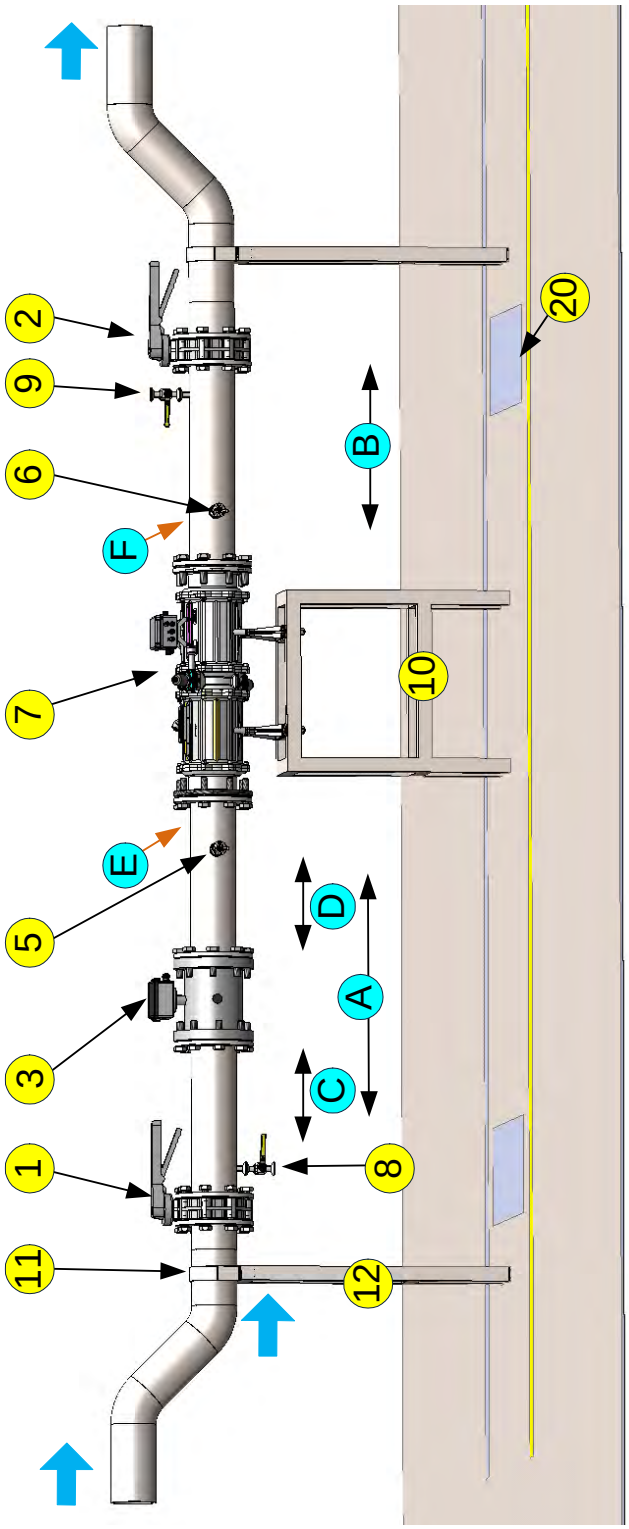


Figure 3-1: Typical Atlantium System Setup (No Bypass)

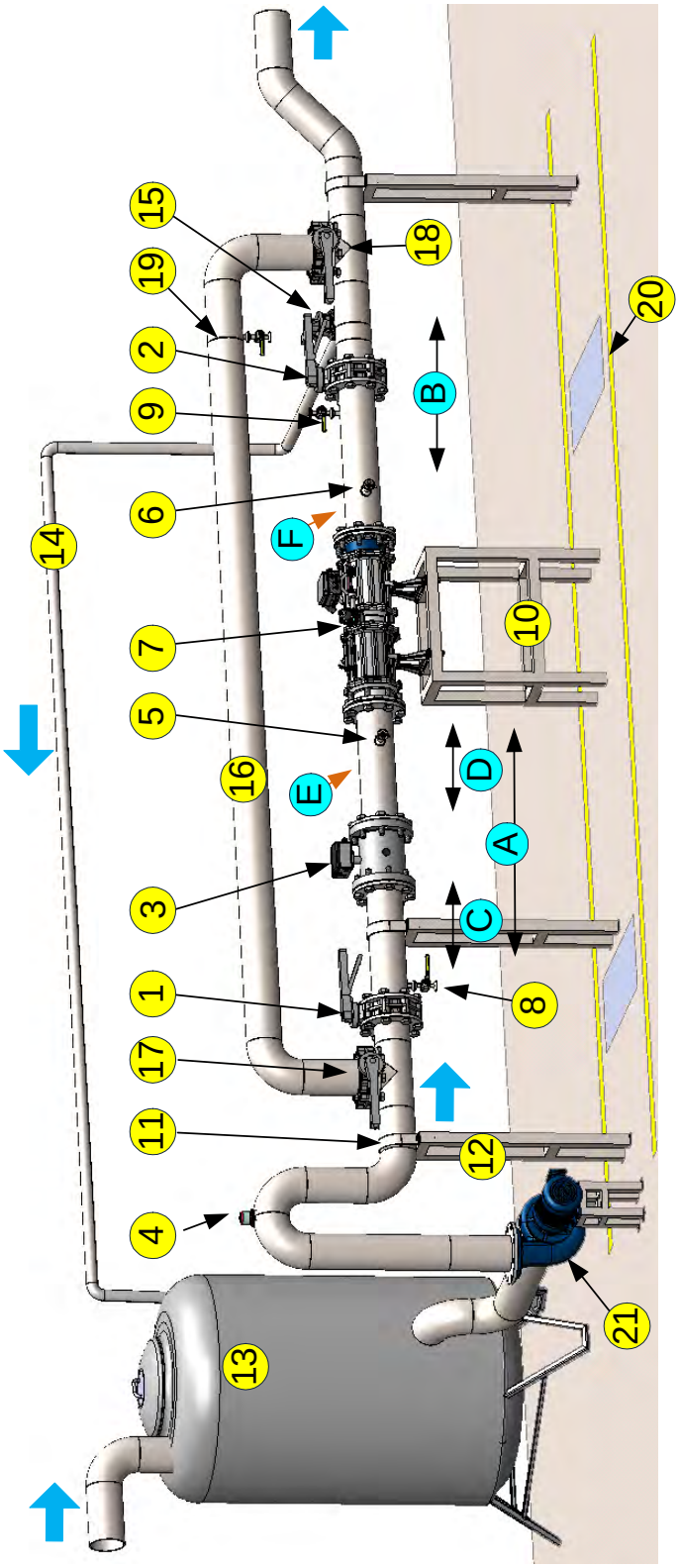


Figure 3-2: Typical Atlantium System Setup (With Bypass)

Table 3-1: Typical Atlantium System Component Setup Key

#	Description		Atlantium P/N or recommended and available at Atlantium
	Direction of water flow in the pipes		-
1	Inlet isolation Valve ■ Isolating Valve on the Inlet side		-
2	Outlet isolation Valve ■ Isolating Valve on the Outlet side		-
3	■ Flow Meter flow measurement device with 4-20mA analogue output signal ■ To be installed on the same line as the Atlantium system, with no branching before or after the flow meter ■ Follow the manufacturer's Installation instructions		Available Flow Meters: KROHNE OPTIFLUX 6000 KROHNE OPTIFLUX 2300
4	Air Release valve - To prevent air bubbles where necessary ■ Automatic Air Release valve is used to release air bubbles trapped within the line ■ Install at a high position to allow air bubbles to accumulate and release from this point		-
5/6	Inlet/Outlet Sampling Points ■ Position aseptic sampling valves **horizontally** (90^0) on the pipe center, within 50cm of the Unit ■ Aseptic sampling valve must be a dedicated valve for sampling only and resistant to alcohol and flame		■ Aseptic Sampling Valve Kit for Stainless Steel Pipe - KTB005700 ■ Aseptic Sampling Valve Kit for Plastic Pipe - KTB0011500
7	Generic Atlantium Unit (shown with flange interface. Systems are also supplied with ferrule interface.)		-
8	Draining/CIP Port - For performing CIP on the Atlantium Unit ■ The Draining port should be positioned close to the Inlet valve and between that and the Unit, ***on the bottom*** of the pipe ■ It is also used as one of the CIP ports and connects to a hose		■ CIP Kit - SAB012900 ■ CIP Valve Kit for Stainless Steel - KT0011600 ■ CIP Valve Kit for Plastic - KT0011700 (the valve itself is not part of the kits) See ***CIP Ports and CIP Kits*** on page **38.**, for information on CIP equipment.
9	CIP Port - For performing CIP on the Atlantium Unit ■ The CIP port should be positioned between the Unit and the Outlet valve, ***on top*** of the pipe, at the ***highest*** point before the isolation valve, in order to allow the pipes to be completely filled with solution		

Table 3-1: Typical Atlantium System Component Setup Key

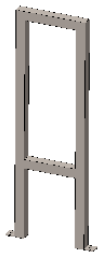
#	Description		Atlantium P/N or recommended and available at Atlantium
10	System Supports ■ Supports are required to assure no mechanical stress on the Atlantium Unit		-
11	Support brackets ■ Brackets are used to hold the Atlantium Unit and pipes attached to the support mounts ■ Is to include internal flexible material to absorb vibrations and assure that the pipes and Atlantium Unit are mounted tightly and securely		-
12	Pipe Supports ■ Pipe Supports are required to assure no vibrations in the pipeline		-
13	Water tank		-
14	Recirculation Line >5m3/h (>20gpm (Optional) ■ A Recirculation line is highly recommended to keep water flowing constantly, thereby assure a high disinfection level		-
15	Recirculation Line Isolation valve (Optional)		-
16	By-Pass (Optional) ■ If the facility uses quartz corrosive CIP chemicals, such as caustic soda, use a By-Pass line to prevent damage to the Atlantium Unit		-

Table 3-1: Typical Atlantium System Component Setup Key

#	Description		Atlantium P/N or recommended and available at Atlantium
17	By-Pass Inlet isolation Valve (Optional)		-
18	By-Pass Outlet isolation Valve (Optional)		-
19	By-pass draining port (Optional) ■ The Draining port should be positioned on the bypass line, at the lowest point of the pipe	 	-
20	Drain ■ It is recommended to install a drain close to the draining valve		-
21	Pump ■ To prevent water hammer, the pump must be able to start up gradually.		-

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
A	Straight horizontal pipe between the Inlet isolation valve and the Atlantium Unit. For most applications, a minimum 10D (1500mm/ 64inch) of straight horizontal pipe between the Inlet isolation valve and the Atlantium Unit. For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
B	Straight horizontal pipe between the Atlantium Unit and the Outlet isolation valve. For most application, a minimum 5D (750mm/32inch) of straight horizontal pipe between the Atlantium Unit and the Outlet isolation valve. For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
C	Some manufacturers require positioning the Flow Meter at a minimum of 5D (750mm/32inch) of straight horizontal pipe immediately preceding it on the Inlet side. Refer to the manufacturer's instructions.

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
D	Some manufacturers require positioning the Flow Meter at a minimum of 2D (300mm/13inch) of straight horizontal pipe immediately following it before the Atlantium Unit. Refer to the manufacturer's instructions.
E	Position the Inlet Sampling Point Maximum Distance from the Atlantium System Inlet .5 meters/1.5ft.
F	Position the Outlet Sampling Point Maximum Distance from the Atlantium System Outlet .5 meters/1.5ft.

3.3 Electrical Requirements

The electrical infrastructure is to be installed and in place prior to the system's delivery.

For information on the footprint of the components of the Atlantium system, see the section, ***The RZ163 series Units*** on page 13. Use the Electrical ICD provided by Atlantium to help you plan the electrical infrastructure. See also ***Power Consumption and Local Voltage Requirements*** on page 19 and ***Connecting the Ballast Module Cables*** on page 65.

Take into account:

For locations with 440Y/254 VAC 3~ +PE 50/60HZ
or 480Y/277 VAC 3~ +PE 50/60HZ

The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

- Evaluate the quality of your power and the likelihood of blackout/brownouts, electrical surges or interruptions. Determine if you need power surge protection and or uninterrupted power supply if any.
- To verify that you order the correct lengths for cables (i.e., Cable Harness, power cables, data cable, etc.), measure the distance between the proposed location of the relevant Atlantium components and between them and the Power source.
  - Planning the placement of the Ballast Modules, **Control Module** and the path of the various cables must take into consideration the length of the **Cable Harness** and the other cables. Cables must be placed in a path that does not interfere with passage ways.
  - The power mains connections must be in the vicinity of the Ballast Modules, Control Module.
  - The power cable and connecting plug are to be obtained locally.
  - An extra-long, 10M (32' 9.7") cable is available for the RS104 system only.
- If your region experience periodic power outages, consider including uninterrupted surge protection devices.
- Prepare cables of sufficient length to connect from the auxiliary equipment (i.e., the flow meter/flow switch, etc.) to the Atlantium system.
- **For Systems Utilizing Seawater**Minimize the corrosion potential of the treated water by ensuring that there is sufficient grounding.
- Do not leave the Atlantium Unit standing with seawater. Either rinse with fresh water and leave the Atlantium Unit standing with fresh water or empty.

3.3.1 Preparing for the Electrical Wiring to the Mains

You must prepare the elements listed below for connecting the Atlantium system to the mains.

For preparing all electrical connections, be sure to use the *Electrical ICD* drawing provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.

- Mains system area circuit breaker according to the total systems' power consumption to be connected to this line. See ***Electrical Usage*** on page **18.**
- For sites with 440/480VAC, plan to include an Over Voltage Protection Device (OVPD) (available from Atlantium) if needed. See ***The OVPD Module*** on page **21.**
- Install the Required Circuit Breakers, as defined in the Electrical ICD.
- If you intend to lengthen or replace the mains power cable connected to the Atlantium Unit with a longer cable, prepare the cables according to your location electrical distribution system requirements.

  - **For locations with 400 VAC: 5-wire system that contains 3 phases, Neutral and Ground (The Neutral and Ground are mandatory)**
  - **For locations with 440Y/254 VAC 3~ +PE 50/60HZ**
    or 480Y/277 VAC 3~ +PE 50/60HZ
    The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

  - For sites with 400VAC 3 phase + Neutral, the Control Module's Mains feeding voltage is 230 VAC (L1 to N or L2 to N or L3 to N).
  - For sites with 440/480VAC 3 phase, the Control Module **cannot** be fed from one phase of the 3 phase supply. The Control Module **must** be connected to a separate mains feeding, which is in the voltage range of 100-240 VAC.

3.4 The Piping Infrastructure

To review how to prepare the infrastructure, read through the typical installation section before you begin and refer to the section, ***About the* RZ163 *System*** on page **12**.

Install the piping infrastructure before the Atlantium system arrives, but to avoid built-in mechanical stresses perform the final pipe adaptations and welds when the unit is actually situated in its permanent position.

ATTENTION!

- When sealing the flanges/ferrules and gaskets, **do not** use sealing material that contains graphite as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- Be sure to align the piping on the inlet and outlet sides of the designated location for the Atlantium Unit with the unit to avoid mechanical stress on the Unit.
- Be sure to secure the piping around the designated location for the Atlantium Unit so that there is no vibration.
- Be sure to position adequate supports on the piping line needed to hold up and stabilize the process piping on the inlet and outlet side of the proposed location of the Atlantium unit.
- Determination of support requirements and design should be made by an on-site mechanical engineer.

3.5 Flow Measurement / Detection

The Atlantium unit requires a flow signal to calculate the dose and operate properly. The flow signal is a measurement of the actual flow or simply a flow/no-flow indicator. The flow signal is collected from either a flow meter or a flow switch.

- A flow meter is required in facilities where the flow rate is variable for any reason.
- A Flow switch can be used in facilities where the flow is at a fixed, steady rate.

The Flow meter or flow switch does not necessarily have to be installed in close proximity to the Atlantium Unit. It may be more remotely positioned, provided that it is located on the inlet side between it and the Unit and there is **no** branching or splits of the water flow.

ATTENTION!

The Atlantium system cannot be operated when there is no water or no water flow. Any damage caused due to operating the system without water or without water flow invalidates the Atlantium warranty.

3.5.1 Flow Meter

In most cases, a flow meter with a 4-20 mA output is required to measure the flow and automatically adjust the output of the UV system. The flow meter must be able to measure the expected maximum water flow of the specified Atlantium unit. The power source for the flow meter must come from an external source according to the manufacturer's instructions.

ATTENTION!

Be sure to provide the power source for the flow meter as part of the preparations for installing the Atlantium Unit.

Install the flow meter in strict accordance with the instructions supplied by its manufacturer.

Flow meters are available from Atlantium, such as:

- **KROHNE OPTIFLUX 6000** flow meter for high-purity water applications
- **KROHNE OPTIFLUX 2300** flow meter – OPTIFLUX 2000 flow meter, includes **KROHNE IFC 300** signal converter

Consult your Atlantium representative.

Examples of Flow Meters

OPTIFLUX 6000 **OPTIFLUX 2300**

Figure 3-3: KROHNE OPTIFLUX Flow Meters

Flow Switch

The flow switch tells the Atlantium Unit when water is or is not flowing. The recommended flow switch is Dry Contact with minimum flow sensing of 0.2m^3/hr (1gpm).

Install it in a vertical (12:00 o'clock) orientation on top of the pipe. Follow the manufacturer's instructions.

Figure 3-4: Examples of Flow Switch

3.6 Sampling Valve Components and Kits

Sampling valve kits are available from Atlantium.

One Accessory kit is required for each Sampling point.

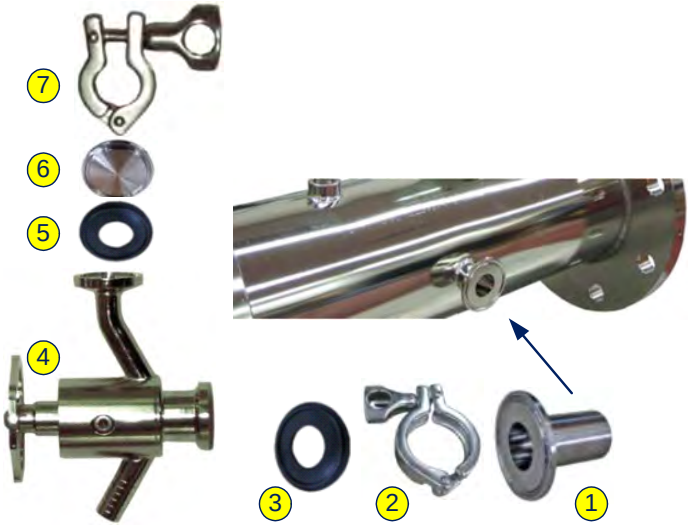


Figure 3-5: Sampling Valve Assembly

The table below details them. For information on installing the Sampling valves, see ***Installing the Inlet/Outlet Sampling Points*** on page **51**.

Table 3-3: Sampling Valve Components and Kits

	Catalog number	Description		Image	Kit for Stainless Steel Pipe KTB005700	Kit for Plastic Pipe KTB005750
1	FAC003700	Long welding ferrule 1” Adapter	For welding the tri-clamp port to the pipe		2 Pcs	-
1	FAB008100	Threaded Male Adapter 1/2” BSP	For connecting the tri clamp port to the plastic pipe		-	2 Pcs
2	FA0000045	Clamp 1.5", ST.ST 304	For connecting the valve to the welded port		2 Pcs	2 Pcs
3	FA0000046	Gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		2 Pcs	2 Pcs
4	COB000500	Aseptic sampling valve, ST.ST 316L	For taking water samples		2 Pcs	2 Pcs
5	FA0000048	Gasket 1/2”, FJ31/ 25-E	For sealing the valve's flushing port		2 Pcs	2 Pcs
6	FA0000049	Solid End Cap, 1/2", ST.ST 304	For covering the valve's flushing port		2 Pcs	2 Pcs
7	FA00000047	Clamp 1/2”, ST.ST 304	For sealing the flushing port of the valve		2 Pcs	2 Pcs
	FA0008300	Round Brush - 15/ 60/150	For cleaning the valve during the process		2 Pcs	2 Pcs

3.7 Draining Valves

It is important to plan for draining the system. Install one draining valve at the lowest point of the pipeline, before or after the Atlantium unit. If you plan to install CIP ports, one of them can double as a draining valve. If there is a Bypass line used occasionally to bypass the Unit in any circumstance, add an additional draining valve to assure that the bypass line does become a potential source of standing water. A bypass line must be empty when not in use.

3.8 CIP Ports and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve. The CIP Cart includes a reservoir, a pump, a supply hose, a return hose, and a pocket to hold these hoses. See the diagram below. Refer to **CIP Cart ICD drawing ICB012900**.

The CIP Cart:

For systems with 400VAC: catalog number: SAB012900

For systems with 440/480VAC: catalog number: SAB012950

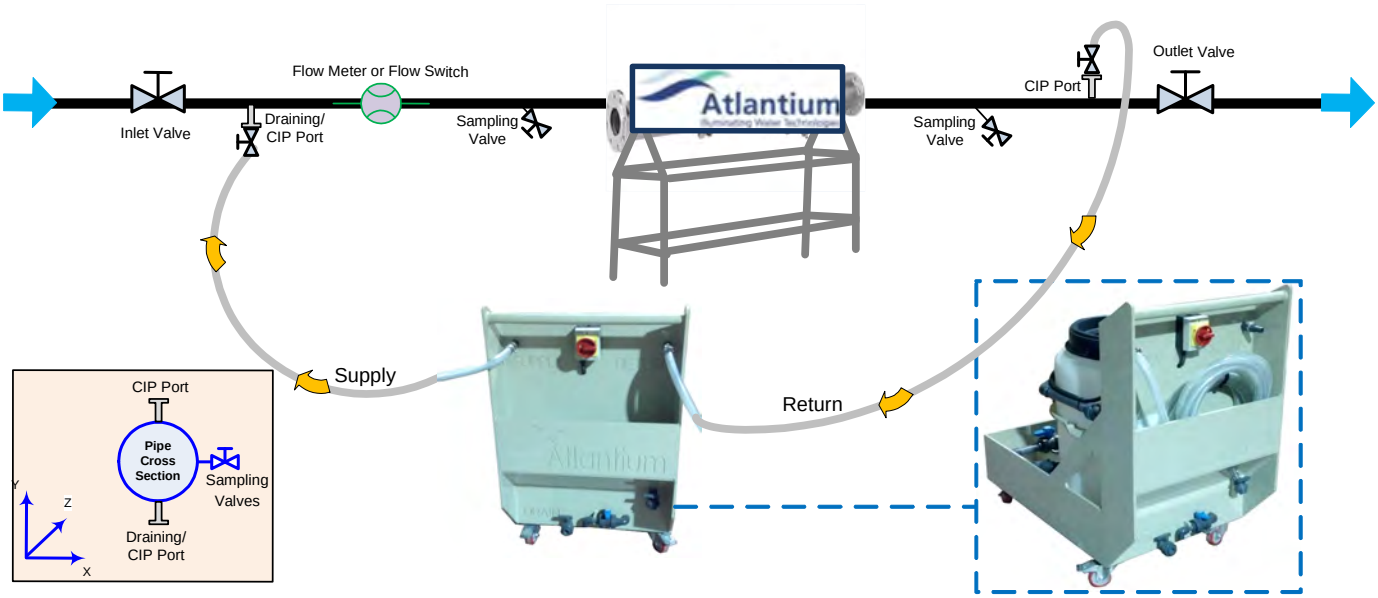


Figure 3-6: CIP Recirculation System Connections

3.9 CIP Accessories and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve.

Two Accessory kits are required, one for each CIP port.

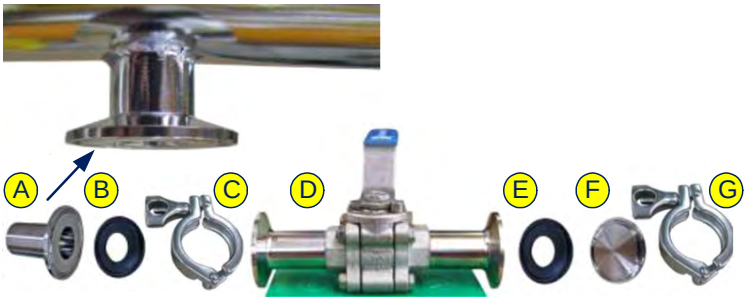


Figure 3-7: CIP Valve Kit

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
A	FAC003700	Long welding ferrule 1” Adapter	For welding the tri-clamp port to the pipe		**2 Pcs**	**-**
A	FAB008100	Threaded Male Adapter 1/2” BSP	For connecting the tri clamp port to the plastic pipe		**-**	**2 Pcs**
B,E	FA0000046	Ferrule Gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		**2 Pcs**	**2 Pcs**
C,G	FA0000045	Clamp 1.5", ST.ST 304	For connecting the valve to the welded port		**2 Pcs**	**2 Pcs**
F	FAC004100	Solid End Cap, 1 1/2", ST.ST 304, EG16A3A 1.54L	For covering the CIP port while not in use		**2 Pcs**	**2 Pcs**
D		Valves	Two high-purity chemical resistant brass ball valve (aseptic ball valves), 1 inch diameter, with tri-clamp fitting connection mechanism (A3 Standard) on both sides.		**2 Pcs Not Included Purchased by Customer**	**2 Pcs Not Included Purchased by Customer**

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
	FAB005700	Hose Adaptor 1” to 3/4"	To attach the CIP hose		**2 Pcs**	**2 Pcs**

3.9.1 CIP Safety Equipment

Required but not included in Atlantium's CIP Kits: protective equipment such as goggles, clothing and (chemical-safe) gloves, according to the safety standards specified at your facility.

3.10 Connection to Facility Control

Options are available to set up control of the Atlantium system via your facility control system. Control setup can be made via any of the following:

- **Modbus** - RTU on RS485 (network) serial connection directly to the Facility control (Appendix A, ***Modbus Communication Protocol*** on page 175)
- **Signal control** - utilizing System Ready, External On/Off, and General Alarm functions, etc. (See ***Cabling Components*** on page 67.)

With any of the above control options, you can still utilize the function of the **Control Module** of the Atlantium system. Whichever control facility is utilized, the system communicates to the other control systems such that the last command issued determines the current status.

4 Safety Overview

The Atlantium System has been designed according to the highest safety standards, ensuring the safety of operating personnel, the environment and the treated water.

DANGER!

Improper use of controls or adjustments or performance of procedures other than those specified herein could result in significant hazards.

Therefore, personnel operating or servicing this system must be thoroughly familiar with all safety requirements and operating procedures and are to adhere to them during use of this system.

To guard against injury, observe basic safety precautions, including the following:

All persons operating the Atlantium system must be aware of proper use and the potential hazards of violating safeguards. Be certain that all personnel carefully review the safety information and the procedures specified in this manual. Only authorized individuals with appropriate training and knowledge, including local regulations, should operate, assist in the operation of or provide maintenance to the system.

All service and repair of the Atlantium system must be performed by Atlantium field service technicians or other factory-authorized personnel ONLY.

UV Exposure: The system generates ultraviolet (UV) light within the Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working. Use the Viewport only. Do not look directly into the lamp enclosure during system operation, examining, or servicing the system's internal components during operation, or when lamp's caps are open, or if energized testing is required.

If the viewport is broken, DO NOT look directly into it. Be sure to request a replacement part immediately. Do not open the viewport's metal cover until the replacement is installed.

Persons potentially subject to UV light exposure – technicians who are servicing the Unit – must wear appropriate eye protection whenever the system is operating.

The Viewport is specifically designed with a UV filter to prevent UV light from escaping so it is safe to look inside. Nevertheless, personnel should not stare at it or any other brilliant light source for any but short intervals to check on Unit operations. The Viewport cover should also be preserved intact.

During maintenance procedures that are conducted with the UV lamps turned on:

- **Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

Protect Hands: **Do not touch the UV lamp with bare hands.**

Wait at least 10 minutes until the lamp is cooled down before touching the lamp or starting the replacement procedure.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

Lamps can reach a temperature of 1,000C° under operating conditions.

Wait ten minutes to allow the lamp to cool down fully before replacing it.

Keep materials that are sensitive to heat, or which contain solvents, a safe distance away from the lamp, Unit, and electrical connections.

Electric Shock: **Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, make sure the switches on the Connection Boxes of the relevant lamps are turned OFF and place a sign on the Atlantium Unit and Control Module alerting others NOT to touch the switches or the screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.**

ATTENTION!

Prevent Damage: **Handle the UV lamp assembly by holding only the contact housings (white ceramic casing).**

When handling the UV lamp, always place it on a flat surface or table, so that it is not accidentally damaged.

If the body of the UV lamp is touched accidentally, clean the fingerprints off with alcohol (at least 70%) and wipe dry with a soft, clean, lint-free cloth (usually provided with the lamp). Do not use cleaning rags or materials that can leave a residue.

Do not use a UV lamp that shows any scratches, cracks, or other damage.

Mercury Poisoning: **Lamps contain mercury, which is a hazardous substance. Always wear gloves and safety glasses when handling or replacing a UV lamp.**

Inhalation of vaporized mercury compounds can be harmful to the lungs, kidneys, and nervous system. Mercury that penetrates the skin or is ingested can also be harmful.

If mercury is inhaled, penetrates the skin, or is ingested accidentally, seek emergency medical treatment immediately.

If a UV lamp brakes during handling and releases mercury, the following precautions should be observed to minimize the risk of exposure to mercury:

- Always wear gloves and safety glasses when handling or replacing a UV lamp.
- Leave the area immediately to avoid inhalation of the mercury vapor.
- Thoroughly ventilate the area for at least 30 minutes or until the mercury vapor concentration is in compliance with applicable federal and local health and safety regulations.
- After handling a broken UV lamp, carefully remove gloves, and then wash hands thoroughly with soap and water. Follow all applicable federal and local health, safety, and environmental regulations.
- Do not turn on or operate the system until all mercury contamination has been cleaned up and removed.
- Establish a Hazard Response Plan to deal with all related hazards and provide it to all personnel associated with the Atlantium system.

Refer the Chemical Safety information at: http://www.inchem.org/documents/icsc/icsc/eics0056.htm

ATTENTION!

Lamp Compatibility: Use original Atlantium UV lamps ONLY. **Other lamps are incompatible.** They **void the warranty** and may damage the system.

Proper Disposal of Lamps

Expired lamps are classified as hazardous waste and must be disposed of appropriately.

- A non-ruptured UV lamp that is no longer used must be disposed of according to specific local or federal environmental and hazardous waste regulations.
- Broken lamps and parts contaminated by the mercury have to be vacuum-packed and disposed of. Broken lamps, as well as contaminated packing material and other parts have to be considered as special waste, which may only be removed by authorized waste disposal companies.

4.1 Electrical Hazards and Safety Considerations

DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS Of THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE **OFF** POSITION AND THAT **NO** ELECTICAL FEED IS LIVE.

DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL **ALL** ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

- **DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!**
- **The Connection Box already contains entry glands for ALL of the cabling needs.**
- **Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside.**
- **Do Not draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.**

The Atlantium Hydro-Optic Water Disinfection system incorporates high-voltage internal components, which can cause serious injury or fatal electrical shock if not used or serviced properly. High voltage components can retain a charge for some period of time even after the system has been turned off.

- Never attempt to remove any system covers or to dismantle any parts, except when performing the maintenance procedures detailed in ***System Maintenance*** on page **100**.
- No portion of the Atlantium system is be opened or removed by anyone other than a trained and authorized technician.
- Do not spray or pour any type of fluid directly into the Unit or electrical components. Moisture causes damage to the equipment and electrical shock may result.
- Do not operate the system if the power cables or harnesses are frayed or otherwise damaged.

CAUTION!

DO NOT enter **Technician Mode** in the **Control Module** unless instructed to do so by Atlantium certified personnel or by an Atlantium System User Manual, for a specific task.

If you are in Technician Mode, **DO NOT WALK AWAY!**

NEVER LEAVE THE ATLANTIIUM SYSTEM WHEN IT IS OPERATING IN TECHNICIAN MODE!!

A SYSTEM LEFT IN TECHNICIAN MODE WITHOUT SUPERVISION CAN RESULT IN PERMANENT DAMAGE!!

Utilize best practices regarding electrostatic discharge (ESD).

Avoid direct contact with the electronic circuit boards of the **Control Module,** which is sensitive to electrostatic discharge (ESD).

Wear an electrostatic discharge (ESD) Wrist Strap when performing procedures where contact with a circuit board is possible.

4.2 Keep the Unit Full of Water

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

4.3 Chemical Use for Cleaning in Place (CIP)

ATTENTION!

- The chemicals selected and the concentrations used must be approved by Atlantium's Application staff, and the ship's quality assurance and safety officials. The CIP process must comply with the written protocols and procedures set in place by those departments. Refer to the section, ***Selecting the Correct Chemicals for CIP*** on page **101**
- Use and handle all chemicals in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- NaOH (Caustic Soda) as it causes irreversible damage to quartz. It is strictly forbidden to use it without a specific analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

Refer the Chemical Safety information at: http://www.inchem.org/documents/icsc/icsc/eics0056.htm

4.4 Anti-Corrosion Spray Safety

WARNING!

When using OKS 2101 Spray:

Refer to the OKS 2101 Safety Data Sheet according to Regulation (EC) No. 1907/2006 - GB.

Use the following protective gear:

- Safety glasses with side-shields conforming to EN166
- Protective gloves that satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it
- Respiratory protection mask with filter type A-P

Follow these precautions for safe handling:

- Do not use in areas without adequate ventilation.
- Do not breathe vapors or spray mist.
- In case of insufficient ventilation, wear suitable respiratory equipment.
- Avoid contact with skin and eyes.
- Keep away from fire, sparks and heated surfaces.
- Persons with a history of skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
- Smoking, eating and drinking should be prohibited in the application area.
- Wash hands and face before breaks and immediately after handling the product.
- Do not get in eyes or mouth or on skin.
- Do not get on skin or clothing.
- Do not ingest.
- Do not use sparking tools.
- These safety instructions also apply to empty packaging which may still contain product residues.
- Pressurized container: protect from sunlight and do not expose to temperatures exceeding 50 °C. Do not pierce or burn, even after use.

4.5 System Safety Features

The Atlantium Hydro-Optic Water Disinfection system includes the following safety features:

4.5.1 UV Protective Seal

Each lamp is enclosed within a sealed lamp chamber that prevents the UV light from escaping. This means that with regular operation there is no ambient UV escaping from the Unit.

While servicing the UV Unit or checking the Intensity sensor, UV blocking safety glasses should be worn by service staff or others working in the area to prevent exposing the eyes to the UV light.

4.5.2 Lamp On/Off Switch

CAUTION!

An On/Off switch is located on the Connection Box on top of the Atlantium Unit.

In the event of an emergency, as well as for maintenance purposes, each lamp can be shut down immediately by turning this switch to the Off position.

To restore operation, rotate the button clockwise and then the system restarts.

4.5.3 Software Safety Controls on Control Module

CAUTION!

When the system is started, the Atlantium software performs a series of internal self-testing routines prior to starting system operation.

If the system **does not** pass these tests satisfactorily, UV lamp operation is disabled until the problem is resolved.

If problems arise during normal operation, appropriate error messages with varying levels of severity are generated on the touch-screen of the Control Module.

Control Module screens that provide setup options and compliance reports are password protected.

ATTENTION!

Control Module has 3 levels of user permission hierarchy **(see Table 7-2):**

- **No User** - enables view of system status screens only
- **Operator** - enables general and maintenance actions
- **Manager** - enables full system settings

All changes in the system settings are logged and tracked in the Settings Report.

DO NOT leave the Control Module unattended, even for a short time, while setting change screens are accessible. An untrained person could damage the operation of the Atlantium system by making unauthorized setting changes.

4.6 Atlantium Training

As part of the Commissioning and Hand-Off process, Atlantium provides hands-on training on all aspects of the Atlantium system for your designated personnel. Should any component be replaced with upgraded features, Atlantium offers training as needed and is ready to provide additional training for new personnel or any other needs related to its systems.

5 Installing the Atlantium System

This section provides details about Installation the Atlantium system.

For preparing all electrical connections, be sure to review the *ICD* drawings provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.

5.1 Installation Overview

Hardware setup to be implemented by your facility personnel:

Phase 1 Unpacking and checking the contents of the Atlantium system packaging (See below)
Phase 2 Installing Sampling valves and CIP ports (See page **51**)
Phase 3 Positioning the Atlantium Unit and connecting to the Piping Infrastructure (See page **54**)
Phase 4 Mounting the **Control Module(s)**, Ballast Module(s), etc. (See page **58**)
Phase 5 Cabling the **Control Module(s)**, Ballast Module(s), etc. (See page **65**)

Commissioning to be implemented by Atlantium personnel:

Phase 6 Starting up the system (See page **77**)
Phase 7 Configuring the **Control Module** (See page **57**)
Phase 8 Tuning the System (See page **151**)
Phase 9 Completing the System Setup and Testing (See page **99**)

The Installation Process Flow diagram below maps out the installation process.

5.2 Installation Process Flow Diagram

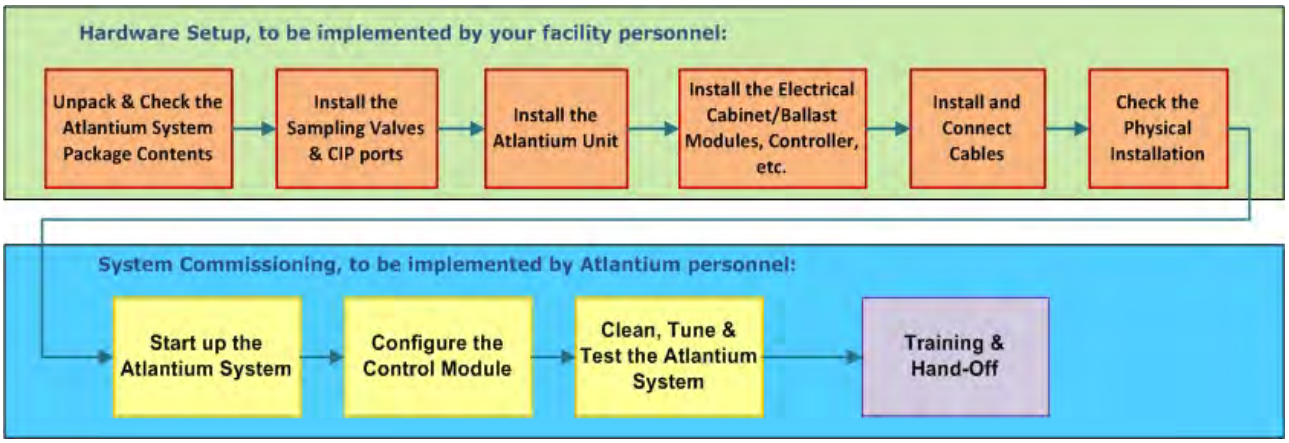


Figure 5-1: Installation Flow Diagram

5.3 Setting the Main Circuit Breakers to OFF Position

Before beginning work to connect the wires/cables of the Atlantium system components, turn the main circuit breakers of the electrical lines feeding

the Atlantium system components to the **OFF** position. **Verify that no electrical feed to the Atlantium system components is live.**

DANGER!

UNDER NO CIRCUMSTANCES
SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE OFF POSITION AND THAT NO ELECTICAL FEED IS LIVE.
DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

5.4 Unpacking And Checking Package Contents

The Atlantium system is packaged and shipped in specially packed crates that contain:

- The Atlantium Unit
- Ballast Module(s)
- Control Module
- 48VDC Module
- OVPD Module (if included)
- Documentation
- Peripheral accessories as per your order and listed on your delivery documents are crated according to their size and weight. Small accessories may be crated together.

Each of the crates is connected to a forklift-friendly wooden pallet. The Atlantium crates must be lifted off the transport vehicle using a forklift or crane by a qualified operator.

➲ To unload and check the system:

Step 1. Using the forklift or crane, lift each of the crates off the transport vehicle and lower them to a level ground surface.

Step 2. Walk around and inspect each crate for damage. Note if there is any separation in the joints or any other indication of any damage in transit. If there is any indication of damage, take a photograph of it and use extra caution when proceeding to the next step.

Step 3. Unclamp and remove the panels of the crates.

Step 4. As you remove the wrapping and shipping protections, verify that the contents of the crates are correct according to the delivery documents. The Atlantium Unit contains a plate with the model number and other product information. Verify that the correct product version is supplied.

- Immediately report to Atlantium and to the insurance carrier any damage to the crates or to the Atlantium system found before opening the crates or during every step of unpacking, installing or testing of the system.
- If you ordered more than one Atlantium system, check the serial numbers against the delivery documents to verify which components comprise each individual system.

5.5 Installing Sampling Points and CIP/Draining Ports

Both the Sampling Points and CIP/Draining Ports must be installed on the inlet and outlet sides of the Atlantium Unit.

5.5.1 Installing the Inlet/Outlet Sampling Points

Aseptic sampling valves are to be located indoors in clean environment without away from direct contact with fan or A/C air flow. They each must be a dedicated valve for sampling only and resistant to alcohol and flame. For information on the Sampling valve's components, see ***Installing Sampling Points and CIP/Draining Ports*** on page **51**. See ***Taking Microbial Samples*** on page **97.** for information on using the Sampling valves.

On the inlet and outlet sides of the water line, connect the aseptic sampling valves.

➲ To install an inlet/outlet sampling point:

Step 1. On the Inlet side, install the **Long welding ferrule** or **threaded male adapter (1).**

- Position aseptic sampling valves **(1 & 2** in **Figure 5-6, on page 53) horizontally** (90^0) on the pipe center, within 50cm of the Unit.
- **For Stainless Steel. Pipes:**
  - Weld the 1 inch **ferrule (1)** clamp fitting at a horizontal orientation (3 or 9 o'clock).

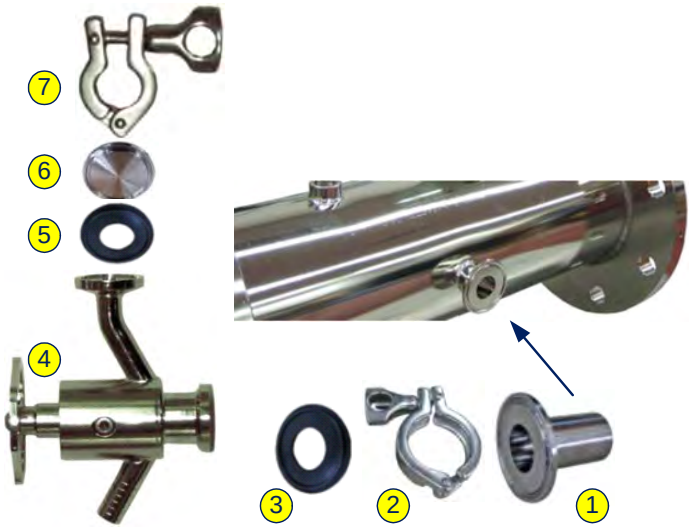


Figure 5-2: Sampling Valve Assembly

- Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
- Passivate the pipes to avoid corrosion.

■ **For Plastic Pipes:**

- Thread the **male adapter (1)** clamp fitting to the port.

■ Repeat for the Outlet side sampling point.

Step 2. Fit the gasket **(3)** into the **welding ferrule** or **threaded male adapter (1)**.

Step 3. Fit the aseptic **sampling valve (4)** on over the gasket **(3).**

Step 4. Fit the clamp **(2)** around the aseptic **sampling valve (4)**/ gasket **(2)**/**welding ferrule** or **threaded male adapter (1)** and close and lock it.

Figure 5-3: The clamp is placed around the aseptic **sampling valve** /gasket /**welding ferrule** or **threaded male adapter**

Closing Off the Aseptic Sampling Valve When Not in Use

Step 5. Into the spout of the sampling valve, fit the 1/2”gasket **(6)** and fit the cap **(7)** over it.

Step 6. Fit the 1/2” clamp **(5)** around the sampling valve spout/ gasket **(6)** /cap **(7)** and close and lock it.

Figure 5-4: The clamp is closed

Figure 5-5: The clamp is closed

5.5.2 Installing the CIP/Draining Ports

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve. For more information on performing CIP, see ***Cleaning In Place (CIP)*** on page 117.

A CIP KIT includes the CIP Cart with water reservoir and a pump, capable of pumping up to 4 cubic meters per hour. It includes a supply hose: ID 0.75" / 19mm (5 meters / 15 feet long) and a return hose: ID 0.75" / 19mm (5 meters / 15 feet long) and pocket to hold these hoses. See the diagram below.

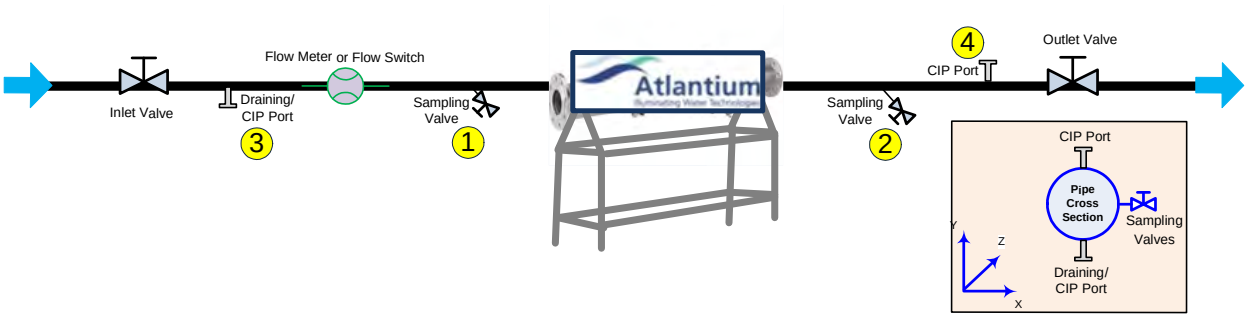


Figure 5-6: CIP Ports and Sampling Valves Orientation

Position the CIP/draining valves **(3 & 4)** on both the inlet and outlet sides of the Atlantium unit. To enable connection of the CIP kit, use high-quality, sanitary equipment, including a 1 inch tri-clamp/ferrule on both sides. A CIP/Draining valve is to be located 1D after the Inlet valve and 1D before the Outlet valve. When not using the CIP [ports. close the valve outlet using the tri-clamp cover.

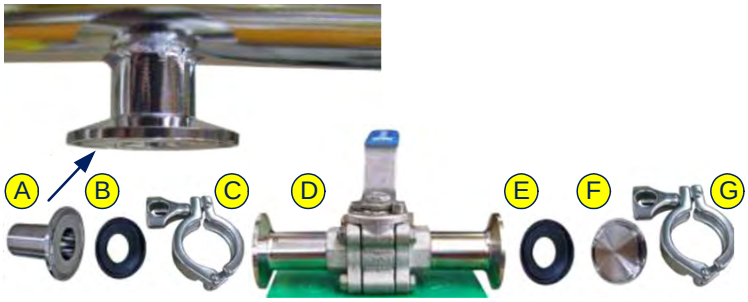


Figure 5-7: CIP Valve Kit

➲ To install the CIP ports:

Step 1. Weld or thread the 1 inch ferrule at a vertical orientation:

For municipal applications, prepare 1” screw ports for the valves and insert the valves:

- **Draining Valve:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
- **CIP Valve:** at a 12:00 o'clock orientation on other side of the Unit.

For all application except municipal:

- **For Stainless Steel. Pipes:**
  Weld the 1 inch clamp fittings:
  - **Draining port:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
  - **CIP port:** at a 12:00 o'clock orientation on other side of the Unit.
  - Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
  - Passivate the pipes to avoid corrosion.
- **For Plastic Pipes:**
  - Thread the clamp the valve fittings to the port..

Attaching the Valve to the ferrule

Step 2. Fit the gasket **(B)** into the **welding ferrule** or **threaded male adapter (A)**.

Step 3. Fit the aseptic **valve (D)** on over the gasket **(B).**

Step 4. Fit the clamp **(C)** around the aseptic **valve (D)**/gasket **(B)**/**welding ferrule** or **threaded male adapter (A)** and close and lock it.

Closing Off the CIP/draining Ports When Not in Use

Step 5. Into the ferrule, fit the gasket **(E)** and fit the cap **(F)** over it.

Step 6. Fit the clamp **(G)** around the CIP port/ gasket **(E)** /cap **(F)** and close and lock it.

Preparing the CIP Hoses

Step 7. On one end of a hose **(A)**, using an appropriate screwdriver, loosen the band **(C)**.

Step 8. Connect the hose to one of the adapters on the cart **(B)**. Push the hose up a far as you can.

Step 9. Using the screwdriver, tighten the band **(C)**.

Step 10. On the other end of a hose **(D)**, using an appropriate screwdriver, loosen the band **(F)**.

Step 11. Connect the hose to one of the hose adapters **(E)**. Push the hose up a far as you can.

Step 12. Using the screwdriver, tighten the band **(F)**.

Step 13. Repeat with the second hose to the adapter on the other side of the cart.

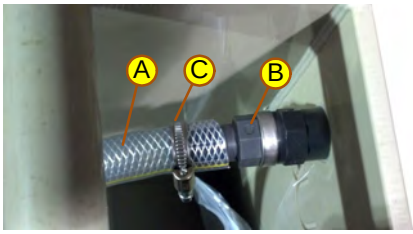


Figure 5-8: Hose Connection to One Side of Cart

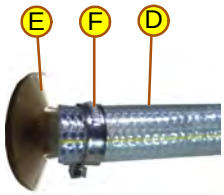


Figure 5-9: Hose Connection to Hose Adapter

5.6 Positioning the Atlantium Unit

Read through this typical installation section ***before*** you begin.

Make sure to use your installation drawing during this procedure. The Unit's triangle support brackets must rest on a strong, stable, level surface and anchor the Unit.Trace the path to be taken to position the Unit to its planned location and clear it of all obstacles.

Atlantium systems can be installed horizontally or vertically in your piping system. For a vertical installation, the Atlantium Unit can be positioned against a wall or on skids.

5.6.1 Required Equipment

Supply the following tools and equipment:

- Lift or crane
- Lifting strap
- Bolts, washers & lock washers
- Supports for inlet/outlet piping and for the Unit as shown on your installation plan

The Atlantium unit is heavy - refer to Table 2-3, ***Ballast Module Physical Details***, on page 18. Two people are required to get the Atlantium Unit to the designated installation location - a qualified lift or crane operator, at least one other person to guide the Unit as it is transported, and others as needed.

To position the Atlantium unit:

Step 1. Note the direction of the water flow that is to occur in the piping process into which you intend to place the Atlantium Unit and the Flow indicator marked on the unit. Use this information to decide at which angle to lift the Unit for transport. This is to help you to prepare for placing the Unit at the location.

Step 2. To lift the Unit from its packing crate, loop a lifting strap around the end fittings on each side of the Unit.

Step 3. Join the straps togethe(1&2)r with a third central strap (3) that is being connected to the lift or crane(4).

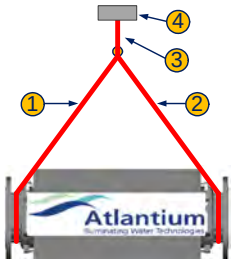


Figure 5-10: Lifting Strap Locations

Step 4. Ensure that all straps are connected securely to the Unit, to each other and to the lift/crane.

Step 5. Using the lift/crane, lift the Unit by the straps, out of the crate and position it for safe transport (near-ground level is preferred).

Step 6. Using the lift/crane and the designated person to guide, carry the Unit to its intended location.

Step 7. Install the Atlantium Unit:

Horizontal Installation

a At the designated location, check the installation plan, including the distances between the triangle support brackets.

b Position the customer-provided supports /surface so that the Unit's triangle support brackets are at least 750mm / 29.5 inch above the floor. (Refer to the **RZB300** ***Systems' Interface Control Drawings*** provided by Atlantium.)

c Carefully lower the Unit to its position.

d Using the appropriate bolts, washers and lock washers, bolt the triangle support brackets to the customer-provided corresponding supports.

Vertical Installation

a Position the Atlantium Unit vertically so that the Flow label is point in up
To verify if the position is correct, check that the :

- **RZ163-11** Unit is positioned so that the view port is on the **bottom** end
- **RZ163-12** Unit is positioned so that the view port is on the **top** end
- **RZ163-13** Unit is positioned so that the view port is on the **bottom** end
- **RZ163-14** Unit is positioned so that the view port is on the **top** end

Figure 5-11: Flow Label Points Up

b Position the Unit's triangle support brackets against the wall or skids.

c Carefully lower the Unit and using the appropriate bolts, washers and lock washers, bolt the triangle support brackets to the customer-provided corresponding supports.

Step 8. To connect the fitting on one side of the Unit to the fittings of the facility's pipes, insert an o-ring between the Unit's and the facility's pipes. In some cases, a reducer or expander may be necessary.

- For systems with flange interface, on one side of the Unit to the flanged fittings of the facility's pipes with eight appropriate **(8)** sets of bolts/nuts/flat and lock washers - do not tighten the hardware.
- For systems with Tri-Clamp fittings, on one side of the Unit, clamp the fittings together - do not tighten the hardware.

Step 9. Repeat the step above for the other side of the Unit.

Step 10. Tighten the fastening hardware so that the torque pressure around the fittings is even, and the Unit is firmly connected to the facility's water line.

Step 11. For installations with multiple units, repeat this procedure for the second unit.

Step 12. To check for any leakage, open the Inlet and Outlet valves to allow water to flow through the newly installed Unit and its surrounding process piping.

Step 13. If you find any leakage, tighten the connections.

Step 14. Close the Inlet and Outlet valves and use the Draining valve, drain the Atlantium system piping from Inlet valve to Outlet valve.

Figure 5-12: Generic Flange fitting to the Water Line

Figure 5-13: Generic Tri-Clamp Fitting

5.7 Installing Surge Protection

Verify that you have a protective device that prevents electrical surges and fluctuations from causing damage. You must have at least one of these three types of surge protection:

- UPS (Uninterrupted Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.
- Surge protection device protects your equipment from variations in electrical current, such as surges and spikes.
- OVPD module (Over Voltage Protection Device) - if you have purchased this from Atlantium, see below for directions on mounting and connection it.

5.7.1 Installing a Surge Protection Device

The diagram to the right shows the generic wiring for a surge protection device. See the manufacturer's installation instructions with the surge protection device to be installed. For more information about surge protection devices available via Atlantium, see ***Surge Protector Devices*** on page 21.

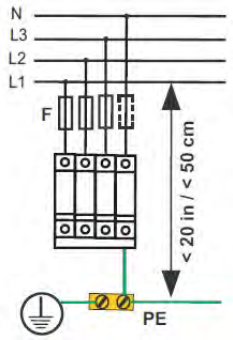


Figure 5-14: Surge protection device wiring - 400VAC

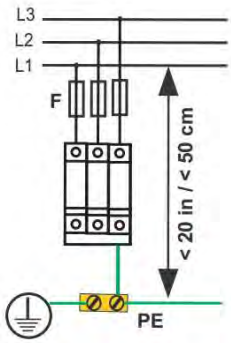


Figure 5-15: Surge protection device wiring - 440/480VAC

5.7.2 Mounting and Connecting the OVPD Module (if included)

If your system includes the OVPD Module (see ***The OVPD Module*** on page 21), mount it in an appropriate location in your electric box and connect it according to the directions in this section. refer to the ***Electrical ICD*** drawing provided to you by Atlantium with this manual.

➲ To connect the OVPD

Step 1. To connect the mains to the OVPD module, thread the mains cables (customer provided) through the openings at the bottom of the OVPD module **(B)** and connect them to the terminal block for the mains **(C)** according to the diagram below.

Step 2. To connect the cables coming from the circuit breaker before the Atlantium Unit, thread the cables through the opening at the top of the OVPD module **(A)** and connect them to the terminal block for these cables **(D)** according to the diagram below.

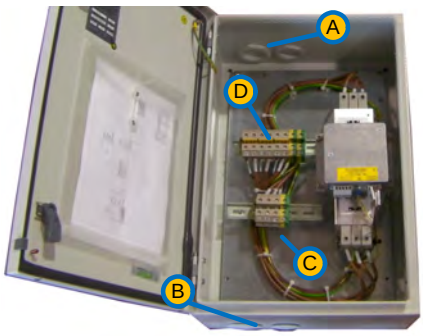


Figure 5-16: OVPD Connections

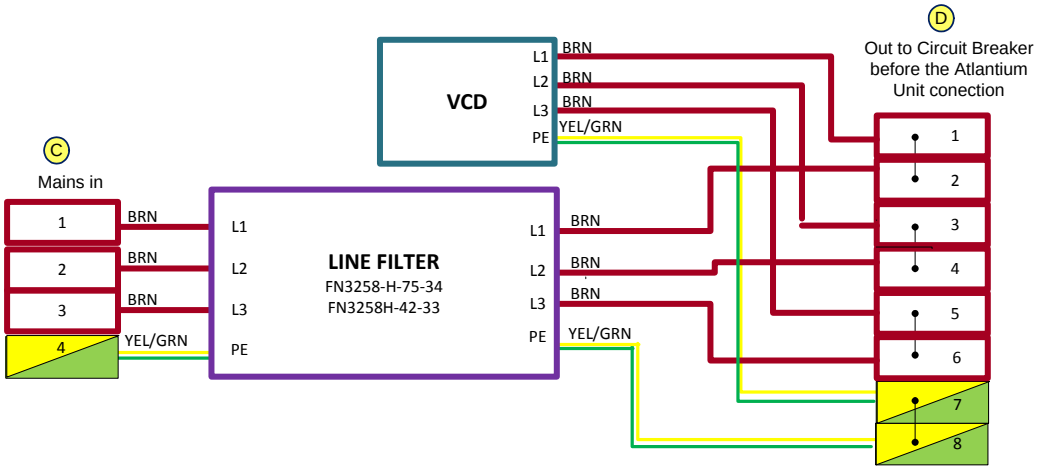


Figure 5-17: Typical Atlantium System Connections for the OVPD

5.8 The Control Module

The **Control Module** contains a jumper marked **11** in the figure below.

Table 5-1: **Control Module** Components

#	Item
1	Mounting Holes
2	Data Cable + Gland
3	Electrical Cable + Gland
4	Electrical Connectors
5	Holes for Peripheral Component Connections
6	Various Component Connectors (See Table 5-7 on page 71)
7	Ethernet Connector
8	Circuit Breaker
9	Wireless Modem (factory installed)
10	Connector for Wireless Modem (factory connected)
11	Jumper
12	Gnd Wire
13	Connection to Screen Display
14	Two USB Ports (top side of the board)

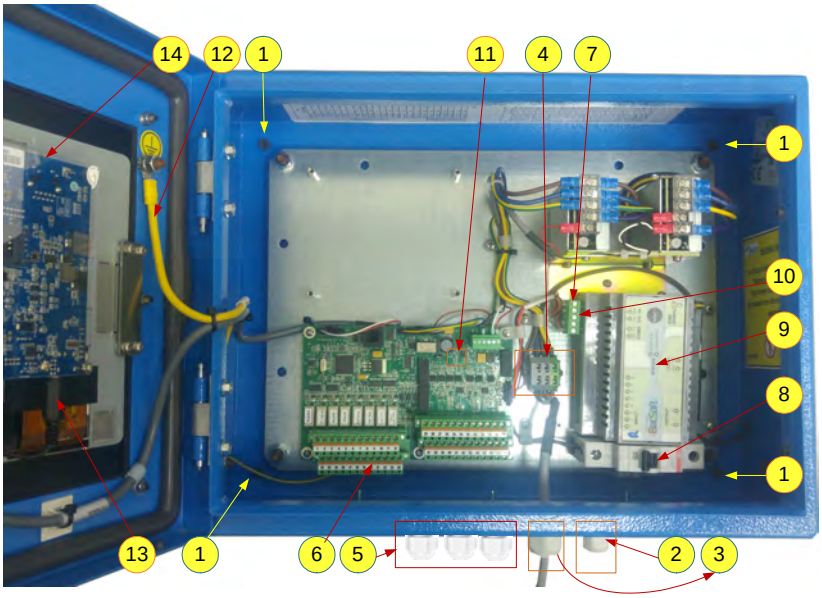


Figure 5-18: Control Module Interior

5.8.1 Mounting the Control Module

The **Control Module** can be mounted on a wall or suitable anchoring point. The **Control Module** comes with an electrical 1.6M/5.25ft. long (1.5m/4.92ft. from the PG gland) and a Data cable (for length options, see the table below). However it can be placed up to 50M/ 164.04 ft. from the Atlantium Unit, in a place that it is easy to reach the touch screen and read the monitoring information. A longer Data Cable can be ordered for this purpose.

ATTENTION!

- When mounting the **Control Module**, take into consideration the length of the cables.
- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

5.9 Mounting the 48VDC Module

The 48VDC Module can be mounted on a wall or suitable anchoring point up to 10 meters (32.8 ft.) from the Atlantium unit. The 48VDC Module can be positioned near a Ballast Module with a minimum of 300 mm (11.8 inches) of clearance on each side for proper ventilation.

The 48VDC cable, Data cable, and Power cable are supplied attached to the 48VDC Module.

ATTENTION!

- **When mounting the 48VDC Mod**ule take into consideration the length of the cables.
- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

To mount the 48VDC Module on the wall:

Step 1. On the wall space designated for the 48VDC Module, make four marks according to the spacing measurements for mounting orientation on the ICD drawing (see the ICD provided by Atlantium).

Step 2. The mounting holes **(A)** are 8.5mm (0.33") in diameter. Using the appropriate drill bit, drill the four holes.

Step 3. In each hole, insert a plastic anchor.

Step 4. Uncrate the 48VDC Module package and carefully lift it out.

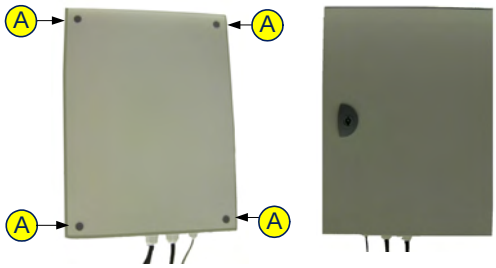


Figure 5-19: 48VDC Module - Back and Front View

Step 5. Open the door of the 48VDC Module and locate the mounting holes **(A)**.

Step 6. Remove the protective covers from each hole.

Step 7. Position the 48VDC Module so that its four holes fit over the drilled holes.

Step 8. Using appropriate screws and with a screwdriver, screw in the four screws in a diagonal pattern (one corner, then the opposite corner).

Step 9. To secure the Unit firmly, return to each screw in the diagonal pattern and tighten the screws securely.

Step 10. Verify that the circuit breakers are on the **Off** position.

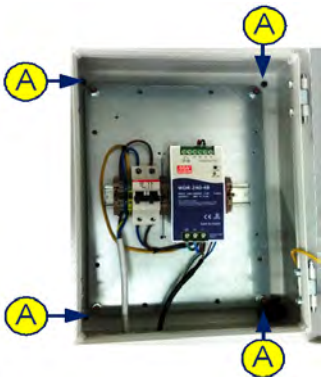


Figure 5-20: 48VDC Module - Generic Inside View

To mount the 48VDC Module on skids:

Step 1. At the location designated for the 48VDC Module, set up the skids with spacing that matches the spacing measurements for mounting orientation on the appropriate ICD drawing (see the ICD provided by Atlantium).

Step 2. The mounting holes **(A)** are 8.5mm (0.33") in diameter. Using the appropriate bolts and nuts, with a screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24') sticking out.

Step 3. Uncrate the 48VDC Module package and carefully lift it out.

Step 4. Open the door of the 48VDC Module and locate the mounting holes **(A)**.

Step 5. Remove the protective covers from each hole.

Step 6. Position the 48VDC Module so that its four holes fit over the corresponding holes on the skids.

Step 7. Using appropriate bolts and nuts and with a screwdriver, screw in the four bolts adding the nuts in a diagonal pattern (one corner, then the opposite corner).

Step 8. To secure the Unit firmly, return to each bolt in the diagonal patter and tighten them securely.

5.10 Installing the Ballast Module(s)

Each lamp chamber has a corresponding Ballast Module. The location for mounting the Ballast Module(s) must take into account the length of the cables to be attached from the Atlantium Unit. If you need to calculate space cooling for the location in which you are installing the Ballast Module(s), refer to the heat load per Ballast Module shown below.

Heat Load per Ballast Module:	
Standard Lamp:	85W
High Power Lamp:	115W
Ultra High Power Lamp:	151W
(For calculating space cooling requirements)	

ATTENTION!

- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

Ballast Modules must be installed vertically and positioned at least one meter above ground level. Ballast Modules must be at least 300mm (12") from each other or from any other object around them to allow proper ventilation. Ballast Modules must not be installed within closed cabinets.

You can mount them next to each other horizontally or one above another vertically. Atlantium recommends you arrange the Ballast Modules in consecutive order from left to right according to the number marked on the bottom of the Ballast Module.

Consult the ***Ballast Module's ICD*** accompanying this manual.

BALLAST MODULE A

(Passive Cooling version)

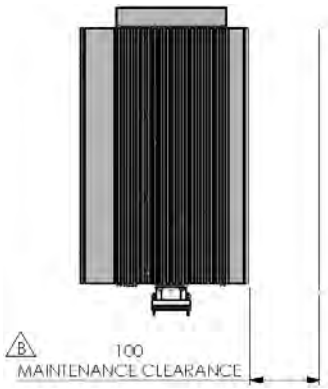


Figure 5-21: Ballast Module A Depth

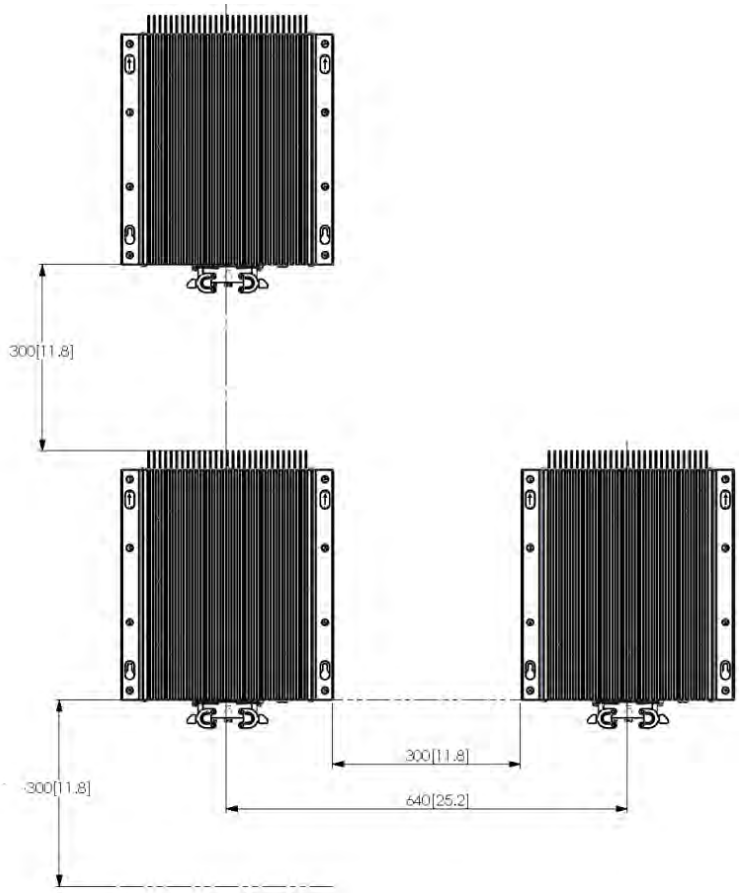


Figure 5-22: Multiple Ballast Module A Installation Diagram

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BALLAST MODULE B

(Active Cooling version)

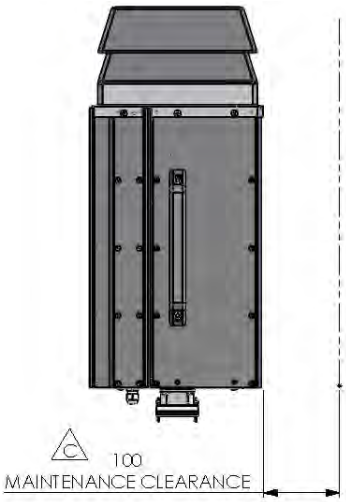


Figure 5-23: Ballast Module B Depth

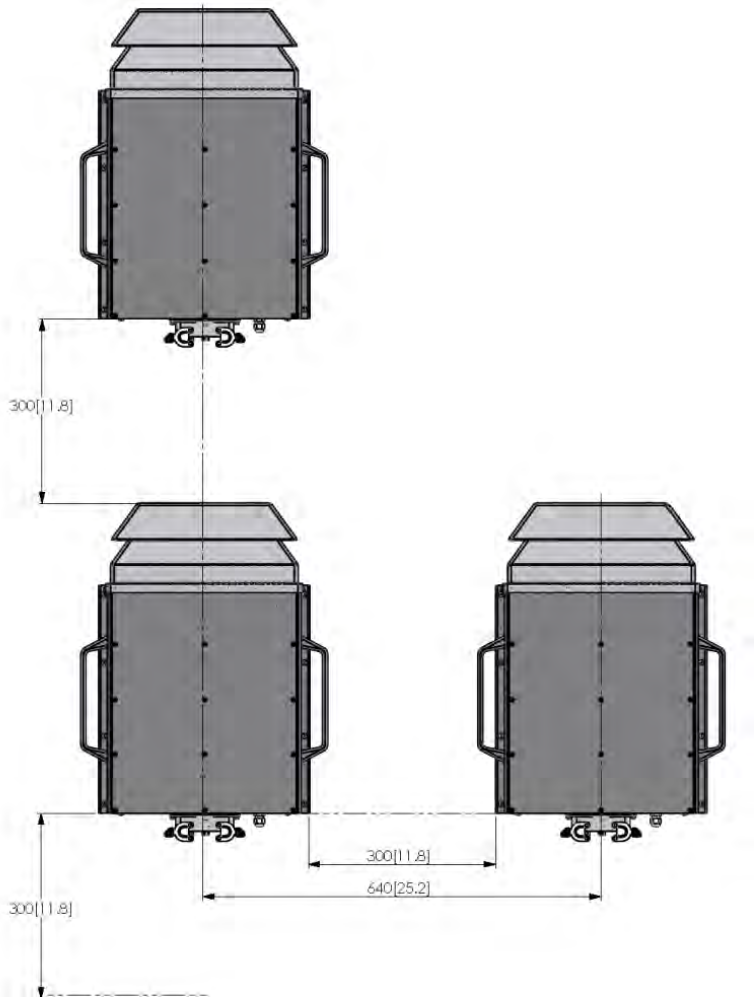


Figure 5-24: Multiple Ballast Module B Installation Diagram

BALLAST MODULE C

(Power Unit version)

Figure 5-25: Ballast Module C Depth

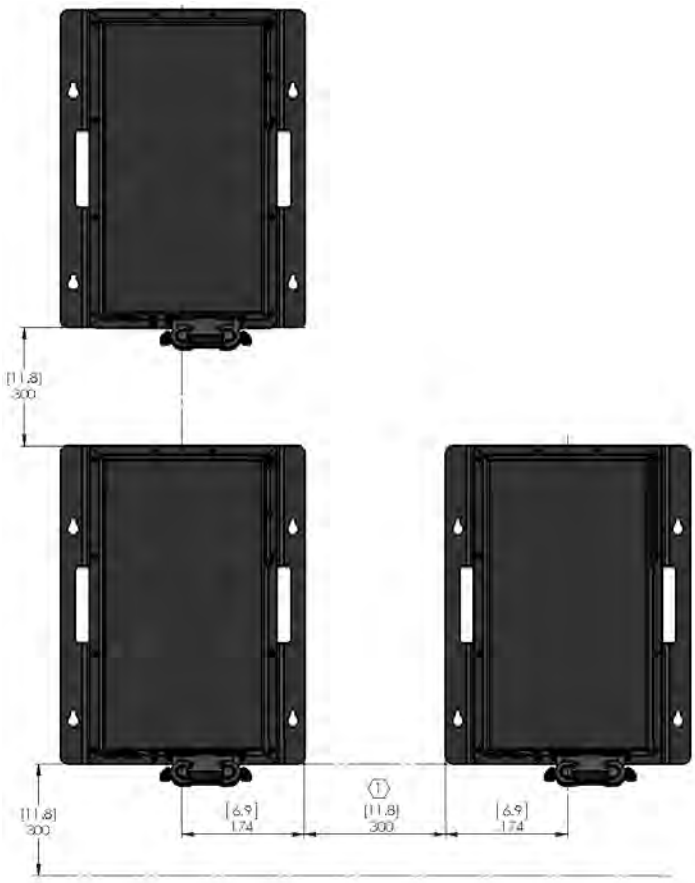


Figure 5-26: Multiple Ballast Module C Installation Diagram

➲ To mount a Ballast Module on the wall:

The Ballast Module is heavy! Be sure that the wall on which you intend to mount the Ballast Module is strong enough to carry its weight.

Step 1. On the wall space designated for the Ballast Modules, make four marks according to the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.

Step 2. The keyhole slots for mounting are 7mm (0.28”) in diameter for screws size M6 OR 1/4” bolts. Using the appropriate drill bit, drill the four holes.

Step 3. In each hole, insert a plastic anchor and with the appropriate screwdriver, screw in the screw leaving around 5.6mm (0.20-0.24’) sticking out of the wall.

Step 4. Uncrate the Ballast Module package and carefully lift out the Unit.

Step 5. Fit the keyhole slots onto the protruding screw heads and slide the Unit downward until the top of the keyhole slots rest on the screws.

Step 6. To secure the Unit firmly, tighten the four screws.

➲ To mount a Ballast Module on skids:

Step 1. At the location designated for the Ballast Modules, set up the skids with spacing that matches the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.

Step 2. The keyhole slot for mounting is 7mm (0.28”) in diameter for screws size M6 OR 1/4” bolts.

Step 3. Using the appropriate screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24’) sticking out.

Step 4. Uncrate the Ballast Module package and carefully lift out the Unit.

Step 5. Fit the keyhole slots onto the protruding bolt heads and slide the Unit downward until the top of the keyhole slots rest on the bolts.

Step 6. To secure the Unit firmly, tighten the four bolts and their nuts.

5.11 Connecting the Ballast Module Cables

One side of the **Cable Harness** is attached to the connection box of each lamp of the Atlantium Unit. Refer to the ***Electrical ICD*** drawing provided to you by Atlantium with this manual for the connection of the other end of the **Cable Harness**.

Table 5-2: Ballast Module Details

#	Item
A	Single Connector with lock latch
B	48VDC for Fan - 440/480VAC only
C	Pressure Relief
D	Fuse Housing (for Atlantium Technicians only)
E	Connector width 13.11cm / 5.159inch.

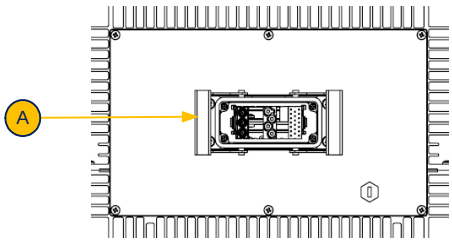


Figure 5-27: Ballast Module A-Waterproof Box

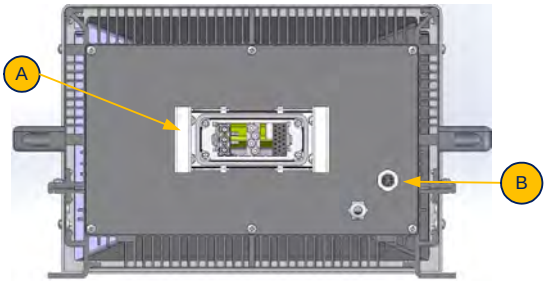
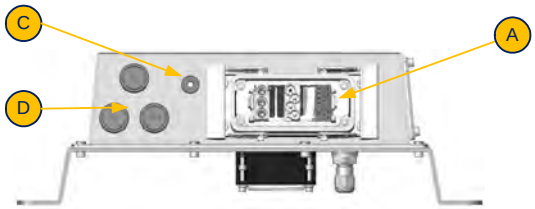


Figure 5-28: Ballast Module A-Waterproof Box

Figure 5-29: Connector Detail



Ballast Module A-Waterproof Box

➲ To connect the cable:

Step 1. According to the electrical drawing of the Installation plan, bring the **Cable Harness** attached to the Connection Box of Atlantium Unit up to the Ballast Module.

Step 2. On the Ballast Module, pull the lock latch of the connector to the open position and remove the protective cover.

Be sure to save the protective cover of the Ballast Module connector in an accessible location for the future in case you need to dismantle the Ballast Module for any reason.

Step 3. Connect the lamp cable lead to the Lamp connection port (**A** in the figures above) (the connector is keyed so it only matches in one position) and pull the lock latch closed to secure the connection.

Step 4. For system with electrical supply 440/480VAC only, connect the 48VDC cable for the fan (**B** in Figure 5-29 above).

5.12 Cabling Components

According to your system configuration, connect the cables of the mandatory flow signal, and optional Inlet, Outlet/Ready and Bypass Valves, etc. to the Connection Box of lamp 1 on the Atlantium Unit. On one side of the Connection box, there are 6 holes (marked **9** in the diagram Figure **5-22** on page **63**) to be used to thread these cables.

WARNING!

DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!

The Connection Box already contains entry glands for ALL of the cabling needs.

Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside. Do Not draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

The directions below are for all **RZ163 series** systems, in which the first Connection Box is labeled number **1**. Where there is reference to the first Connection Box, for systems that contain only one Connection Box the single Connection Box is the first Connection Box.

5.12.1 Connection Box Card 1300+1400

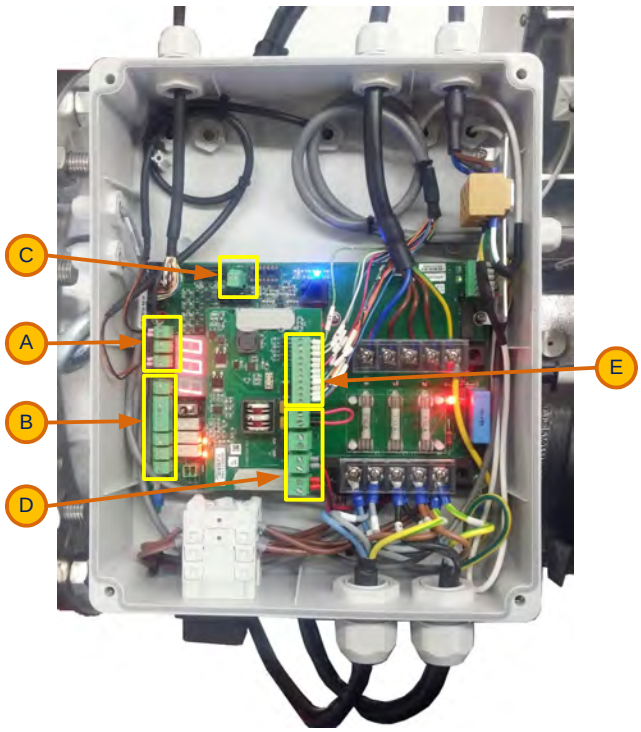


Figure 5-30: Connection Box Card 1300+1400

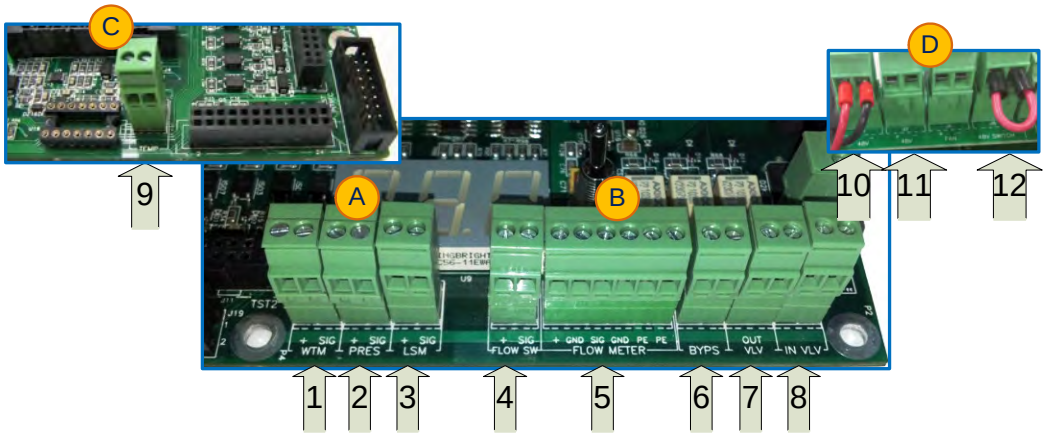


Figure 5-31: Connectors on Connection Box Card 1300+1400

Table 5-4: Connections Key - Connection Box Card 1300+1400

#	Text on the Circuit Board	Description	Signal type	Value	Notes
A					
1	WTM	UVT Analyzer input	Analog	4-20mA	already assembled
2	PRES*	Pressure Transmitter input (Optional)	Analog	4-20mA	Optional
3	LSM	UVIS Sensor input	Analog	4-20mA	already assembled

Table 5-4: Connections Key - Connection Box Card 1300+1400

#	Text on the Circuit Board	Description	Signal type	Value	Notes
B					
4	FLOW SW*	Flow switch input	Digital	Dry contact / Open collector	Either Flow Switch or Flow Meter Optional *
5	FLOW METER*	Flow meter input	Analog	4-20mA	
6	BYPS STR*	Bypass valve output	Dry contact		Optional
7	OUT VLV*	Output valve output	Dry contact		Optional
8	IN VLV*	Inlet valve output	Dry contact		Optional
C					
9	TEMP	Temperature sensor input	Analog		already assembled
D					
10	48V	48V power connection for DPM			already assembled
11	48V	Daisy chain connector for additional lamps for 48V power connection for DPM			already assembled
12	48V Switch	Jumper			already assembled
E	Various	DPM connectors			already assembled

➲ To cable signal components:

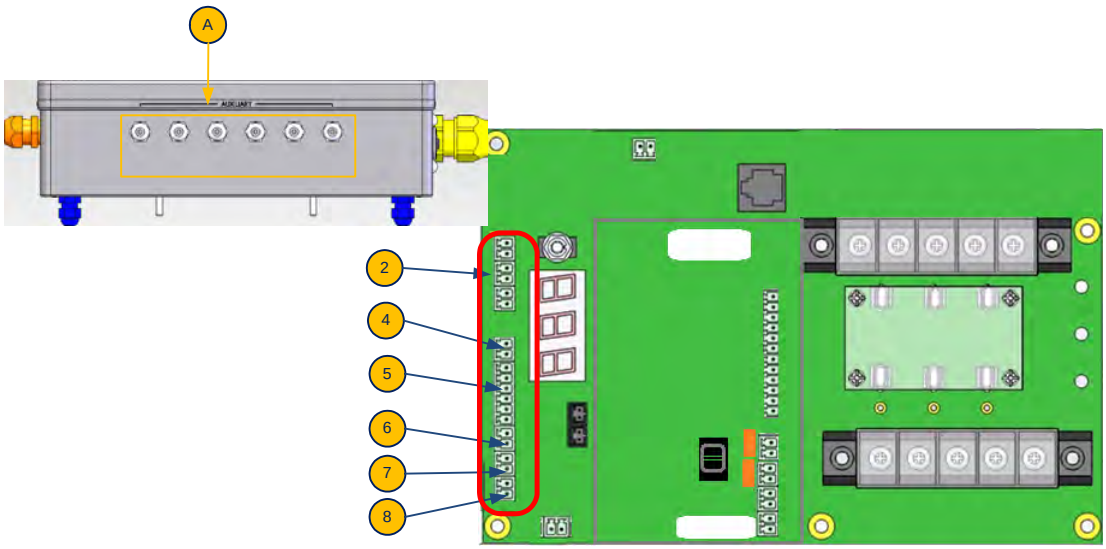


Figure 5-32: Connection Box 1300+1400 Side and Inside View-Auxiliary Components Connections

Step 1. On the first **Connection Box**, remove the covers of holes (marked **A** in the diagram below) as needed for the auxiliary components in your system configuration.

Step 2. Add plastic glands to each hole to be used.

Step 3. For the (optional) **Inlet Valve** connection, the maximum voltage is **24V max**. The maximum current is **2A max**. The maximum switching capacity is **50W max**. On the first Connection Box, insert the **Inlet Valve** cable into one of the holes marked **A** and thread it up to the Dry Contact connectors labeled **IN VLV** (marked **8** in the diagram above.)

Step 4. For the (optional) **Outlet Valve** connection, the maximum voltage is **24V max**. The maximum current is **2A max**. The maximum switching capacity is **50W max**. On the first Connection Box, insert the **Outlet Valve** cable into one of the holes marked **A** and thread it up to the Dry Contact connectors labeled **OUT VLV** (marked **7** in the diagram above.)

Step 5. For the (optional) **Bypass Valve** connection, the maximum voltage is **24V max** The maximum current is **2A max**. The maximum switching capacity is **50W max**. On the first Connection Box, insert the **Bypass Valve** cable into one of the holes marked **A** and thread it up to the Dry Contact connectors labeled **BYPS STR** (marked **6** in the diagram above.)

Step 6. To connect the **Flow Signal:**

- For a **Flow Meter**, on the first Connection Box, insert the cable into one of the holes marked **9** and thread it up to the connectors labeled **FLOW METER** (marked **5** in the diagram above.) the standard current is 4-20mA. Connect the wires according to the manufacturer's instructions. The connectors are marked **PE**, **PE**, **SIG**, **GND**, and **+**. Typically, the cable is comprised of three wires and the shield. A wire connects to the **+** port and a second wire connects to the **SIG** port. The GND wire connects to the **GND** port. The Shield connects to one of the **PE** ports.

ATTENTION!

Use the **+** pin for the signal source (current loop) **ONLY**.

Do not use the **+** pin as the power source for the flow meter. For more information see ***Flow Measurement/ Detection*** on page **35**.

- For a **Flow Switch**, on the first Connection Box, insert the cable into one of the holes marked **9** and thread it up to the connector labeled **FLOW SW** (marked **4** in the diagram above.) Connect the wires according to the manufacturer's instructions. The ports are marked **+** and **SIG**.

Step 7. Fasten the plastic nut of each of the cable glands.

5.13 Connecting the 48VDC Module

The 48VDC cable from the first Connection Box of the Atlantium Unit must be connected to the 48VDC Module.

➲ To connect the 48VDC cable:

Step 1. Thread the 48VDC cable from the first Connection Box of the Atlantium Unit through the gland of the 48VDC Module **(A)**.

Step 2. Connect the wires to the connectors:

- Black wire to the black terminal connector **(B)**
- Red wire to the red connector **(C)**

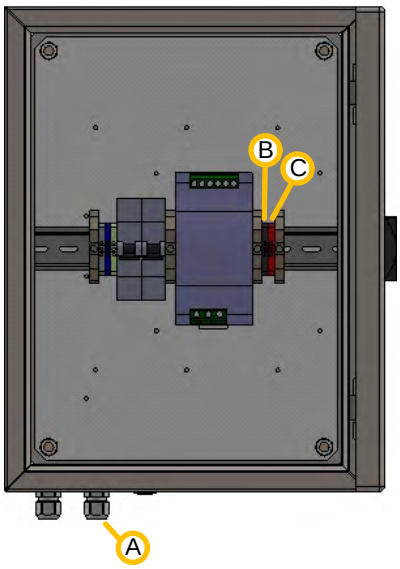


Figure 5-33: Connecting the 48VDC Cable

➲ To connect peripheral equipment:

Step 1. Using the attached key, open the screen panel of the **Control Module**.

Step 2. Use the holes, adding plastic glands as needed, on the bottom **(B)** to insert cables.

Step 3. Use the hole **(C)**, adding plastic glands as needed, on the bottom to insert the data cable from the Atlantium Unit.

Step 4. Thread the cables to the appropriate connector **(A)** according to the table below.

Step 5. For connecting to an external facility control via Modbus, see Appendix **A**, ***Modbus Communication Protocol*** on page **175**. Add jumpers to any unused pins as shown in the diagram below.

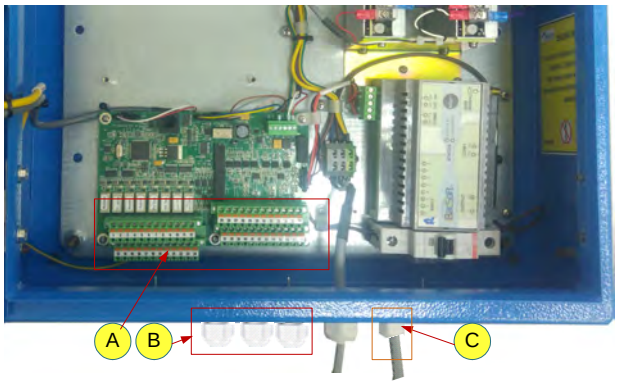


Figure 5-34: Control Module Interior Generic

5.14 Modbus Communication Connections

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks. Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

The Atlantium system can be controlled directly from your facility control system via Modbus - RTU on RS485 (network) serial connection.

To accomplish this:

- The Modbus Settings must be configured in the **Control Module**. **68.**
- The Facility control system communication properties must be configured to match that of the Atlantium system. See ***Setting Up the Facility System Communication Properties*** on page **72.**
- For the terminal block connectors for the Modbus in the **Control Module,** see above.

- Signalling communication incoming to the facility control system is according to the Modbus Registers. Appendix B, ***System Messages*** on page 181.

ATTENTION!

To ensure communications are operational, be sure to connect the cables **TX to RX** and **RZ to TX**.

5.14.1 Setting Up the Facility System Communication Properties

To properly configure communications between your Atlantium system and control network, set up the communication properties and **COM1** port settings at the facility control system as follows:

Node Address:	***
Available Commands	Function codes
Function Codes: *	0x03
Supports IEEE Floating Point: **	NO (register/xxx
Baud Rate:	4800, 9600, 19200, 38400, 57600, 115200 ***
2 Wire	
Parity:	None
# of Stop Bits:	1
# of Data Bits	8

(*) Read all the registers in the system at once with **0x03** (includes coils, discreet coils, holding registers and input registers).

(**) Divide the desired parameter by xxx to calculate the correct signal point. See the Divide column in the table in the section, ***Modbus Communication Protocol*** on page 175.

(***) Must be the same in the **Control Module** setting

To view the Modbus Registers, which govern the signaling communication from the Atlantium system to the facility control system, see Appendix B, ***System Messages*** on page 181.

5.15 Connecting the Atlantium System to the Mains

DANGER!

ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENT MUST BE SET TO THE OFF POSITION AND LOCKED WITH AN APPROPRIATE LOCK TO PREVENT ACCIDENTAL CHANGE TO THE ON POSITION.

DO NOT REMOVE THE LOCK OR SET ANY MAIN CIRCUIT BREAKER TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED.

UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

The infrastructure of the electrical connections is to have been prepared according to the ***Electrical ICD*** drawing provided to you by Atlantium with this manual.

For installations that require the OVPD unit (440-480VAC only), the customer must provide the Power cable to the mains.

5.15.1 Mains Connection Phases

Phase 1 Connect the **Control Module** to the mains.

Phase 2 Connect the Atlantium Unit's 48VDC Module to the mains (if exists)

Phase 3 Connect the Atlantium Unit's Power cable to the mains

ATTENTION!

- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

5.15.2 Connecting the Control Module's Power Cable

The **Control Module** comes with a power cable of 2m/6.5ft. length. For the electrical specifications, See ***The Control Module*** on page **23.** Prepare the appropriate plug for the electrical standards of your region. Considering the distance between the **Control Module** and your electrical box, you may want to replace the electrical cable with a longer one.

➲ To connect the Control Module's power cable:

- Connect the appropriate plug for the electrical standards of your region.
- **Do not** plug in the **Control Module** to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** on page **77.**)

5.15.2.1 Replacing the Control Module's Power Cable

You can order a longer cable from Atlantium or acquire one that conforms to the specifications in Table 5-2 on page 55 from a local source.

Avoid direct contact with the electronic circuit board which is sensitive to electrostatic discharge (ESD). Utilize best practices regarding ESD when replacing the power cable.

➲ **To connect a longer power cable to the Control Module:**

5.15.3 Connecting the 48VDC Module to the Mains

If your system includes the 48VDC Module, the voltage is 180 to 550VAC. The customer-supplied cables are to be connected either L-L or L-N.

➲ **To connect the mains cable to the 48VDC module:**

Step 1. Thread the mains cable through the gland of the 48VDC Module **(D)**.

Step 2. Connect the wires to the connectors:

- Brown wire to the grey terminal connector L1 **(E)**
- Blue wire to the blue terminal connector N/L2 **(F)**
- yellow wire to the yellow terminal connector GND **(G)**

Step 3. When you are ready to activate the system, turn the circuit breakers to the **On** position.

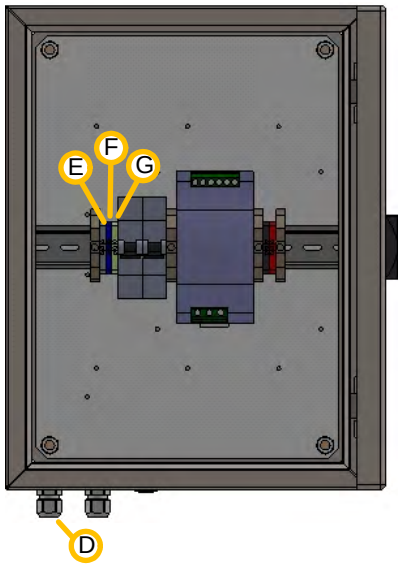


Figure 5-35: Connecting the Mains to the 48VDC Module

ATTENTION!

- **Please follow the Electrical ICD drawing provided to you by Atlantium with this manual**

5.15.4 For Shipboard Installations

- When the Atlantium Unit is installed shipboard, all metal components must be grounded to the nearest possible grounding point on the ship's metal structure. Be sure to prepare a ground point next to the designated location for the Atlantium Unit.

5.15.5 Connecting the Atlantium Unit's Power Cable

The Atlantium Unit comes with a power cable of (5m/15ft.) or 10m/30ft (according to you order.) Prepare the appropriate cables and plugs for the facility mains connection according to the electrical standards of your region. Considering the distance between the Atlantium Unit and your electrical box, you may want to replace the electrical cable with a longer one.

- **A certified electrician is required to perform these procedures. Follow your local regulations.**
- **Do not** connect the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** on page **77.**)
- **Do not** connect the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** on page **77.**)

➲ To connect the Atlantium Unit's power cable:

- Connect the appropriate plug for the electrical standards of your region and connect it to a mains outlet.

- The power cables supplied with the unit has 5 wires: 3 Phases (L1, L2, L3), Neutral (N) and Ground (PE). However, for the 440/480V installation, the Neutral wire is redundant and is not to be connected on either side of the cable.

5.16 Installation Check List

When you are ready to schedule the Atlantium technician to commission the Atlantium system, complete the check list below.

Send this check list together with digital photos of every angle of the installed system to your Atlantium representative. It generally requires at least 2 weeks after the check list has been completed to schedule the Atlantium commissioning.

Table 5-5: Atlantium System Installation Check List

Item	Done
■ The System is installed: ❑ Vertically - Water is flowing upward ❑ Horizontally	
■ The Atlantium Unit, Ballast Module and other components are located in their final position	❑
■ The Atlantium Unit and all of its components are not located where they may be in contact with direct water or chemical dripping	❑
■ Ballasts are installed in a vertical position.	❑
■ All Ballast Module connections are complete.	❑
■ The electrical cables between components are not stretched.	❑
■ All of the System components are securely installed and have sufficient service clearance on all sides (according to the ICDs).	❑
■ All power preparations are complete and Power connects are wired as instructed in this guide.	❑
■ Correct voltage was measured where the Power connects to the Ballast Module	❑
■ Voltage rate _____ Number of phases _____	

Table 5-5: Atlantium System Installation Check List

Item	Done
■	❑
■	❑
■ The Atlantium Unit is connected to the water pipes according to the Installation plan developed by facility engineers and Atlantium personnel. Proper pipe lengths (Inlet - 10D, Outlet - 5D) (Inlet - 10D, Outlet - 5D) are installed.	❑
■ Water flows through the Atlantium Unit - The Unit has been checked while full of water and no bubbles or leaks exist in the system. The system is balanced, supported and pressure stable (no vibration, etc.). No known water hammers or pressure above 10 bars are on the line.	❑
■ The **Control Module** connects is complete. All supporting features (e.g., General Alarm, External On/Off, etc.) and wiring were installed according to the Installation plan based on the instructions in this guide.	❑
■ All Connection Box connections are complete. All supporting features (e.g., flow meter or flow switch, auto valves open/close) and wiring were installed as instructed in this guide.	❑
■ For systems with **TRACS**, network connection was installed between **TRACS** the **Control Module**.	❑
■ **You are scheduling an electrician to be on site and available during the System commissioning (a must or the installation cannot be completed).**	❑
■ To be sent to Atlantium with this checklist, attach photos of the following: ❑ Atlantium Unit ❑ Ballast Module(s), **Control Module** and other components ❑ Accessories (Flow meter or flow switch, CIP ports, Sampling valves, etc.)	
Customer Approval Name of Customer's authorized representative: _____ Position: _____ Signature of Customer's authorized representative: _____ Date: _____	

6 Initializing the Atlantium System

The Atlantium system starts up with the initialization of the **Control Module.**

➲ To start up the Atlantium system:

Step 1. Check that the Ballast Module power cable is connected to the main power source.

Step 2. Repeat above step for each electrical cabinet in system.

The system powers up and the **Control Module's** touch-screen monitor is activated, displaying the **Main Operation** screen . The **Lamp Power** button on is on the **Off** position.

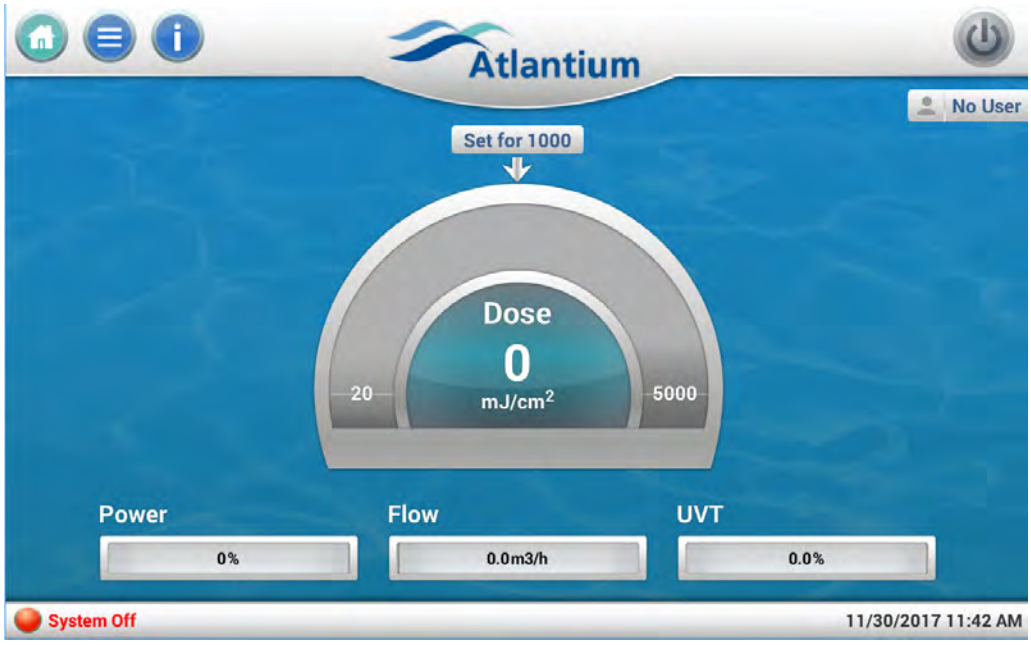


Figure 6-1: Initial Main Operation Screen

Step 3. Press the Lamp Power button to ignite the lamps.

7 Configuring the Control Module

7.1 The Control Module HMI

The **Control Module** provides the Human Machine Interface (HMI) for control of operations and dose measurements of the Atlantium system.

The regulations applying to the type of application determine sizing as well as software defaults and operational control options. This is set up at the original manufacturing facility and completed on site by the Atlantium technician.

The **Control Module** provides a real-time display of operational data. A Ready indicator turns green when the system is ready for operation.

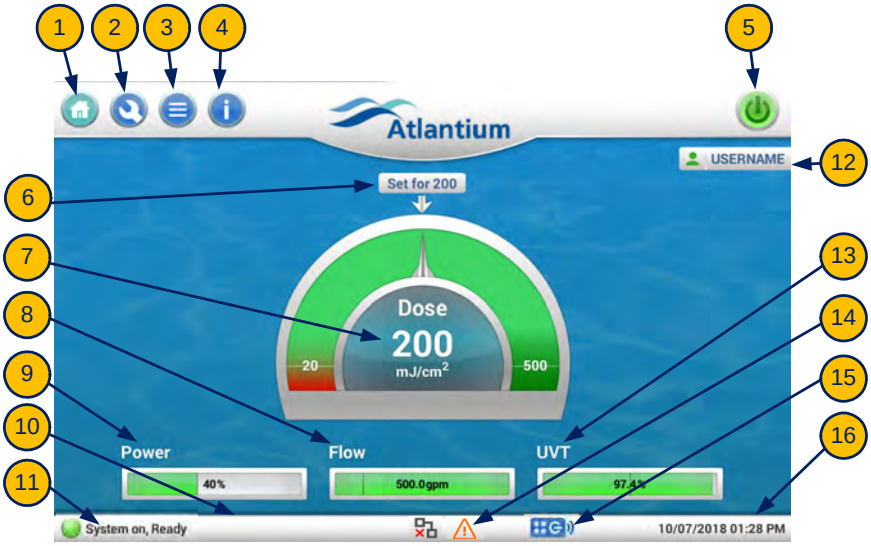


Figure 7-1: Initial Main Operation Screen*

Table 7-1: Control Module Main Screen Details

#	Item	Icon	Description
1	**Return to Main screen**		To return to the Main screen.
2	**Settings Screen Access**		To access the Settings screen
3	**Event Log Access**		To the Events screen.
4	**System information Screen Access**		Includes the Generic Info tab that displays general system information and lamp status and the Info Monitor tab that displays a summary of system status information.

Table 7-1: Control Module Main Screen Details

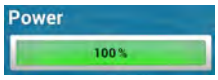
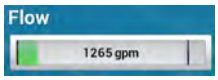
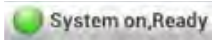
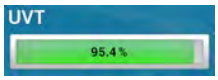
#	Item	Icon	Description
5	**Main Unit Operation Button**		ON/Off toggle button turns on/off the Atlantium unit components (lamp(s), UV power, etc.) in an orderly sequence.
6	**Set For**		The user should adjust the **Set for** dose to the desired operating dose. Confer with your Atlantium Representative, per the original sizing specifications. In some cases, this is 10-20% over the regulated or required dose. In other cases it depends on the variability of local conditions and the best strategy to ensure reliable and effective operation.
7	**Dose Gauge**		Represents the system dose range. The red area of the gauge represents off-spec low dose. The dark green area represents the user's set maximum allowable dose. The current dose (mJ/cm^2) is displayed in the middle of the gauge. The minimum and maximum allowable dosage levels are user-defined.
8	**Power**		Scale shows the current Power level.
9	**Water Flow**		Scale shows the current rate of water flowing through the Unit The scale color changes to red when the flow is **either** less than minimum flow setting **or** above the maximum flow setting. When there is **no** flow the scale goes gray.
10	**Bottom Footer**		Displays certain system events for a short period of time.
11	**Ready LED**		Displays the basic system status. When the system is ready for operation (all the lamps are ignited and the dose is above the minimum settings, the system displays the **green** Ready icon. If the system startup fails or the system is no longer working at or above the minimum dose, the system displays the **red** icon.
12	**User Login**		The Login button enables access to the login dialog box and User Management. See ***Adding Users*** on page 95.
13	**UVT (Transmittance)**		Scale shows the amount of UV light transmitted through the water. If the UVT percentage factor is high, the transparency of the water to UV light is high, and vice versa. When there are several UVT sensors in one system, this Scale displays the average of the UVT values measured by each sensor. The Scale color becomes red when the UVT is lower than minimum UVT setting. The color changes even if the UVT alarm is set to None.
14	**Event Icons**		A relevant icon appears when a critical event occurs. It disappears after the event has been acknowledged in the log.

Table 7-1: Control Module Main Screen Details

#	Item	Icon	Description
15	**Remote On/Off indicator**		The Remote-On/Off indicator appears if marked in the Signals tab settings.
16	**Date and Time**	11/30/2017 11:46 AM	Date and time are displayed according USA or European conventions
17	**Lost Network Communications Indicator**		This Icon appears when communication is lost.

The specific configuration for your Atlantium system is based on the Application type set for your industry and according to the parameters required for your facility's process.

Configuring the **Control Module** begins with logging in and accessing the Settings screen and follows the general order of the screen components.

7.2 Log In and Password Change

Users may log in, change their password, and reset their password if they forget it.

Users can login to the **Control Module** to access screens according to the permission level assigned to them. When logging on as the Facility Administrator for the first time, use the user name and initial password assigned to you during system commissioning.

Atlantium recommends that the initial Facility Administrator log on with this initial password, then change the password and make a record of the new password in a secure location, in the event that it is forgotten. The Facility Administrator can then define additional users according to facility needs.

For information on adding additional Users and permission levels, see ***Adding Users*** on page **95**.

➲ To login:

Step 1. Tap the User Indicator

. The Log In dialog box appears.

Step 2. From the **User Name** list, select your user name (for first-time use, the user name assigned to you during system commissioning).

Step 3. Tap the **Password** field and from the virtual keyboard that is displayed, enter your password (for first-time use, the password assigned to you during system commissioning).

Step 4. Tap **Log In**. Your user name appears in the User Indicator box on the upper-right of the main screen.

Figure 7-2: Login Dialog Box

➲ To change your password:

Step 1. Tap the User Indicator

. The User Administration Tasks dialog box appears.

Step 2. Tap **Change Password**. The Change Password dialog box appears.

Step 3. In the **Password** field, enter your current password.

Step 4. In the **New Password** and **Confirm New Password** fields, enter the new password.

Step 5. Tap **Save New Password**. The password is changed.

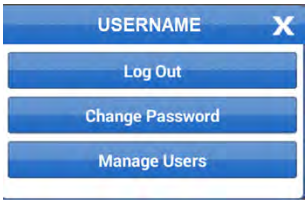


Figure 7-3: User Administration Dialog Box

Figure 7-4: Change Password Dialog Box

7.2.1 Login Timeout

To prevent unauthorized system changes, the Control Unit reverts automatically to No User mode 10 minutes after the last tap on the screen by the current user. In the No User mode, system information may be viewed but no changes can be made (see Table 7-2). To make any changers, an authorized user must login again.

7.2.2 If You Forget Your Password

Generally, users with Manager Profile may configure other users and their passwords. If no user with Manager Profile is available, Atlantium Customer Support can help using the following procedure:

➲ To reset your password

Step 1. In the Login dialog box, tap **Forgot Password?**. The **Forgot Password** dialog box appears.

Step 2. In the **User Name** field, enter your user name.

Step 3. Tap **Generate Code**. A code appears in the field below.

Step 4. Send the User Name and Code to Atlantium Customer Support.

Step 5. In the **Confirmation Password** field, enter the **Confirm Password code** received from Atlantium Customer Support.

Step 6. Tap **Reset Password to Default.** The password is reset to the default **1234**. The user should then change it to a secure password.

Figure 7-5: Forgot Password Dialog Box

7.3 The Settings Configuration Screen

The settings screen and individual tabs are visible according to the permission level of the User and are usually accessed during the installation and setup of the Atlantium system and when there are hardware changes made in peripheral equipment.

During normal operations, an authorized facility technician may need to access certain functions, or view certain information, that appears on the some of these tabs. For information on which tabs are visible according to permission levels, see ***Adding Users*** on page **95**.

WARNING!

Take care to ensure that the system configuration made via the Settings screen is not tampered with! Unauthorized changes may have a serious impact on the function of the Atlantium system. **Only trained and authorized** facility technicians are to access the Settings screen.

Screens display values in metric or US units, according to customer localization.

➲ To access the Settings screen: at the top of the screen, tap

Step 1. The main Settings screen is displayed with the Monitor tab open:

Lamp ID	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp kW	Lamp V	Lamp A	Temp °F	On Off	Lamp Enabled
1	100	100	97.0	17.8	18.0	1.70	220.00	7.72	77		✓
2	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		✓
3	100	99	0.0	17.8	0.0	1.70	220.00	7.72	77		✓
4	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		✓
5	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
6	100	100	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
7	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
8	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓

Figure 7-6: Settings Screen, showing Monitor Tab (generic example with 8 lamps)

Step 2. Tap the **Signals tab**. The **Signals Tab** appears.

Figure 7-7: Signals Tab on Settings Screen

7.3.1 Configuring the Signal Settings

The Signals tab displays the settings for the system's dry-contact Input and Output connections: either **Normally Closed (NC)** or **Normally Open (NO) see** Figure 7-7 on page 83.

The following is an explanation of Input and Output signals.

7.3.1.1 Signal Outputs

Inlet valve/System on - The facility's electronically-controlled **Inlet** and **Outlet valves**, that control the water supply to the Atlantium unit, as well as the **Bypass valve** (if a bypass is implemented), must each be configured to receive control signals from the **Control Module**. These valves affect to various facility processes relevant to the Atlantium system, such as:

- Managing high flow conditions
- Bypassing and streaming the water to a different place if the water quality has degraded
- Bypassing for cleaning chemicals that are not compatible with the quartz chambers

- When the system is activated (either manually via pressing the **Operation** button on the **Control Module**'s main screen, or via an external signal utilizing the **External On/Off)**, the **Control Module** sends a signal, within 10 seconds, that opens the **Inlet valve**. This signal can also be used to provide feedback that the system is activated.
- When the all the lamps are ignited and ready to use, the **Control Module** sends a signal to the **Outlet valve** to open. This signal can also be used to notify that the system is **ready**. The **Control Module** also sends a signal to the **Bypass valve** to close.
- Similarly, when the system is deactivated, (either manually via pressing the **Operation** button on the **Control Module**'s, or via an external signal utilizing the **External On/Off)**, the **Control Module** sends a signal that closes the **Inlet** and **Outlet valves**, and also opens the **Bypass valve**.

Signals to Alarms - To set off a General Alarm mechanism, such as a flashing light, alarm, bell, or a call or message to a telephone, a system alarm signal must be connected to a SCADA control system. See

General Alarm - requires connection of a General Alarm mechanism to receive a signal from the **Control Module**. Once configured, the alarm is triggered by the signal.

- **High Temperature** - requires connection of a General Alarm mechanism to receive a To connect a mechanism to receive a signal when system temperature has exceeded the configured maximum.
- **Low UVT** - To connect a mechanism to receive a signal when the system's UVT has fallen below the configured minimum.
- **Lost Comm** - When the **Control Module** AIO lost communication with UV unit (all cards disconnected).
- **Shutdown -** To connect a mechanism to receive a signal when an event causes the system to shutdown.

7.3.1.2 Signal Inputs:

- **Flow Switch** - If your system utilizes a Flow Switch, you must configure its electrical polarity in the **Control Module**. For the Flow Meter, see ***Configuring the Flow Settings*** on page 85.
- **External On/Off** - A switching mechanism to synchronize the Unit with the facility processes. This remote PLC turns the Atlantium Unit on/off. For details about the External On/Off signal wiring, see ***System Messages*** on page 181.

For example, the facility's SCADA (supervisory control and data acquisition) control system can send a signal to the Unit to start or stop its operation as needed. Using this setting, the Atlantium system recognizes this signal and turn the Unit on or off, as appropriate, without the need for manual intervention.

When the **External On/Off** sends a signal to activate the system, the Atlantium **Control Module** sends a response within **10 seconds** that the command was received. If no response is received, it is likely that the **External Operation Button** on the **Control Module** main screen is manually set to the **Off** position. Check this button and if needed manually tap it to reinstate the **On** position, so that the **External On/Off** can control the system activation.

For regular wiring, set according to the table in ***External On/Off - Additional Notes*** on page **59**:

- For **IN+** select **Normally Closed**
- For **IN-** select **Normally Open**

- **Low/Normal Mode** - for systems with an external signal for switching to Low Flow mode.

7.3.1.3 Other Settings

- **Maximum Temperature Warning** - The System shuts down automatically if the water temperature exceeds the maximum set value. You can set the maximum temperature threshold for sending a warning notification that alerts you to take action to prevent the water from reaching the maximum temperature that triggers shutdown. It is recommended to set this threshold to 5% below the maximum water temperature allowed for your system.
- **System Auto Power On** - You can also set the Atlantium system to restart automatically when power is restored after a facility-wide power failure, so that it reactivates according to the configured **Auto Power On** settings.

➲ To configure the Signals settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

The states of the **Outputs** and **Inputs** are set via toggle switches that by default are set to **NC (Normally Closed)**. To change any of them to **NO (Normally Open)**, tap the corresponding toggle switch. It changes to **NO**. Tap again to change it back to NC.

Step 2. Under **Output**, set the electrical polarity for:

- **Inlet valve/System On**
- **Bypass Valve**
- **Outlet valve/Ready**
- **General Alarm**
- **High Temperature**
- **Low UVT**
- **Lost Communication** (not implemented in this version)
- **Shutdown**

Step 3. Under **Input**, set the electrical polarity for:

- **Flow Switch** (if installed)
- **External On/Off**

Step 4. To enable the External On/Off, mark the **External On/Of Enabled** checkbox.

7.4 Configuring the Flow Settings

If a flow meter is utilized, the **Control Module** data must be configured to match the flow meter at the time the system is originally commissioned, but may need to be changed if a flow meter is changed, etc. The Max Flow setting provides a threshold to compare to the actual flow rate. If the actual flow rate exceeds this set Max Flow value, the system sends notification and triggers an alarm.

- **About the Low Flow Mode**

Set the Atlantium system's response to a low water flow situation via the Low Flow mode. This unique Atlantium feature minimizes the on/off cycles by allowing the system to operate at an extremely low flow value. The Atlantium Application Engineer defines this value according to the specific application. In some cases, this value can be zero flow for a predefined period of time. The **Low Flow Mode** options are:

  - Restart
  - Shutdown

- **Restart** – (Default) This option assumes that low flow is not necessarily a sign of a fault and ensures that the system is ready to come up to full dose or power immediately when the flow rises.

Atlantium allows the system to continue operating at a low or even zero flow rate up to a max time interval, the **Flow Delay,** as defined by the Atlantium application engineer. With this setting, we reduce the amount of on/off and maintain the integrity of the UV operation, while preventing the flow of non-disinfected water into the line.

  - As long as the flow rate stays below the **Min Flow** for less than the amount of time set in **Flow Delay**, the Unit continues to operate in standby mode, generally at the lowest possible power.
  - However, if the water flow rate drops below the set **Min Flow** rate for a time that is **longer** than the **Flow Delay**, then:
    - The **Low Water Flow** warning is triggered
    - The lamps are automatically shut down
    - An event notification is written to the event screen and log file
  - The **Restart** mode causes the system automatically restart when the water flow rate returns to a level above the **Min Flow**. The Unit automatically and instantly returns to full operation, without any operator intervention or connection to a higher control level, and without any delay or cool down/ warm up period.

The default values for the minimum flow rate and maximum flow delay appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but **be sure to coordinate** with the Atlantium representative.

- **Shutdown** – In this mode, in the event of low flow, which may be a sign of an upstream malfunction, the system shuts itself down in an orderly fashion, but operator intervention is required to start it up again.
  - As long as the flow rate stays below the **Min Flow** for less than the amount of time set in **Flow Delay**, the Unit continues to operate in standby mode, generally at the lowest possible power.
  - However, if the water flow rate drops below the set **Min Flow** rate for a time that is **longer** than the **Flow Delay**, then:
    - The **Low Water Flow** warning is triggered
    - The lamps are automatically shut down
    - An event notification is written to the event screen and log file
- In the **Shutdown mode,** the operator is required to turn on the Atlantium system, once the water is back to the proper Flow rate.

To configure the Flow Meter settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **Flow tab**. The **Flow Settings** tab appears. The **Current Flow** indicator displays the current rate of water flowing through the Atlantium system in **GPM** (gal/min) or **M3H (M^3H)** units. The **Main Flow Input** indicator displays the **Flow Meter (FM)** or **Flow Switch (FS)** in **GPM** or **M3H (M^3H)** units.

- For Flow Meter:
  - **Flow Meter** appears as the **Main flow input** source,
  - The **Current flow** is displayed on the right.

Figure 7-8: Flow Setting Screen - Flow Meter

Step 3. For **Min Flow**, using the **-/+** buttons, enter a value for the minimum water flow rate threshold. Below this threshold, the system enters **Low Flow Mode**.

Step 4. For **Low Flow Mode**, tap the arrow and from the drop-down list, select either **Restart** or **Shutdown**.

Step 5. The default value for maximum **Flow Delay** appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but be sure to coordinate with the Atlantium representative. If you need to change this value, set the desired value using the **-/+** buttons.

Step 6. For **mA at 0 Flow (Flow Factor)**, to ensure that the system does not operate on the basis of a discrepancy in the Current Flow readings, turn off the flow and then check the value for Current Flow. If it is not zero, adjust the value, by using the **-/+** buttons, until the Current Flow value is zero. Then resume the water flow.

Step 7. For **Flow x 100 at 20mA (Offset)** , on the flow meter's datasheet, find the meter's maximum-allowed flow rate.

Step 8. In the **Flow x 100 at 20mA** field, using the **-/+** buttons, enter the meter's maximum-allowed flow rate divided by 100. (If the max flow rate is 100 m^3/h, enter the value **1**. Convert GPM values into m^3/h values to make this calculation.)

Step 9. For **Max Flow**, using the **-/+** buttons, enter a value for the maximum water flow rate threshold. If the actual flow rate exceeds this threshold, the system delivers a notification message and triggers an alarm.

Step 10. To set the action to be performed if the water flow rate exceeds the value set in the **mA at 0 Flow** field. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution**- One or more caution messages are issued.

- **Warning** - One or more warning messages are issued.
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm.

For Flow Switch:

Flow Switch appears as the **Main Flow Input**.

The current flow is displayed on the right.

Figure 7-9: Flow Setting Screen - Flow Switch

Step 11. For **FS Target Val**, using the **-/+** buttons, enter the maximum value expected for the flow in m^3/h.

Step 12. For **Min Flow**, using the **-/+** buttons, enter a value for the minimum water flow rate threshold. Below this threshold, the system enters **Low Flow Mode**.

Step 13. For **Low Flow Mode**, tap the arrow and from the dropdown list, select either **Restart** or **Shutdown**.

Step 14. The default value for maximum flow delay appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but be sure to coordinate with the Atlantium representative. If you require changing this value, for **Flow Delay**, using the **-/+** buttons, set the desired value.

7.5 Configuring Dose Settings

The system is set at a particular dose. The Atlantium system maintains the designated dose as conditions allow.

➲ To configure the Dose settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **Dose Settings tab**. The **Dose Settings** screen appears.

Figure 7-10: Dose Settings Screen - Power Mode

Step 3. For **Operation Mode**, tap the arrow and from the dropdown list, select the desired mode:

- **Power mode**
- **Dose mode**

Step 4. In Power mode, use the **-/+** buttons to set the desired power level. The system is set to operate at this power level as a constant. Most operational settings used in Power mode provide information or shut-down the system under specified conditions. The system operates at the prescribed power level and does not change power level to achieve dose settings.

In **Dose mode**, continue with the rest of the settings:

Figure 7-11: Dose Settings Tab- Dose Mode

Step 5. For **Minimum Dose Val**, using the **-/+** buttons, set the lowest dose in **mJ/cm^2** units under which you want the needle to move to the red zone to send a warning or shut down.

Step 6. For **UV Dose Set for**, using the **-/+** buttons, set the general operating dose in **mJ/cm^2** units. When the system is operating in that dose range the needle is in the center of the green zone.

Step 7. For **Maximum Dose**, using the **-/+** buttons, set the maximum UV dose limit in **mJ/cm^2** units, above which you want the system to send a warning or take other action. When the system operates above the maximum dose, the needle is in the dark-green zone.

Step 8. For **Action** corresponding to each of the above settings, tap the arrow and from the dropdown list, select the desired corresponding system response in the event that the set power level pushes the dose above or below the set values. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution**- One or more caution messages are issued.
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm.

The system responds as designated here when, under the current operating conditions and required power level, the dose exceeds or falls below the set values.

Step 9. For **Target Pathogen**, tap the arrow and from the dropdown list, select the desired Pathogen either **Virus**, **Giardia**, or **Crypto (Cryptosporidium)**.

Step 10. For **Log**, tap the arrow and from the dropdown list, select the appropriate Log Inactivation:

- **3 Log**
- **3.5 Log**
- **4 Log**

Step 11. For systems with bypass valve, for **Upstream Delay**, using the **-/+** buttons, set the maximum time it takes for water to pass between the Atlantium system and the bypass valve + 20%. For systems with no bypass valve, set this value to 0.

Step 12. For **Validation Factor**, mark the **Validation Factor** checkbox. For no **validation factor**, leave the checkbox unchecked.

Step 13. For **Lpeq**, mark the **Lpeq** checkbox. For no Lpeq, leave the checkbox unchecked.

7.6 Configuring the General & Connection Settings

You can set a number of general settings, as well as the connection settings, via the General tab.

The following General Settings are to be configured:

- Date and Time
- Format for units of measurement (US or Metric)
- Maximum Temperature Warning - Since the System shuts down automatically if the water temperature reaches the maximum allowed. You can set the maximum temperature threshold for sending a warning notification so that you can take action to prevent the water from reaching the maximum temperature that triggers shutdown. It is recommended to set this threshold to 5% under the maximum temperature allowed for your system.
- Automatic System Auto Power On in the event of restored power after power failure.
- Assign System Unit identification (PWSID (Public Water System IDentification), Plant, and Train)
- Minimum UVT threshold value for the UV transmission below which the system is to issue an alarm, warning of UVT degradation and the action to be taken accordingly.

The Connection Settings consist of the Modbus node address assigned to this system is configured in the **Control Module**, as well as the appropriate baud rate for this system.

➲ To configure the General & Connection settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **General tab**. The **General Settings** screen appears.

Figure 7-12: General tabtab

General Settings

Step 3. To set the date and time, tap . The **Set Date And Time** dialog box appears.

Step 4. Swipe the date and time elements to toggle to the correct settings.

Step 5. Tap **Set**. The correct time appears on the **Output / Input** Tab and is set for the system.

Figure 7-13: Set date And Time Dialog Box

Depending on the format selected (see below) the Metric format is day first and a 24-hour clock and the US format is month first AM/PM.

Step 6. To select the **Units format**, tap the arrow and from the dropdown list and select the desired format, imperial for **US** units or **Metric** for metric units.

Step 7. When the maximum temperature threshold is reached, a warning notification is sent so that you can take action to prevent the water from reaching the temperature that triggers shutdown. For setting the maximum temperature:

- **a** Set the system water flow to the minimum possible.
- **b** Wait a few minutes for the system to stabilize.
- **c** Check the **Control Module's** Monitor screen's temperature reading.

Step 8. Under **Maximum temperature**, using the **-/+** buttons, set the maximum temperature threshold to the temperature reading in **Step 7/c** above plus **5°C/41°F.** When this threshold is reached, a warning notification is sent.

Step 9. To enable the system to restart automatically when power is restored, mark **System auto restore On** checkbox.

Step 10. For each Unit defined in you system configuration, you can assign the PWSID, Plant, and Train identification. From the **Unit Number** dropdown list, select the Unit number to be identified.

Step 11. For **PWSID**, **Plant**, and **Train**, tap the field. A virtual keyboard appears. Use it to enter the appropriate identification.

Connection Settings

Step 12. For **ModBus Address**, using the **-/+** buttons, set the Modbus address node assigned to this system as a slave while the remote PLC is the Master.

Step 13. For **Baudrate**, select the appropriate baud rate for this system. The options are:

- 115200
- 57600
- 38400
- 19200
- 9600

Minimum UVT

Step 14. For **Minimum UVT**, using the **-/+** buttons, set the threshold value for the UV transmission below which the system is to issue an alarm, warning of UVT degradation.

Step 15. For **Action**, from the dropdown list, select the action to be performed if the factor falls below the Min UVT and becomes off-spec. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution** - The caution alerts you to an anomaly based on the user preferences.
- **Warning** - One or more warning messages are issued. \

- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm.

7.7 Configuring the Cleaner

If your system includes a cleaner, it is pre-configure to run automatically (Enabled) according to a frequency that you can also configure.

➲ To configure the cleaner:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **Cleaner tab**. The **Cleaner** screen appears.

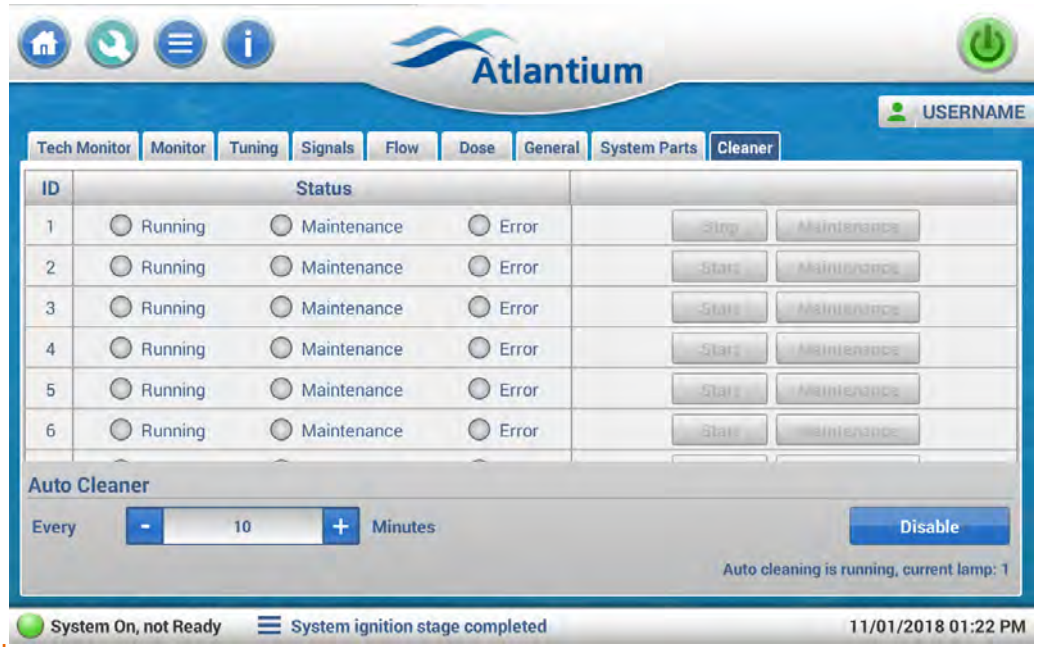


Figure 7-14: Cleaner Screen

Step 3. At the bottom of the screen, under **Auto Cleaner**, for **Every**, using the **-/+** buttons, set the frequency of the cleaning process in minutes.

Step 4. To set the duration of the cleaning process, for **For**, using the **-/+** buttons, set the duration in minutes.

7.7.1 Disabling the Cleaner

If needed, you can disable the cleaner.

➲ To disable the cleaner:

Step 5. At the bottom-right of the screen, tap **Disable**, the cleaner is disabled.

■

7.8 Importing/Exporting a System Configuration

After setting up a system configuration in the **Control Module**, you can export it to save as a backup or to use for configuring additional **Control Modules** by importing a saved system configuration.

7.8.1 Exporting a System Configuration

You can export a system configuration.

Prepare the following:

- USB Drive

To export a system configuration:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **General tab**. The **General Settings** tab appears.

Figure 7-15: General tab

Step 3. Open the **Control Module** and on the back of the door, insert the USB drive into one of the USB ports at the top of the circuit board, as shown in the figure below.

Figure 7-16: Control Module USB Port Location

Step 4. On the **General tab**, tap **Export Configuration**. The system configuration is copied to the USB drive. confirmation appears.

Step 5. Remove the USB drive and keep it in a safe place.

7.8.2 Importing a System Configuration

You can import a system configuration.

Prepare the following:

- USB Drive containing a saved system configuration

➲ To import a system configuration:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **General tab**. The **General Settings** tab appears.

Step 3. Open the **Control Module** and on the back of the door, insert the USB drive (containing the saved system configuration) into one of the USB ports at the top of the circuit board, as shown in Figure **7-16** above.

Step 4. On the **General tab**, tap **Import Configuration**. A popup appears asking you select the source.

Step 5. Tap **USB Flash**. The system configuration is copied to the **Control Module**.

Step 6. You are prompted to restart the **Control Module**. Tap **Yes**. The **Control Module** restarts with the imported system configuration.

7.9 Adding Users

Typically, the designated facility manager defines user names for the relevant personnel. However, each User can add addition Users with the same authorization type or in the hierarchy under him/her.

All-In-One screen access is according to the table below.

Table 7-2: User Hierarchy and Permissions

Section	Screen	Manager	Operator	No User Logged On
	Event Log	X	X	X
	System Info	X	X	X
	Monitor	X	X	X
	Lamps	X	X	
	Monitor	X	X	
	General	X	X	
	Cleaner	X	X	
	Tech Monitor	X		
	Tuning	X		
	Signals	X		

Table 7-2: User Hierarchy and Permissions

	Flow Setting	X		
	Dose Settings			
	Settings System Parts	X		

To add a User:

Step 1. Tap the User Indicator

. The User Administration Tasks dialog box appears.

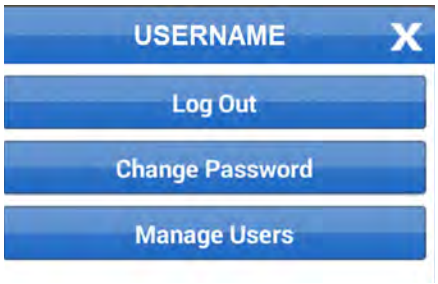


Figure 7-17: User Administration Tasks Dialog Box

Step 2. Tap **Manage Users**. The Manage Users dialog box appears.

Figure 7-18: Manage User Dialog Box

Step 3. Tap **Add User.** The Add User dialog box appears.

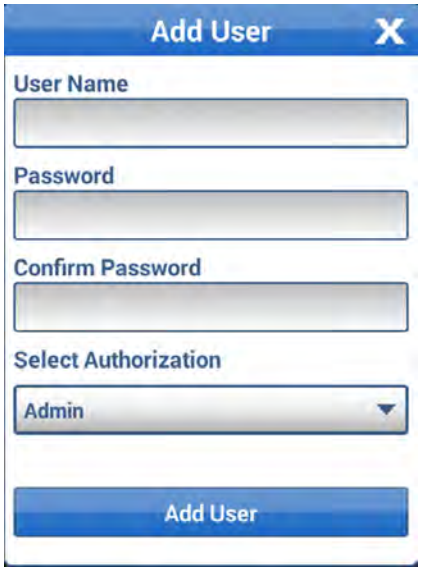


Figure 7-19: Add a User Dialog Box

Step 4. Tap the **User Name** field. A virtual keyboard appears. Type in the user name.

Step 5. Tap the **Password** field. A virtual keyboard appears. Type in the password.

Step 6. Tap the **Confirm Password** field. A virtual keyboard appears. Type in the same password to confirm it.

Step 7. From the **Select Authorization** dropdown list, select the desired User type

Step 8. Tap **Add User.** The new User is added to the system and appears on the list of Users.

7.10 Editing / Deleting Users

Each User can reset the password to the default password **1234** of Users with the same authorization type or in the hierarchy under him/her or delete any of those Users.

➲ To edit / delete a User:

Step 1. Tap the User Indicator

. The User Administration Tasks dialog box appears.

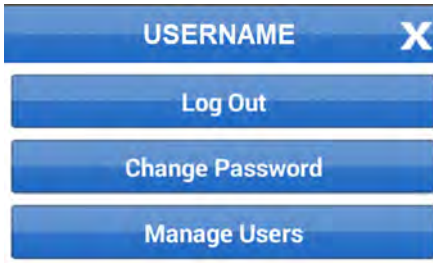


Figure 7-20: User Administration Dialog Box

Step 2. Tap **Manage Users**. The Manage Users dialog box appears.

Figure 7-21: Manage User Dialog Box

Step 3. From the **Select User** dropdown list, select the desired User. The Manage User dialog box appears displaying additional options.

Figure 7-22: Add a User Dialog Box

Step 4. To Reset the password of the selected User, tap **Reset Password To Default**. The password is reset to **1234**.

Step 5. To delete a User, tap **Delete User.** The User is deleted.

8 First Time System Activation

Following **Control Module** configuration, completing the system commissioning includes the following major steps:

- Cleaning the Atlantium Unit (See below)
- Filling unit with water (See below)
- Igniting the UV Lamps (See below)
- Initiating the Water Flow (See page **100**)
- Adjusting the Auxiliary Equipment (See page **101**)
- System Tuning (See page **101**)
- Installation QA (See page **101**)

8.1 Cleaning the Atlantium Unit

The inner surfaces of the Atlantium Unit are to be cleaned via the CIP process at installation, before starting up and proceeding with the commissioning phase. Follow the directions in ***Cleaning In Place (CIP)*** on page **101**.

8.2 Filling unit with water

The unit must be filled with water before full activation.

Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. **Make sure that water hammer is NOT present in your facility's water line.**

To prevent water hammer, consult your system engineer and check your facility system procedures, including:

- When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
- Sudden valve closures can cause water hammer. Make sure that valves close gradually enough to avoid this problem.
- Ensure that all pumps used with the Atlantium system employ soft start-up procedures.

➲ To introduce water:

- Open the **Inlet valve** and, using the **Viewport** on the Unit, check that the quartz chamber fills with water completely.

8.3 Igniting the Lamps

The lamps of the Atlantium Unit are ignited to commence live system operation.

- **Temporarily, the caps on the lamps must be left open to allow any condensation water to evaporate. During this time, set up Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

➲ To ignite the lamps:

Step 1. **Temporarily place Caution floor signs at a distance of 1m/40 inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**

Step 2. Open each of the lamp breather caps (2 on each side of a lamp). For the location of the lamp breather caps, see Figure **10-3** on page **107**.In the **Control Module**, log in and at the top of the screen, tap . The **Monitor Tab** is displayed.

Lamp ID	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp kW	Lamp V	Lamp A	Temp °F	On Off	Lamp Enabled
1	100	100	97.0	17.8	18.0	1.70	220.00	7.72	77		✓
2	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		✓
3	100	99	0.0	17.8	0.0	1.70	220.00	7.72	77		✓
4	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		✓
5	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
6	100	100	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
7	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
8	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓

Figure 8-1: Monitor Screen*

**** This screen (as well as all screens shown in this manual) is a generic example. The number of lamps shown on the actual screen are configured according to the number of lamps in your Atlantium Unit.***

Step 3. Verify that all of the lamps in your system are enabled.

Step 4. At the top of the screen, tap the **Event Log** screen is displayed.

Step 5. Verify that an Event log item showing that the lamps have been ignited is displayed. The event may take a few seconds to appear on the screen.

Step 6. Let the system operate for at least one hour.

Step 7. Continue with Section **8.4** and Section **8.5** below while the lamps complete the first hour of operation.

Step 8. Close all of the lamps' caps and remove the Caution floor signs.

8.4 Initiating the Water Flow

When initiating the water flow, the water may be set to drain out or collect in a tank until the system is fully operational. Once the entire installation and QA testing is complete, the water is set to flow normally as part of the facility operation.

➲ To initiate the water flow:

Step 1. Open the relevant valve that allows water to flow through the Atlantium unit. This may be the draining valve or bypass valve.

Step 2. Open the viewport and verify that the Unit is completely filled with water and that no air gaps and no air bubbles exist.

8.5 Adjusting the Auxiliary Equipment

All relevant auxiliary equipment , i.e., flow meter, automatic valves, PLC, etc., must be adjusted at this time. Refer to the manufacturers' user documentation of the relevant equipment.

If your configuration includes a flow meter, check that the value of the flow displayed on the **Control Module** matches the reading on the flow meter.

8.6 System Tuning

The Atlantium system must be tuned via the **Control Module**. Follow the instructions in ***Replacing/Checking a UVT Analyzer or UVIS Sensor*** on page **119**.

8.7 Installation QA

Atlantium personnel performs a quality assurance procedure following which the Atlantium system is ready for regular facility operation.

9 Operating the Atlantium System

This section details how to operate the system and handle basic and periodic routines to be carried out by Atlantium trained the local staff.

For system without **Deposit Prevention Mechanism (DPM) Cleaning system**, disregard the irrelevant sections.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

9.1 Operational Guidelines

Follow these guidelines for smooth Atlantium System operations:

- **Incoming Water Supply** - Be alert about incoming water supply. If you do not have pre-treatment, be alert to reports of contamination by upstream slugs of mud, particles, oil, etc.

  If the water source was not previously in operation or the line is new, - flush the line prior to introducing water into the Atlantium system
- **Avoid chemicals that are harmful to UV systems** - Avoid caustic soda and other corrosive chemicals that can degrade gaskets and o-rings. For information on approved chemicals for CIP use, see ***Selecting the Correct Chemicals for CIP*** on page 115
- **Prevent Water Hammer** - Water hammer can cause damage to any hydraulic system, including the Atlantium unit. Take steps to ensure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check you facility system procedures for following:
  - Avoid filling empty lines with water without keeping valves open for releasing the air from the pipes and the system.
  - Avoid rapid closure of water lines by sudden closure of valves.
  - Before introducing the water, ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
- **Avoid vibrations on upstream and downstream piping** - Make sure that the piping to be attached to the Atlantium Unit is reinforced to protect the Unit against vibration.
  - A high vibration level can cause damage to any hydraulic system, including the Atlantium unit.
  - High vibration can be caused by an unbalanced pump or by sudden valve close.
- Do not leave the Atlantium Unit filled with stagnant seawater for a long time. Either rinse with fresh water and leave the Atlantium Unit standing with fresh water, or empty. Leaving the Unit empty is a less preferable option, since when refilling the system you must avoid air collection in the Unit and avoid water hammer, as mentioned above.

9.2 Basic Operational Tasks

- Shutting down the Atlantium system - See below
- Starting up the Atlantium system - See page 103
- Taking microbial samples - See page 109

9.2.1 Shutting Down the Atlantium System

You can shut down the Atlantium system via the **Control Module**.

➲ To shut down the Atlantium system:

Step 1. In the **Control Module**, on the top of the screen, tap . The button turns grey. Wait approximately three minutes for the **Ready Lamps On** signal to transmit.

Step 2. Check the main screen of the **Main Operations** and verify that the system parameter for **Power** is **0%**.

Step 3. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

9.2.2 Starting up the Atlantium System

You can start up the Atlantium system via the **Control Module** and then check the water and system parameters.

➲ To start up the Atlantium system:

Step 1. Check that the power cable of the Ballast Modules is connected to the main power source.

Step 2. Connect the power cable of the **Control Module** to the main power source.

Step 3. Using the attached key, open the screen panel of the **Control Module**.

Step 4. Locate the circuit breaker and turn it to the **On** position.

Step 5. In the **Control Module**, on the top of the screen, tap . The button turns green. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 6.

Step 7.

Step 8.

Step 9. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 10. Check the **Control Module** Screens (See ***Configuring the Control Module*** on page **71.**) and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range (see the table below to enter your expected values).

Notify the executive maintenance engineer in case the values are out of range.

Table 9-1: Abbreviation of Table in Appendix C. Checking the System Parameters

	Parameter	Expected Value	
		Minimum	Maximum
1	Power		
2	Flow Rate		
3	UVT		
4	UV Dose		
5	Lamp Status		

For periodic checks on Atlantium system parameters, a convenient check list is supplied in Appendix **C**, ***Checking the System Parameters*** on page **184**.

Viewing System Information

You can view general information about the system in the **System Info** screen.

➲ To view generic System information:

- At the top of the screen, tap . The **System Info** screen is displayed, containing the following information:
  - Power in %
  - Status in % (Average system status)
  - UVT in %
  - Flow in m^3/h
  - Temperature
  - Dose in gpm or h^3/m
  - Software version (of the **Control Module**)
  - ISO version (of the **Control Module** card)
  - IP Address (Optional for connecting to the Control Module via remote PC to generate history Excel files)
  - External Storage (SD card for backing up system data, i.e., event log, system configuration, system parts info, history)
  - Total Space
  - Total Free Space

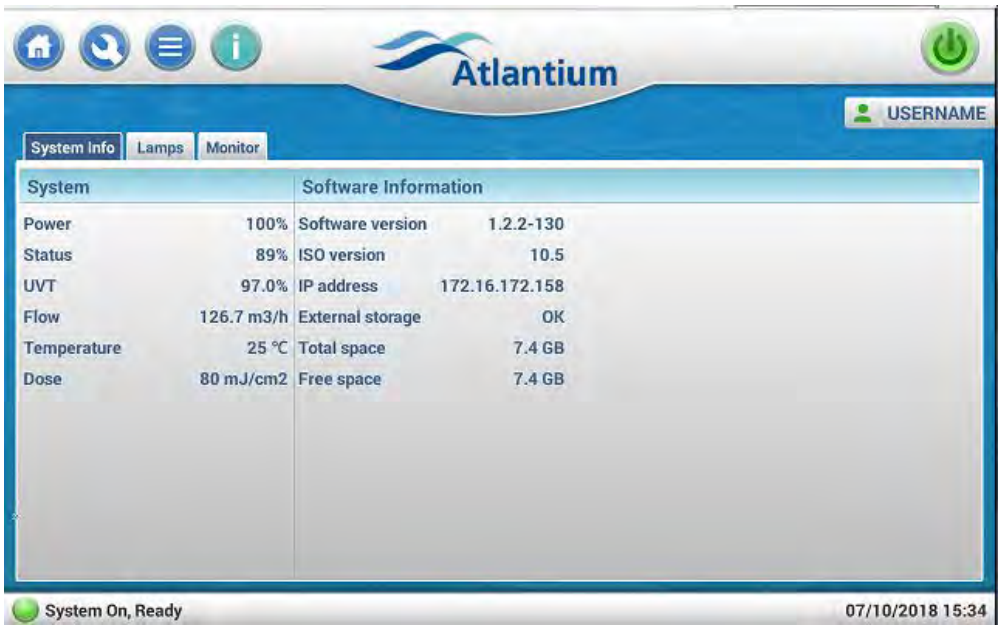


Figure 9-1: System Info Screen

9.3 Viewing Lamp Information

You can view information about the lamps in the **Lamps** screen.

To view information about the Lamps:

Step 1. At the top of the screen, tap . The **System Info** screen.

Step 2. Tap the **Lamps** tab. The Lamps screen is displayed, containing the following information:

- Lamp ID
- Per lamp
  - Status in %
  - Age in hours
  - Number of ignitions
  - State: On, Ignition, or Off

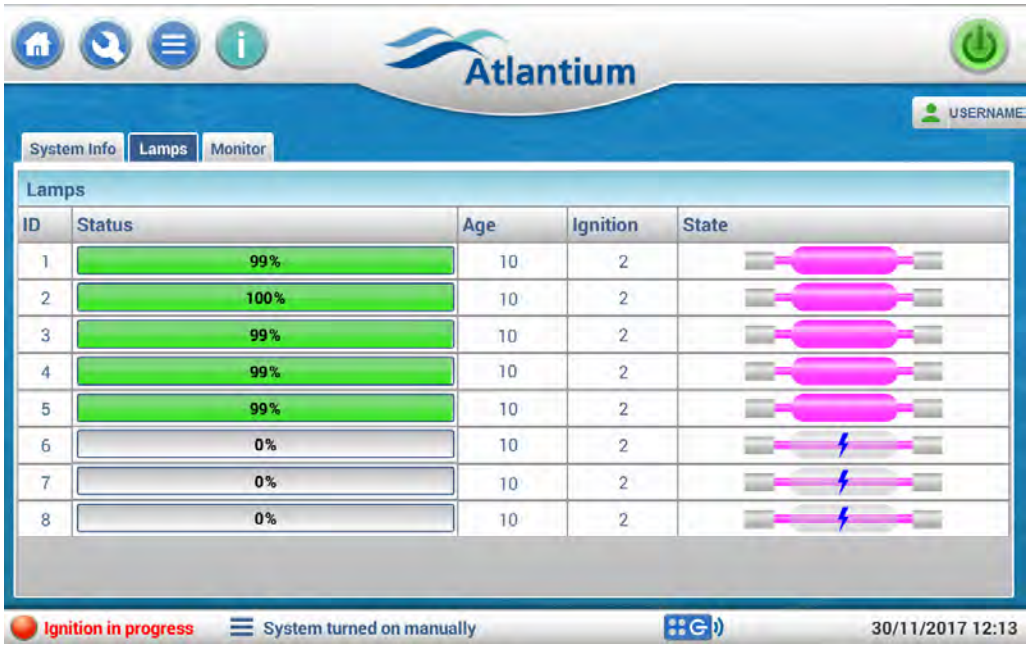


Figure 9-2: Lamps Screen

9.4 Viewing the System Monitor

You can view information about the system status in the **Monitor** screen.

To view the System Monitor:

Step 1. At the top of the screen, tap . The **System Info** screen.

Step 2. Tap the **Monitor** tab. The Monitor screen is displayed, containing the following information:

- Lamp ID - The ID number of the lamp.
- LM Card SW - The software version of the Connection Box card to which the lamp is connected.
- Power % - The Power supply's power in percentage.
- Status % - The Real time efficiency status of the lamp in percentage.
- UVT% - The quality of the UV transmittance through the water in percentage.
- UVIS mA -
- UVTS mA -
- Lamp kW -
- Lamp V -

- Lamp mA -
- Temp - The lamp's temperature reading in either ^{O}C or ^{O}F.

Lamp ID	LM Card SW	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp kW	Lamp V	Lamp mA	Temp °C
1	78.9	100	99	91.9	17.9	17.8	4.20	400.00	10.50	25.0
2	78.9	100	99	0.0	17.8	0.0	4.20	400.00	10.50	N/C
3	78.9	100	99	0.0	17.8	0.0	4.20	400.00	10.50	N/C
4	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
5	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
6	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
7	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
8	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C

Figure 9-3: Monitor Screen

9.5 Viewing the Event Log and Acknowledging Event Alerts

Every defined event is written to the event log of the **Control Module**. Events with alerts also trigger an alert icon at the bottom of the screen. You are can acknowledge each event alert or acknowledge all together.

➲ To view the Event log and acknowledging event alerts:

Step 1. At the top of the screen, tap the **Event Log** screen is displayed.

Step 2. To navigate through the log, swipe down the Event list.

Figure 9-4: Event Log Screen

Step 3. To view more information about a single event, tap the event. The event item expands to display the full information, including reading for System Power, Dose, Flow, Status, UVT, and Temperature at the time the event occurred. When expanding an event with an alert, the alert is acknowledged and relevant icons at the bottom of the screen disappear.

Figure 9-5: Event Log Screen - Event Information Examples

Step 4. To acknowledge all alerts, tap the **Acknowledge all** button. The relevant icons at the bottom of the screen disappear.

9.6 Taking Microbial Samples

The designated aseptic sampling valves (i.e., EGMO ESV valves) are the **Inlet** and **Outlet Sampling Points** located on the sides of the Atlantium Unit. Before taking a microbial sample, a valve is sterilized internally with Ethanol 70%, as well as with flame.

Required Equipment

- Sampling containers that are clean, laboratory-grade, sampling bottles with a volume no less than 100ml (3.38oz).
- Ethanol 100-70%
- Bunsen Burner and lighter
- Permanent marker
- Sterile gloves
- Proper Sampler attire according to local laboratory protocol

CAUTION!

- Use gloves during sampling and treat them with Ethanol 70%.
- Use **sterile disposable** sampling containers.
- Prevent contact between the sampling containers and the valves' surfaces.
- Be sure that no foreign matter or liquids fall into the sampling containers.
- Perform the sample collection as fast as possible to minimize environmental contamination of the sample.
- **Use open flame to treat the surrounding air while sampling.**

➲ To take a microbial sampling:

Step 1. Open the Sampling valve and allow water to flow for 2-3 minutes. Then close the valve.

Step 2. If the sampling valve contains a cover at the top, open its cover and place it in a clean place.

Step 3. Spray all over the Sampling valve and inside it thoroughly with Ethanol 70%.

Step 4. Using a Bunsen burner, apply the flame comprehensively over the entire Sampling valve surface from its connection to the main pipe to its spout.

Step 5. Spray the cap thoroughly with Ethanol 70% and place it on the Sampling valve.

Step 6. Open the valve and allow the water to flow for 2 minutes. If you have applied the flame properly, vapor emerges from the spout. (If there is no vapor, application of the flame was insufficient. Repeat the previous step.)

Figure 9-6: Cleaning the Sampling Valve with Ethanol

ATTENTION!

Make sure never to touch the opening at the Sampling container's neck.
While the Sampling container is open, do not turn the cap upside down.

Step 7. On the Sampling container, with one hand, remove its cap carefully, avoiding touching the inside of the cap with fingers or any other object. **Do not turn the cap upside down.**

Step 8. With your other hand position the Sampling valve under the water stream and take a water sample. Make sure never to touch the valve's opening or the opening at the container's neck. Fill the sampling container, leaving ample air space in bottle (at least 2.5 cm).

Step 9. Carefully replace the container's cap and tighten it to avoid cross contamination.

Figure 9-7: Taking a Sample

Step 10. Label the Sampling container with the following data:

- Sampling site
- Sampling point
- Sampler name

Step 11. Close the Sampling valve.

Step 12. Spray the sampling container thoroughly with Ethanol 70%.

Step 13. Store the water samples in a dark, refrigeration unit at a temperature of 2-8°C (35.6-46.4^{o}F) until they are examined.

9.6.1 Microbial Analysis Guidelines

The microbial analysis is to be conducted according to the company guidelines which are used in the local microbiology laboratory.

- Atlantium's recommendation is to use the microbial filtration method for the testing.
- If a number of consecutive anomalous microbiological results are obtained in one of the sampling points, take the samples only after performing CIP with a disinfection chemical (peracetic acid is recommended). It is important to let the chemicals clean and sanitize the aseptic valves, by letting the cleaning solution flow through them. See ***Cleaning In Place (CIP)*** on page **114.**
- In the event that system outlet counts are higher than the inlet counts, the results are to be considered as a sampling error.
- In the event that the microbial counts are not proportional in different dilutions, the results are to be considered as a laboratory error.
- For further guidance regarding microbial sampling and analysis, contact the Atlantium Application department.

9.7 Viewing Reports

The reports contain information for the components listed in the table below.

Table 9-2: Report Contents

Component Type	Details Provided
Lamps	■ Unit number ■ Lamp number ■ Last operator who installed or modified the component ■ Date of Installation ■ Date of Replacement (if replaced) ■ Reason for Replacement (if replaced) ■ Age in hours ■ Number of Ignitions ■ Serial number
UVI Sensor	■ Unit number ■ Associated lamp number ■ Last operator who installed or modified the component ■ Date of Installation ■ Date of Replacement (if replaced) ■ Reason for Replacement (if replaced) ■ Age in hours ■ Serial number
UVT Sensor	■ Unit number ■ Associated lamp number ■ Last operator who installed or modified the component ■ Date of Installation ■ Date of Replacement (if replaced) ■ Reason for Replacement (if replaced) ■ Age in hours ■ Serial number

Table 9-2: Report Contents

Component Type	Details Provided
Ballasts (Modules)	■ Unit number ■ Associated lamp number ■ Last operator who installed or modified the component ■ Date of Installation ■ Date of Replacement (if replaced) ■ Reason for Replacement (if replaced) ■ Age in hours ■ Number of Ignitions ■ Serial number
Cleaner (if installed)	■ Unit number ■ Lamp number ■ Last operator who installed or modified the component ■ Date of Installation ■ Date of Replacement (if replaced) ■ Reason for Replacement (if replaced) ■ Counter ■ Type
UVI Reference Sensor	■ Unit number ■ Last operator who installed or modified the component ■ Date of Installation ■ Reason ■ Expiration date ■ Serial number

➲ To view a report:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **System Parts tab**. The **System Parts** screen appears.

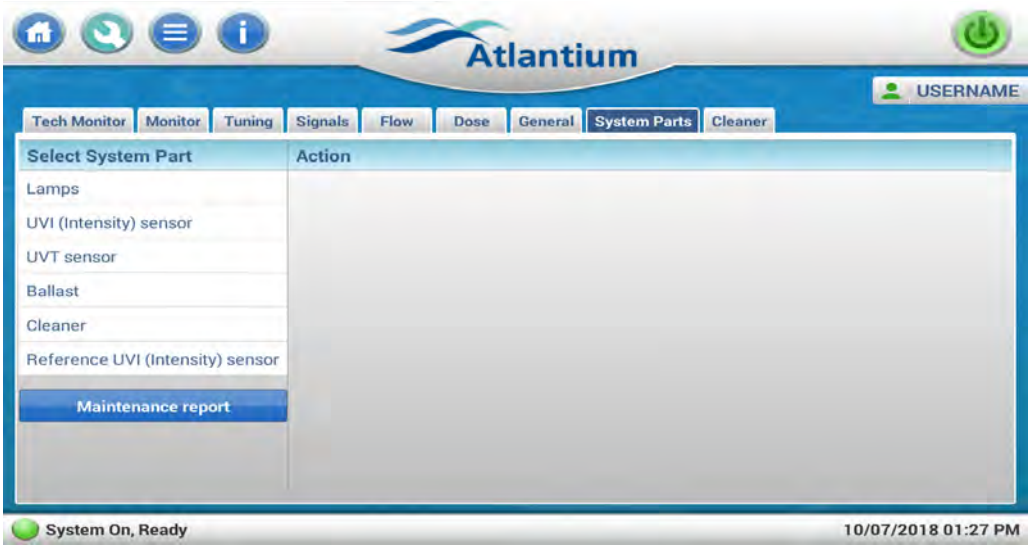


Figure 9-8: System Parts Screen - Lamps Selected

Step 3. Tap Maintenance report. The **Maintenance Report** screen appears.

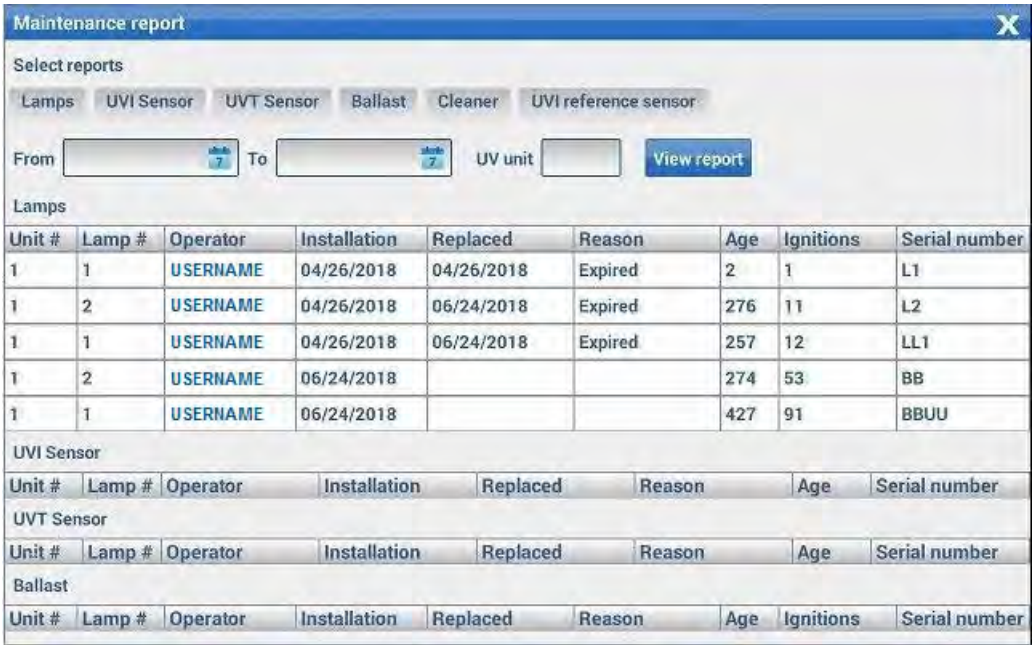


Figure 9-9: System Parts Screen

By default, all of the tabs are selected and all of the component lists appear.

Step 4. To produce a report with a list of components from less than all of the components, tap the components you do not want to include. They are de-selected and their lists do not appear in the report.

Step 5. To produce a report that list the components installed in the system between specific dates, For the **From** and T**o** dates, tap the Calendar icon and from the date picker that appears set the date range.

Step 6. If you have more than one Atlantium Unit installed, For **UV Unit**, select the desired Unit number.

Step 7. Tap **View Report**. The report displays accordingly.

10 System Maintenance

To keep the Atlantium system in peak form, preventive and periodic routine maintenance is essential. During commissioning, Atlantium trains your local staff how to do the tasks on a prescribed schedule or as needed.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

Before you begin a procedure, carefully read through it so that you can anticipate the steps efficiently.

The periodic routine maintenance schedule includes:

- Weekly Maintenance Tasks:
  - Walk-Around Inspection Tour - see below
- Monthly Maintenance Tasks:
  - Checking the Ballast Module Cooling Fans (if relevant)
  - Checking the Temperature Sensor Reading (See page **122**)
- Yearly Maintenance Tasks:
  - Cleaning the Quartz Sleeve (or Replacing as needed) and Replacing the Quartz Sleeve O-rings (See page **134**)
  - For DPM: Replacing Dynamic Seal, Base O-ring, and Wiper (See page **154**)
  - Anti-Corrosion Maintenance (See page **150**)
- Maintenance Tasks Performed as Needed:
  - Cleaning In Place (CIP) (See page **114**)
  - Replacing a Lamp (See page **123**)
  - Checking/Replacing a Sensor (See page **140**)
  - Testing System Lamps (See page **145**)
  - Tuning the Atlantium System (See page **146**)
  - Checking/Replacing the Quartz Sleeve (See page **134**)
  - Cleaning/Replacing a UVIS Wave Guide (See page **132**)
  - Replacing/Editing System Parts in the **Control Module** (See page **148**)
  - DPM Cleaner Maintenance
    - Cleaning the DPM's Opto-Coupler (See page **152**)
    - Replacing the DPM Belt (See page **153**)
    - Replacing Dynamic Seal, Base O-ring, and Wiper (See page **154**)
  - Temperature Sensor Maintenance
    - Replacing the Temperature Sensor (See page **159**)

10.1 Walk-Around Inspection Tour

Perform Weekly

Walk-Around and inspect the Atlantium system once a month to check for water leaks, visible damage to the cables and harnesses, or any other part of the system. Report any problem you find to an authorized technician.

➲ To perform a Walk-Around inspections:

Step 1. Walk around the Atlantium unit and inspect it. Look for leakage at the incoming and outgoing pipes.

Step 2. Inspect the **Cable Harness** and cables running between the Unit and the Ballast Module(s) and **Control Module** for signs of fraying and any other visible damage.

Step 3. Inspect the Ballast Module(s) and **Control Module** for any signs of damage.

Step 4. Report any leakage or damage to an authorized technician immediately.

10.2 Cleaning In Place (CIP)

Performed as needed

The CIP procedure consists of two parts:

- Cleaning to remove the accumulated deposit (particles) that collect on the quartz sleeve over time
- Disinfectant cleaning to sterilize the quartz sleeve (not applicable to some Municipal installations)

Clean the Atlantium Unit's inner surfaces and quartz sleeve generally on a periodic basis that is determined by observing that the performance is on target and the UVT Sensor readings are stable. You can clean using this process less frequently, depending on the specific chemistry of your water.

A CIP recirculation system is available from Atlantium as an optional CIP Kit. Connect the recirculation system to the CIP ports to create a closed-loop flow line that pumps a diluted solution through the Unit.

- In facilities where CIP is performed on the plant level, circulating though all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.
- In facilities that utilized sea water, sea water may be used in the CIP process provided that the chemical manufacture of the cleaning/disinfectant solution approves.

The estimated time for a CIP process is 1.5 hours.

Prepare the following:

- Protective chemical-safe goggles, clothing and gloves
- An external pump-driven recirculation system with a built-in 100 liter reservoir (26.5 US gallons) or Atlantium's CIP Kit including its accessories (See ***Sampling Valve Components and Kits*** on page **36.**)
- Scalant cleaning solution
- Disinfectant cleaning solution (if disinfection is planned)

10.2.1 Selecting the Correct Chemicals for CIP

It is **critically important** to select the correct cleaning and disinfecting chemical for the CIP process. It must be powerful enough to clean the inner surfaces of the Atlantium unit, while at the same time:

- Does not harm the components of the Unit that it comes into contact with (stainless steel 316, quartz and Viton® o-rings).
- Does not contaminate the product for which the water being treated by the system is used.

WARNING!

- The chemicals selected and the concentrations used must be approved for use by the facility's quality assurance and safety officials, and must comply with the written protocols and procedures set in place by those departments.
- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- Use of NaOH (Caustic Soda) may cause irreversible damage to the quartz. Use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

- All chemicals must be obtained locally.
- Atlantium does not supply chemicals - get them locally from commercial chemical suppliers.

Recommended Scalant Cleaning Solutions

The following chemicals have been tested and found to be very effective for cleaning and are recommended for use when cleaning the Atlantium system:

- Phosphoric acid (H_3PO_4) 2% diluted solution
- Nitric acid (HNO_3) 2% diluted solution
- HCl 2% diluted solution - recommended for metal based (rust-colored) deposits
- Sulfamic acid (H_3NSO_3) 1.5% diluted solution - recommended for calcium based deposits

Recommended Disinfectant Solutions (not applicable to some Municipal installations)

(The Disinfectant CIP process is not applicable to some Municipal installations)

The following have been tested and found to be very effective for disinfection and are recommended for use when disinfecting the Atlantium system:

- Peracetic acid-based chemicals - 30 min. contact time, for example:
  - P-3 Oxonia Active 150 - 0.5% diluted solution according to manufacturer's recommendation
  - Divosan Forte - 0.5% diluted solution according to manufacturer's recommendation
- Hot water at 85°C (185°F) for 30 min.

WARNING!

- **It is the responsibility of the Atlantium system's owner/operator to use and apply the correct chemicals.**
- **The use of any other cleaning and disinfection chemicals requires Atlantium's approval.**

10.2.2 Preparing the Chemical Solution

For both cleaning and disinfecting, the liquid in the reservoir must contain the desired percentage of the chemical additive (see above). Therefore, the purchased chemical solution must be diluted and added to the reservoir. Calculate the correct amount of chemical solution to be used for the CIP process.

Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

- The concentrations refer to room temperature.
- The diluted solution values denote the required dilution of the chemical circulating through the Atlantium Unit.
- While calculating the diluted volumes, consider all volumes including the Unit, pipes and the recirculation reservoir.

➲ To calculate the correct amount of chemical solution:

Step 1. Calculate the total volume you need for the CIP* by adding the water volumes in:

- The Atlantium Unit volume (measurements in liters/gallons) - Refer to the ***Physical Specifications*** Table in ***Atlantium System Footprint*** on page 16
- The pipes between the Atlantium Unit and the **Inlet** and **Outlet Valves**
- The CIP recirculation system
- The hoses between the Unit and the recirculation system

Step 2. Calculate the amount of chemical additive to be added to the reservoir to achieve the desired dilution. Use this formula:

$$C = \frac{A * V_t}{B}$$

A - The target concentration for the liquid (chemical substance + water) in %

B - The concentration of the chemical substance to be added in %

C - Volume of the chemical substance to be added in liters/gallons

V_t - Volume of the water in the Unit + piping + CIP Reservoir (as described above*) in liters/gallons

OR:

Assume that you need a 2% diluted solution in the CIP process and assume that your purchased solution is of a 35% concentration in a 4-liter container.

Calculate according to this calculation: [200 liter X 2%] / [35% X 4 liter] = [400] / [140] = 2.9 = 3 containers.

CAUTION!

* If you intend to collect the solution into the CIP reservoir or other container, the total volume of the cleaning solution must not exceed the volume of the CIP reservoir or container.

Be sure to obtain from an engineer the correct calculation for the amount of chemicals to be used for the process you intend to use.

10.2.3 Performing CIP

CAUTION!

The anti-scaling procedure is be performed only if necessary. Atlantium recommends that anti-scaling always be followed by disinfection.

The CIP process is performed utilizing fresh water. For systems with sea water, drain the system and rinse before preforming CIP.

- In facilities where CIP is performed on the plant level, circulating though all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m^3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.

This section details the CIP process with recommended steps for carrying it out. The procedure is to be adapted according to the specific circumstances of your installation.

The CIP process is to follow these major steps:

(Refer to the next section for the detailed description on the CIP process.)

Step 1. Close the inlet and outlet valves and immediately shut down the Atlantium system

Step 2. For systems with sea water, if needed, drain the system

Step 3. Remove the caps covering the CIP ports

Step 4. Connect the supply hose to the bottom CIP port and connect the return hose to the top CIP port

Step 5. Fill the CIP reservoir with fresh water or sea water

Step 6. Starting the circulation

Step 7. Adding the cleaning solution (See ***Selecting the Correct Chemicals for CIP*** on page **115.**)

Step 8. Running the CIP circulation for at least 30 minutes or as specified according to the chemical manufacturer

Step 9. Rinsing the system and the CIP cart

Step 10. Performing CIP Disinfection with the disinfection solution

Step 11. Ending CIP

The CIP diagram below is followed by the full recommend CIP procedure.

CIP Setup

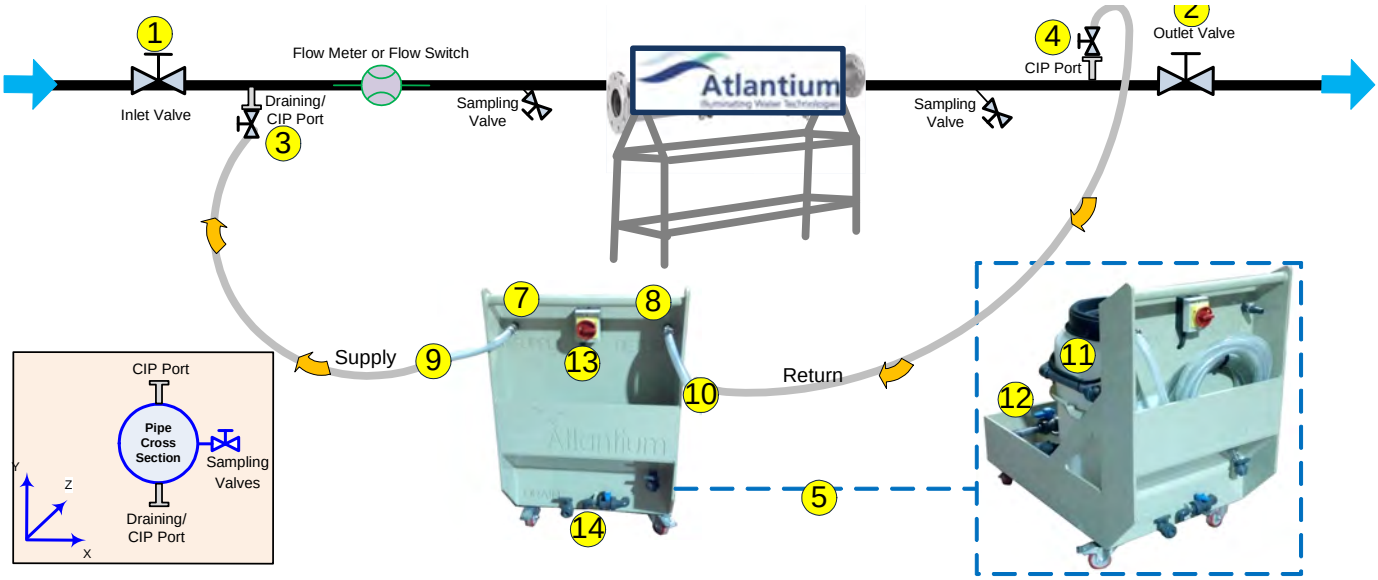


Figure 10-1: Recommended CIP Setup

If the installation is vertical, the water flow is from bottom to top. The Draining/CIP port **(3)** is on the bottom, under the Atlantium Unit and the CIP port **(4)** is above the Atlantium Unit.

➲ To Perform CIP:

ATTENTION!

If you are only performing disinfection: Follow Steps 1 through 20. Then resume with Step 22.
If you are performing both procedures: Follow all of the steps below.

Shutting Down the System

Step 1. Close the Unit's inlet isolating valve **(1)** and outlet isolating valve **(2)**. Immediately proceed to the next step.

Step 2. Shut down the Atlantium system, on the **Control Module**, by pressing the **Main Unit Operation Button Operation** button in the upper-right corner.

Step 3. On each CIP port, remove the cap's clamp and remove the caps covering the CIP ports **(3 & 4)**. Place the caps in a safe place for reuse later. Verify that the gasket is still inserted into the ferrule.

ATTENTION!

Make sure that the hose fittings are secure and tight.
Water must fill the entire path through the piping on both sides of the Atlantium Unit and the Unit itself, as well as the Supply and Return hoses and the CIP Kit's reservoir.

Connecting the CIP Hoses

Step 4. Remove the hoses from their pocket on the Cart.

Step 5. Using the CIP cap's clamp, connect the CIP Kit's Supply hose to the bottom CIP port **(3)**. Verify that the gasket is present.

Step 6. Using the CIP cap's clamp, connect the CIP Kit's Return hose **(10)** to the top CIP port **(4)**. Verify that the gasket is present.

Filling the CIP Reservoir

Step 7. To fill the CIP reservoir **(A)** with water:

- **Filling option 1:**
  - Slightly open the Unit's inlet isolating valve **(1)** and open the top CIP port on the outlet side **(4)** and allow the facility's water to fill the CIP Kit's reservoir **(11)**.
  - There is a full line indicator on the reservoir **(11)**. Make sure the water in the reservoir is up to the 100 liter (26.5 US gallons) level.
    - Close the Unit's inlet isolating valve **(1)**.
    - Close the CIP port **(4)** valve.
- **Filling option 2:**
  - Open the top cover of the CIP Kit's reservoir **(11).**
  - Using a hose of an external water source, fill the reservoir with 100 liter reservoir (26.5 US gallons of room-temperature tap water (±25°C/75°F).
  - There is a full line indicator of the reservoir. Make sure the water in the CIP Kit's reservoir **(11)** is at the 100 liter reservoir (26.5 US gallons level.
  - Close the cover.

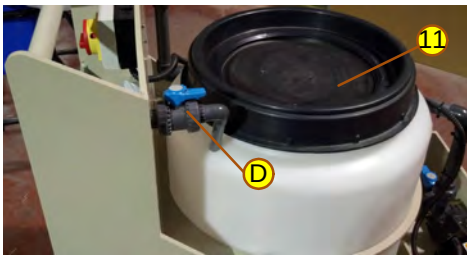


Figure 10-2: CIP Kit's Reservoir

Figure 10-3: CIP Cart's Inlet and Drain Port

WARNING!

Observe chemical safety rules! Follow all safety recommendations in the chemical manufacturer's product information and MSDS publications!

Adding the Chemical Solution

Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

Step 8. Open the top cover of the CIP Kit's reservoir **(11)** and add the prepared chemical solution to the water in the reservoir. (See ***Selecting the Correct Chemicals for CIP*** on page **115.**)

Starting the CIP Circulation Process

ATTENTION!

The Atlantium Unit must be absolutely full for the solution to reach all of the inner surfaces.

Step 9. If the bottom and top CIP ports **(3 & 4)** are valves, open them.

Step 10. To start the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **On** position and allow the solution to circulate through the Unit for **the period of time specified in the chemical's Use instructions, but not less than 30 minutes**.

Step 11. Verify that the Atlantium Unit is full of water and that the water is circulating correctly.

Step 12. During the circulation period open and close the sampling valve several times to assure chemical contact with the CIP chemicals.

If the draining valve is separate from the CIP ports, open and close the draining valve.

Rinsing the System and CIP Kit

After the chemical solution has circulated through the Unit for a sufficient duration, the chemicals solution must be rinsed from the system.

Step 13. Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system. See ***Alternatives for Draining the Water with Cleaning Solution*** on page 121.

Step 14. Turn on the Atlantium system, on the **Control Module**, by pressing the **Main ON/OFF Operation** button in the upper-right corner.

ATTENTION!

Before performing the next step, verify that the lamp(s) are on and the system has stabilized and indicates it is ready. If not, restart the lamp(s) again and wait until the system has stabilized and indicates it is ready.

Step 15. Close the inlet CIP port **(3)**, stop the recirculation pump **(12)** and open the CIP recirculation system's drain valve **(C)**. Immediately slightly open the inlet valve **(1)** to rinse the Atlantium system thoroughly.

Allow this rinsing process to operate until at least three cycles of water volume have gone through the Atlantium Unit to remove all of the scalant cleaning/disinfectant solution.

Step 16. To assure that all the chemicals are flushed out, use a measurable indicators, such as measuring the pH, or other indicators depending on the chemical being used (consult your chemical supplier).

Step 17. Slightly open the outlet valve **(2)** and close the CIP port on the outlet side (4). To avoid hammer effect if there is air trapped in the system, carefully open the inlet valve **(1)** completely. Resume normal operation.

Rinsing the CIP Reservoir

After the chemical solution has circulated through the Unit for a sufficient duration as specified by the chemical supplier (typically 30 minutes), the chemicals solution must be rinsed from the system by the steps below.

Step 18. Open the CIP cart's drain valve **(C)** and completely drain the CIP reservoir **(11)**.

Step 19. Using an external water source, rinse the CIP cart’s reservoir, letting the rinse water drain out of the CIP cart's drain valve **(C)**.

Step 20. Close the CIP cart's drain valve **(C)**.

Performing CIP Disinfection

Step 21. If you are planning to perform an additional CIP process using the disinfectant solution, refer to ***Preparing the Chemical Solution*** on page 116 and follow the CIP process again from ***Filling the CIP Reservoir*** in **Step 7.** above.

Or,

To return to normal operating status, proceed to the next step.

Ending CIP

Step 22. To drain out any water inadvertently collected in the CIP cart's storage pocket or base, use the CIP cart's drain valves **(A** and **B)**.

Step 23. Roll up the CIP Kit's hoses and return them to their storage pocket **(F)**.

Refilling the System

ATTENTION!

Make sure that the Atlantium Unit is filled with water before powering it up and turning on the lamps.

Step 24. Turn on the Atlantium system at 100% power and wait until the system has stabilized and indicates that it is ready.

Returning Power to the Lamps

Step 25. On the Connection Box next to each lamp, turn the switch to the **On** position.

Step 26. Turn on the Atlantium system, on the **Control Module**, by pressing the **Main ON/OFF Operation** button in the upper-right corner.

Alternatives for Draining the Water with Cleaning Solution

- **If local regulations permit, and a physical drain setup exists**, drain the cleaning solution to the facility's sewage system:
  - a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
  - b With the Atlantium System operating, sightly open the Inlet valve.
  - c Close the inlet bottom CIP port **(3).**
  - d To stop the CIP recirculation, on the CIP cart, turn the switch **(13)** to the **Off** position.
  - e Open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
  - f Connect the CIP cart's drain valve **(C)** to the facility's sewage system.

Or,

- **According to local regulations, use the Draining/CIP port to drain the solution in the system into a external container.**

Or,

- **Use the CIP reservoir (11) to collect the chemical solution**.
  - a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
  - b With the Atlantium System operating, close the inlet bottom CIP port **(3).**
  - c To stop the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **Off** position.
  - d Allow the chemical solution to drain entirely into the CIP reservoir **(11)**.
  - e Dispose of the contents according to regulations. Connect a drain pipe to the CIP cart's drain pipe and open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
  - f Disconnect and close the CIP cart's drain valve **(B)**.

10.3 Checking the Cooling Fans

Perform Monthly

For Ballast Module B, once a month the cooling fans of the Ballast Modules must be checked to verify that the air is flowing out.

To check the cooling fans:

- Check that air is being blown out of the fan on top of the Ballast Module.

10.4 Checking the Temperature Sensors' Readings

Perform Monthly

Once a month both temperature sensors must be checked to verify that they are working properly.

To check the temperature sensors' readings:

Step 1. At the top of the screen, tap . The **System Info** screen.

Step 2. Tap the **Monitor** tab. The Monitor screen is displayed.

Lamp ID	LM Card SW	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp kW	Lamp V	Lamp mA	Temp °C
1	78.9	100	99	91.9	17.9	17.8	4.20	400.00	10.50	25.0
2	78.9	100	99	0.0	17.8	0.0	4.20	400.00	10.50	N/C
3	78.9	100	99	0.0	17.8	0.0	4.20	400.00	10.50	N/C
4	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
5	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
6	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
7	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C
8	78.9	100	99	0.0	17.9	0.0	4.20	400.00	10.50	N/C

Figure 10-4: Monitor Screen

Step 3. In the middle-left, check the value for **Temp**.

Step 4. Verify that the temperature readings are within acceptable temperature range.

10.5 General Lamps, Sleeves, & DPM Maintenance Tips

The following are some tips that are meant to help you understand maintenance requirements:

- Lamps and quartz sleeves traverse the width of the Unit and are held in place by holders and connectors at both sides. For maintenance, you can access the lamps and quartz sleeves from either side. Depending on the constraints of the installation (sometimes one side is less accessible) you can usually decide from which side to access the them.
- You can perform maintenance on the quartz sleeves and their o-rings without performing any tasks on the DPM Cleaner.
- To perform maintenance on the DPM's dynamic seal, wiper and related o-rings, you must also perform maintenance on the quartz sleeves and there o-rings.
- To access the quartz sleeve, you must remove the lamp housed within it.
- Generally, a yearly maintenance is performed on the quartz sleeves and their o-rings, as well as on the DPM's dynamic seal, wiper and related o-rings.
- On rare occasions, quartz sleeves and their o-rings may need to be replaced because of a maintenance issue that arises.
- If the Unit is installed in an environment that causes more than usual wear on the DPM's components, they may need to be replaced more often.

10.6 Replacing a UV Lamp

The UV lamp can be used as long as it provides the required dose and efficient service. (Generally, a lamp service status of 80% or more is considered efficient service.) A lamp should be replaced ONLY when necessary.

Each lamp's performance is measured directly by a dedicated UV Intensity Sensor. Replace a lamp when:

- Its status or performance declines and the lamp no longer provides the required performance at the appropriate electrical cost
- Once its rated hours expire, be alert to any changes in efficiency that could indicate the need to change the lamp

In the **Control Module**, you can set a margin to assure notification as status declines and a warning is sent to the operator at both the user setting level and the internally coded minimum.

For systems with multiple lamps, the system may continue to operate while any one of the lamps is being replaced, however, precautions must be carried out to protect you from the UV.

ATTENTION!

- If your system uses High-Power or Ultra High-Power (UHP) lamps, be sure to use replacement lamps of the same type.
- Note that the High Power/UHP lamps are distinguishable by the label on the lamp marked in red. See the example below.

Figure 10-5: High Power Lamp Label in Red

Before commencing this procedure:

CAUTION!

- Allow the old lamp at least 10 minutes to cool down before starting.
- Read carefully the instructions provided in the new Lamp package.
- Wear white cotton gloves.
- Before installing a new lamp, verify that it is clean and free of grease.
  If not, rub it gently with the provided lens tissue. Never touch the lamp with your bare hands - hold it only by the end-connectors. Oily residues from fingerprints can damage the lamp.

WARNING!

- The UV lamps are designed with high internal positive pressure. Wear protective eye-wear while replacing a lamp. Pay strict attention to the safety warnings and precautions in the ***Safety Overview*** on page **41** in the front of this manual.
- In case of lamp breakage, refer to the Safety instructions in ***Safety Overview*** on page **41**.

When replacing a lamp while the rest of the system is still in operation, you must follow these safety requirements:

- Shutdown the electricity supplied to the lamp to be replaced.
- **During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

CAUTION!

While carrying the steps of this procedure, replace all of the O-rings on the components, such as the lamp housings and holders.

Prepare the following Information for the replacement lamp:

- lamp's serial number
- system model
- the number of hours on the lamp usage clock when you changed the lamp
- the number of ignitions of the lamp
- the status of the lamp when you changed it

10.6.1 Lamp Replacement Options

- Replacing a Lamp after System Shutdown (See page 125)
- Replacing a Lamp while System is Operating ()

Generally, a lamp is replaced after shutting down the entire Atlantium system. However, in a facility where there is no redundant UV purification, and you require minimizing system downtime, you can replace a lamp while other lamps continue operation provided you follow these safety requirements:

Replacing a Lamp after System Shutdown

Prepare the following:

- Replacement lamp
- Lamp cleaning cloth (comes with the lamp)
- White cotton gloves
- Eye protection
- #2.5 Allen wrench
- #3 Allen wrench
- #4 Allen wrench
- Caution signs

10.6.2 Shutting Down the Entire System

Generally, a lamp is replaced after shutting down the entire Atlantium system.

➲ To shut down the entire system:

Step 1. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

Step 2. Check the main screen of the **Control Module** and verify that the system parameter for **Power** is **0%**.

Step 3. On the top-right of the screen, tap . The button turns grey. Wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Step 4. On the corresponding Connection Box associated with the lamp to be replaced, turn the switch to the **OFF** position.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's switchswitch before you proceed to replace the lamp.

Step 5. Open the **Control Module** using the attached key and locate the circuit breaker. Turn the circuit breaker to the **OFF** position.

Step 6. For the lamp that is to be replaced, on the connection box, turn the lamp switch to the **Off** position.

Step 7. Continue with the procedure, ***Replacing the Lamp*** on page 126.

10.6.3 Replacing a Lamp while System is Operating

If you are replacing a lamp while other lamps are still in operations, follow these safety requirements:

- Shutdown the electricity supplied to the lamp to be replaced.
- **During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards. The UV lamps are designed with high internal positive pressure.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

➲ To shut down electricity to a single lamp:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Lamp ID	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp kW	Lamp V	Lamp A	Temp °F	On Off	Lamp Enabled
1	100	100	97.0	17.8	18.0	1.70	220.00	7.72	77		✓
2	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		✓
3	100	99	0.0	17.8	0.0	1.70	220.00	7.72	77		✓
4	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		✓
5	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
6	100	100	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
7	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓
8	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		✓

Figure 10-6:

Step 2. Under **Lamp Enabled**, for each lamp you want to disable, tap the checkbox to remove the check box. The corresponding lamp is disabled.

- A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp before you proceed to replace the lamp.

Step 3. Continue with the procedure, ***Replacing the Lamp***, below.

10.6.4 Replacing the Lamp

Once the electricity is turned off for the system or for the specific pair of lamps (at least one of which is to be replaced) and preparations are made (see above) you can replace the lamp.

➲ To replace a lamp:

Step 1. If you are replacing a lamp while other lamps are still in operations, follow these safety requirements:

- **During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into the lamp enclosure during the entire lamp replacement procedure.**

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

Positioning the DPM Wiper Assembly

If you are accessing the lamp on the upper side, it is recommended to leave the DPM wiper in the parking position at the top. Is so, skip to Step 5. below.

If you are accessing the lamp on the lower side, it is recommended to position the wiper in Maintenance mode, which is at the lower end of the Unit.

Positioning the DPM Wiper for Maintenance Mode

Step 2. For systems with DPM, at the top of the **Control Module** screen, tap . The **Monitor Tab** is displayed.

Step 3. Tap the **Cleaner tab**. The **Cleaner** screen appears.

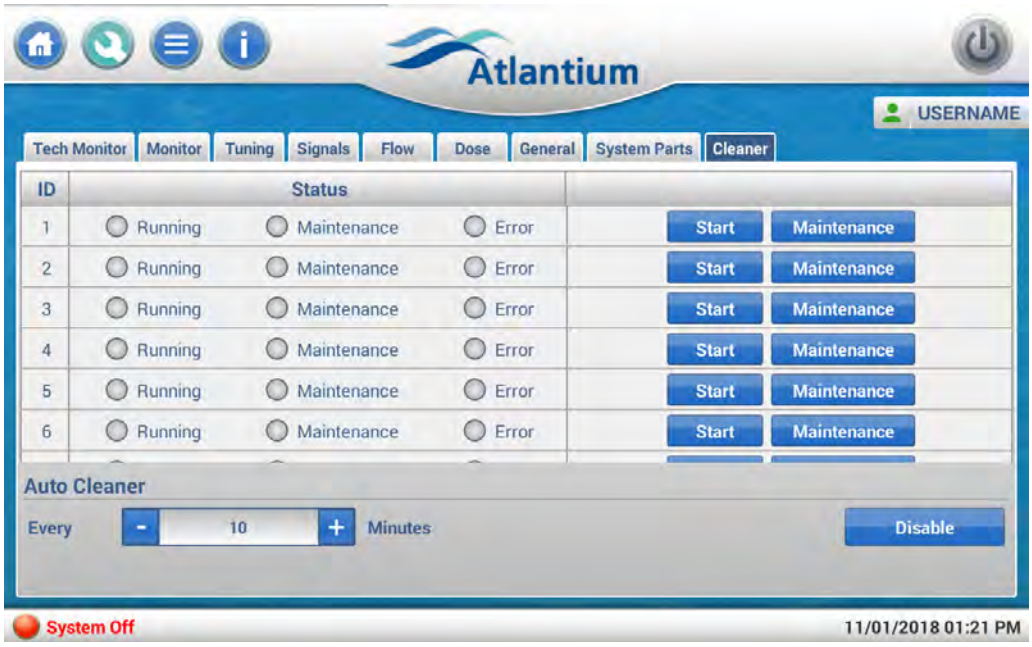


Figure 10-7: Cleaner Screen - Maintenance Mode

Step 4. For the desired lamp, on its line, tap **Maintenance**. White the wiper assembly moves to the **Maintenance** position, the **Start** button become the **Stop** button and The is **Maintenance** greyed. When the wiper assembly moves to the **Maintenance** position, the **Maintenance LED** displays green and the button becomes the **Home** button.

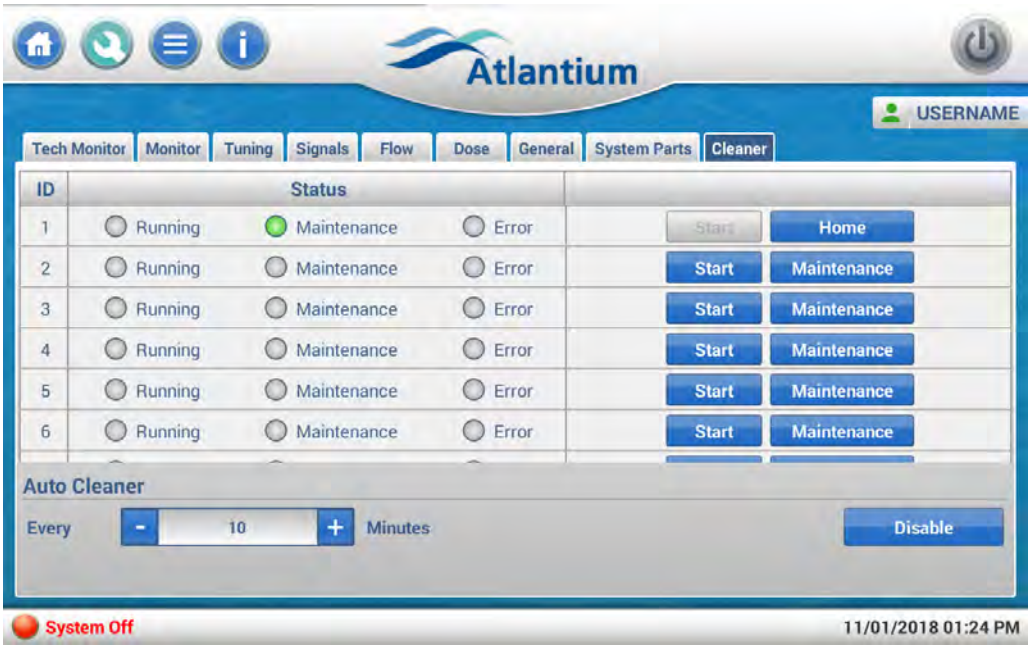


Figure 10-8: Cleaner Screen - Maintenance Operating on Lamp 1

Turning Off the Electricity to the DPM

Step 5. .Open the 48VDC Module door and set the circuit breaker **(A)** to the **Off** position.

Step 6. On the Connection Box, set the switch to the **Off** position.

Accessing the Lamp

ATTENTION!

For systems that contain wave guides, **be sure to access the lamp from the side that contains the sensors**. The wave guide is fragile and may be damaged easily if the lamp is inserted from the opposite site.

Step 7. On the Connection Box, set the switch to the **Off** position.

Accessing the Lamp

Step 8. Using the #2.5 Allen wrench, on each side of the DPM on the lamp to be replaced, remove the eight screws **(B)** of the DPM cover.

Step 9. Using a #3 Allen wrench, remove the four Allen screws **(C)** that secure the lamp's connector assembly. Each screw has a washer.

Step 10. **To prevent any damage to the lamp, make sure you pull the lamp's connector assembly straight out and not at an angle.** Gently set the connector assembly aside on a stable surface.

Figure 10-9: Removing the Lamp's Connector Assembly

Figure 10-10: 48VDC Module Circuit Breaker

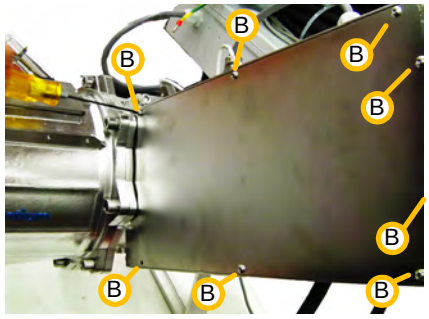


Figure 10-11: DPM Cover's eight screws

Step 11. On upper side of the quarts sleeve, using a #4 Allen wrench, remove the four Allen screws, opening them in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) **(D)** that secure the sleeve's holding ring to the Unit. Loosen one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the sleeve (2 turns).

Step 12. Continue to alternate until both sleeve holders are loose, and then remove the screws completely. Each screw has a washer.

Step 13. **To prevent any damage to the lamp**, make sure you pull the lamp's connector assembly straight out and not at an angle.

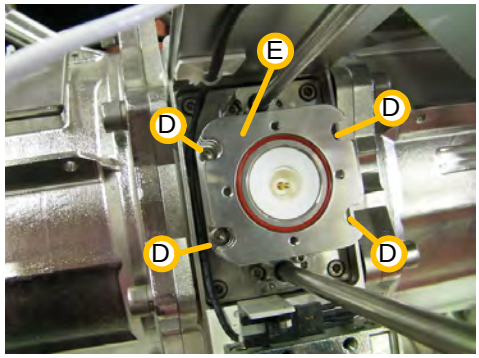


Figure 10-12: Remove the Lamp's Housing

Step 14. Grasp the lamp's housing (**E**) and rotate it clockwise and counterclockwise until it loosens. Make sure the housing is not tilted against the quartz sleeve while rotating it.

(The O-Ring on the sleeve stopper between the holder and the stopper may cause the two parts to stick due to the heat. The slight rotation releases it).

Step 15. **To prevent any damage to the lamp and quartz sleeve,** make sure you pull the lamp's housing (**E**) straight out and not at an angle.

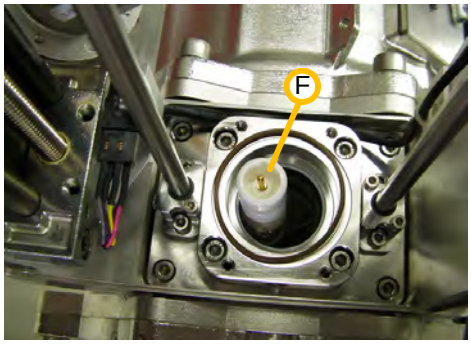


Figure 10-13: Access to the Lamp

Removing the Faulty Lamp

The lamp is hot! DO NOT touch the lamp with bare hands, particularly the quartz portion.
Wait at least ten minutes until the lamp has cooled down or use an appropriate pair of long-nose pliers to carefully pull out the lamp.

Step 16. Wearing appropriate cotton gloves and skin covering, and using a suitable long-nose pliers, pull the lamp straight out of the quartz sleeve in the Unit and set it aside on a stable surface.

Step 17. Check the inner surface of the quartz sleeve for residual moisture or high humidity. If any visible moisture exists, using a lens cleaning cloth, dry it.

Installing the Replacement Lamp

Step 18. Unpack the new UV lamp from its vacuumed sealed package.

Step 19. Make a note of the serial number of the new lamp.

Figure 10-14: Removing the lamp

Step 20. Clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp.

Step 21. Insert the new lamp **(F)** into the quartz sleeve, all the way until it stops. Ensure that the male contact point **(G)** on the end of the lamp has meshed with the female contact point **(G)** in the lamp's housing on the other side of the Unit.

Step 22. Check that the lamp housing's O-ring is in place, clean, dry and undamaged. Replace if necessary.

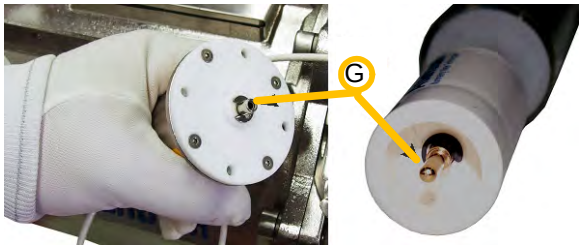


Figure 10-15: Lamp, Connector Assembly and connectors

Step 23. Reinstall the lamp's housing and, using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Step 24. Reinstall the lamp's connector assembly pushing it straight in. Ensure that the male contact point **(G)** on the end of the lamp has meshed with the female contact point **(G)** in the lamp's housing on the other side of the Unit.

Step 25. Using the # 3 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Reinstating the Electricity to the DPM

Step 26. On the Connection Box, set the switch to the **On** position.

Step 27. Open the 48VDC Module door and set the circuit breaker **(A)** to the **On** position.

Step 28. On the Control Module, return to the Cleaner screen.

Step 29. In the **Control Module**, Cleaner Settings screen, tap **Home** . The Maintenance mode is deactivated and the **Home** button becomes the **Maintenance** button again. The DPM Cleaning mechanism (inside) moves to its position at the top.

Reactivation the Lamp

Step 30. For the lamp that is replaced, on the connection box, turn the lamp switch to the **On** position. Check that the lamp ignites properly.Figure 10.6

Allowing the Atlantium Unit to Run with Caps Open

Step 31. **Temporarily place Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**

Step 32. On the side without the DPM, unscrew and remove the two caps **(H)** on each side of the lamp. (See the figure to the right.)

Step 33. Operate the system without the lamp's caps for one hour.

Step 34. Return the lamp's caps and remove the Caution floor signs.

Perform system tuning according the instructions on page **144**

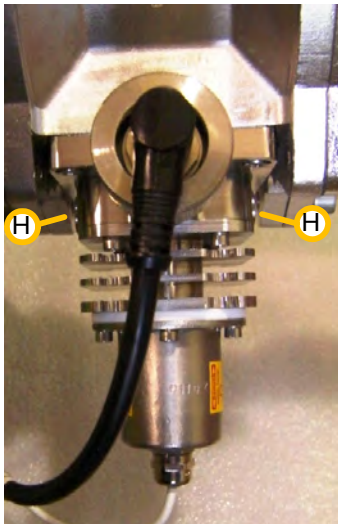


Figure 10-16: Lamp's Caps (Generic View)

For All

Step 35. Check that the lamp ignites properly.

Step 36. Operate the system without the lamp's caps for one hour.

Step 37. Return the lamp's caps **(G)**.

System Tuning

Step 38. Perform system tuning according the instructions on page **146**.

Observing Lamp Efficiency

Step 39. Observe the system's efficiency for **100 hours**. If after this period there is a performance degradation of more than 5%, remove the UV intensity sensor and check for humidity. Clean if necessary. For the adjustable UVIS sensor, see the instructions on page **141**.

Step 40. Perform **System Tuning** according the instructions on page **146**.

10.7 Cleaning / Replacing the UV Intensity Wave Guide

Performed as needed

If your system includes wave guides:

For each UVIS sensor in the Atlantium Unit, there is an internal wave guide for directing UV light into the sensor. The UV Intensity Sensor's wave guide must be cleaned every time the lamps are replaced.

Prepare the following:

- Replacement UV Intensity Sensor's Wave Guide
- Wave Guide tool
- White cotton gloves
- Alcohol/isopropanol and lint-free lens cleaning paper
- Appropriate Phillips screwdrivers

- Allow the old lamp at least 10 minutes to cool down before starting.

- Be very careful when handling the wave guide. The wave guide is made of quartz and is fragile and easily broken.

➲ To clean / replace the wave guide:

Accessing the Wave Guide to be Replaced

Step 1. On the side of the Unit where the UVIS sensors are located, identify the lamp whose wave guide is to be cleaned/replaced.

Step 2. Follow the directions for removing a lamp. See ***Replacing a UV Lamp*** on page **123.**

Removing the Wave Guide

Step 3. Screw the Wave Guide tool **(L)** onto the wave guide socket **(J)**.

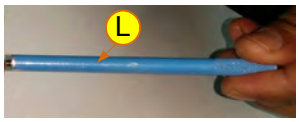


Figure 10-17: Wave Guide Tool

Step 4. While holding the Wave Guide tool **(L)** now attached to the wave guide **(G)**, using an appropriate Phillips screwdriver, **(M)** unfasten each of the screws **(K)**.

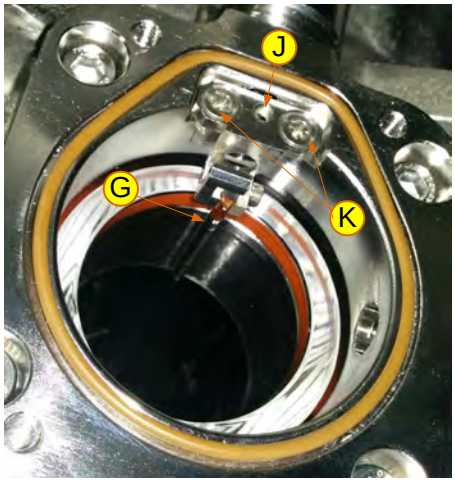


Figure 10-18: Wave Guide Elements (Generic View)

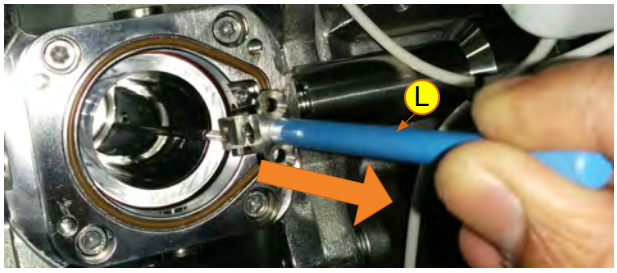


Figure 10-19: Removing the Wave Guide (Generic View)

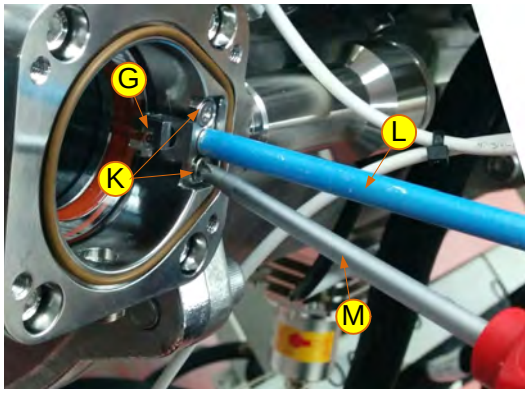


Figure 10-20: Unfastening the Wave Guide (Generic View)

Step 5. With the Wave Guide tool **(L)**, carefully pull out the wave guide.

Step 6. Unscrew the Wave Guide tool **(L)** and detach it from the wave guide **(G)**.

Figure 10-21: Wave Guide Tool and Wave Guide

Cleaning the Wave Guide

Step 7. While wearing white cotton gloves, carefully clean the wave guide with isopropanol and lint-free lens cleaning paper.

Step 8. Ensure that the wave guide is completely dry before returning it to its place.

Step 9. Examine the wave guide for marks and damage. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the wave guide's surface is damaged, it must be replaced.

Preparing the Replacement Wave Guide

Step 10. Screw the Wave Guide tool **(L)** onto the wave guide socket **(J)**.

Step 11. With the Wave Guide tool **(L)**, carefully insert the wave guide in place.

Step 12. While holding the Wave Guide tool **(L)** now attached to the wave guide **(G)**, using an appropriate Phillips screwdriver, **(M)** fasten each of the screws **(K)**.

Step 13. Check that the wave guide is parallel to the side of the quartz sleeve and not pointing at an angle. If it is at an angle, the quarts sleeve is not installed straight. Remove the wave guide and remove and reinstall the quartz sleeve following the directions in ***Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings*** on page **134**. When complete, install the wave guide again.

The correct position of the wave guide is parallel to the side of the quartz sleeve and a distance of 0.75mm from the quartz sleeve.

Step 14. Follow the directions to reinsert the lamp and refasten the lamp's housing and connector See ***Replacing a UV Lamp*** on page **123.**

To access the mirror, remove the lamp connector assembly from the side without the DPM cleaner. The Lamp housing on this side has the UVIS sensor connected. See page **123** for instructions on accessing the Lamp housing.

10.8 Cleaning / Replacing a Lamp's Quartz Sleeve and O-Rings

Perform Yearly

Performed as needed

The quartz sleeve is cleaned as needed according to the UVT Sensor readings and when deposit collects on the quartz sleeve. **The quartz sleeve's rubber O-rings must be replaced when dismantling the quartz sleeve**.

Depending on the quality of the water being treated by the Atlantium Unit, the exterior surface of the quartz sleeve that houses a lamp may require periodic cleaning or replacement if after the optional Cleaning system and the CIP process the quartz sleeve is still not sufficiently clean. This is to be determined by the quality control officer at the facility.

Under normal environmental conditions, the quartz sleeve's rubber O-rings (as shown in Figure 10-19 on page 136) must be replaced once a year even if they pass visual inspection. For environments where the water temperature is above 70°C (160°F), the quartz sleeve's O-rings must be replaced yearly.

Yearly maintenance required that the Quartz Sleeve be checked, o-rings be replaced, and, for systems with DPM, the dynamic seal and wiper be replaced. See ***Replacing the DPM Dynamic Seals, Base O-Ring, & Wiper Assembly Wipers*** on page 154.

Prepare the following:

- Replacement quartz sleeve (if replacing)
- Two O-rings (See view B in Figure 10-34 below)
- #2.5 Allen wrench
- #3 Allen wrench
- #4 Allen wrench
- Protective gloves
- Appropriate cleaning cloth. A mild, household cleaning solution
- Alcohol

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

➲ To clean or replace the quartz sleeve:

Powering Down the System

Step 1. On the **Control Module**, tap . The button turns grey. Wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Positioning the DPM Wiper Assembly

If you are accessing the lamp on the upper side, it is recommended to leave the DPM wiper in the parking position at the top. Is so, skip to Step 5. below.

If you are accessing the lamp on the lower side, it is recommended to position the wiper in Maintenance mode, which is at the lower end of the Unit.

Positioning the DPM Wiper Assembly for Maintenance Mode

Step 2. For systems with DPM, at the top of the **Control Module** screen, tap . The **Monitor Tab** is displayed.

Step 3. Tap the **Cleaner tab**. The **Cleaner** screen appears.

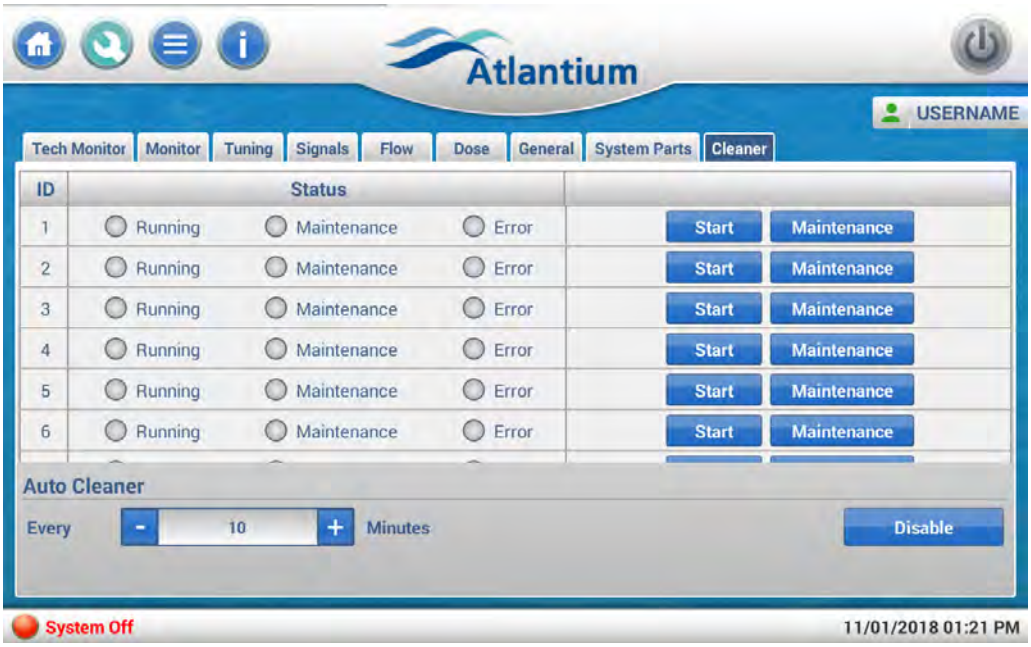


Figure 10-22: Cleaner Screen - Maintenance Mode

Step 4. For the desired lamp, on its line, tap **Maintenance**. The wiper assembly moves to the **Maintenance** position. The **Maintenance LED** displays green. For the desired lamp, on its line, tap **Maintenance**. White the wiper assembly moves to the **Maintenance** position, the **Start** button become the **Stop** button and The is **Maintenance** greyed. When the wiper assembly moves to the **Maintenance** position, the **Maintenance LED** displays green and the button becomes the **Home** button.

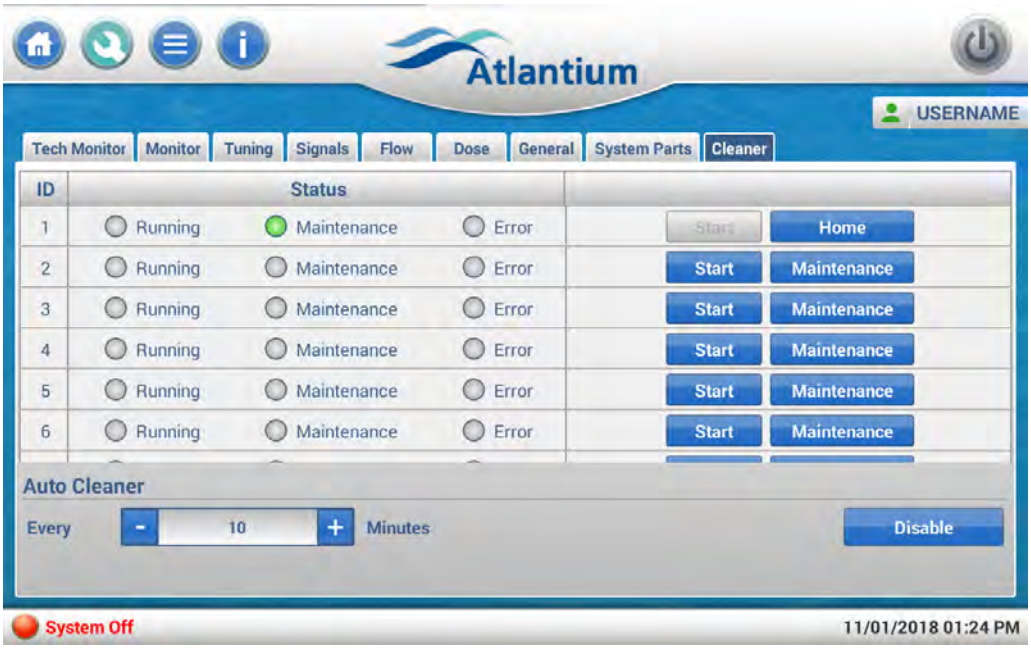


Figure 10-23: Cleaner Screen - Maintenance Operating on Lamp 1

Turning Off the Electricity to the DPM

Step 5. Open the 48VDC Module door and set the circuit breaker **(A)** to the **Off** position.

Step 6. On the Connection Box, set the switch to the **Off** position.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's before you proceed to replace the lamp.

Step 7. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

Step 8. Check the main screen of the **Control Module** and verify that the system parameter for **Power** is **0%**.

Isolating the Unit

Step 9. To isolate the Unit, close the Inlet and Outlet valves.

Step 10. Using the Drain valve, drain the water from the Unit.

Accessing the Quartz Sleeve

Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) and the lamp be removed, as well as the quartz sleeve holder. For systems with DPM, the DPM frame must be removed.

For Systems With DPM - Removing the DPM Frame

If your systems does not have DPM, proceed to **Step 17.**

Step 11. Using the #2.5 Allen wrench, on both sides of the DPM, remove the covers. Each side has 8 screws **(A)**. The top cover has 6 screws.

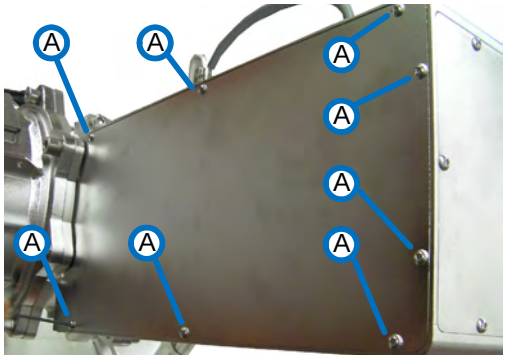


Figure 10-24: DPM Cover's 8 screws

Lowering the Rods

Step 12. Using a #3 Allen wrench, remove the screws **(B)** that hold the rods and push down the rods.

Step 13. Release the DPM cable and Lamp Cable **(C)** at the back so that it is loose.

Removing the Lamp Connector Assembly

Step 14. Follow from **Step 9.** on page **129** and remove the lamp connector assembly.

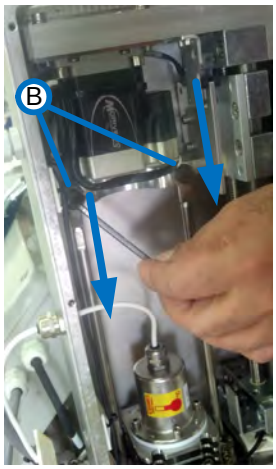


Figure 10-25: Unfastening the Rod

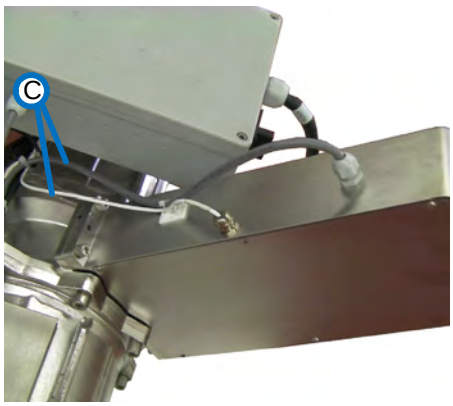


Figure 10-26: Release the cables at the back

Removing the DPM Frame

Step 15. To remove the DPM frame, unfasten the four bolts (two are show **(D, E)** and there are another two on the other side.

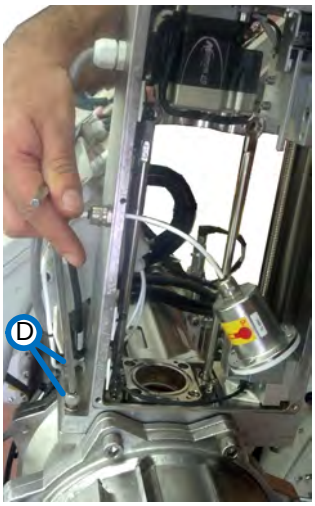


Figure 10-27: Unfastening the DPM Frame - Upper Bolts

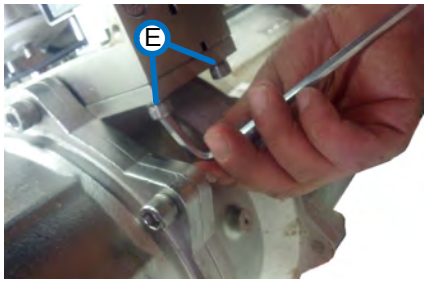


Figure 10-28: Unfastening the DPM Frame - Lower Bolts

Step 16. Carefully remove the DPM frame **(F)** and lay it on top of the nearby Connection Box **(G).**

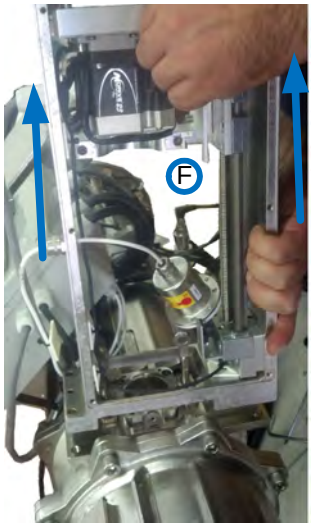


Figure 10-29: Removing the DPM Frame

Figure 10-30: DPM Frame Resting on the Connection Box

Step 17. Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) for systems that have them, and the lamp be removed. In the section, 10.6.4. Replacing the Lamp, follow from **Step 2.** on page **132** and remove the lamp connector assembly and lamp housing **from both sides of the Unit,** as well as the lamp, placing them in a secure location until they need to be reinstalled.

For systems with wave guide, while reinstalling the sleeve holder, DO NOT tilt it and DO NOT insert it at an angle. the wave guide is fragile. Take care not to damage it when inserting the lamp housing.

Step 18. To remove the quartz sleeve's holder on upper side of the quarts sleeve, using a #4 Allen wrench, remove the four Allen screws, opening them in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) **(A)** that secure the sleeve's holding ring to the Unit. Loosen one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the sleeve (2 turns).

Step 19. Continue to alternate until both sleeve holders are loose, and then remove the screws completely. Each screw has a washer.

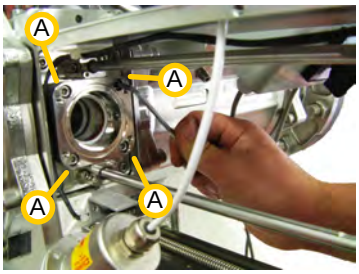


Figure 10-31: Removing the Quartz Sleeve's Holder

While pulling the quartz sleeve's holder away, DO NOT tilt it and DO NOT pull it away from the lamp/sleeve axis at an angle.

Step 20. On the lower side of the Lamp chamber, remove the cable from the UVIS sensor **(B)**.

Step 21. To remove the quartz sleeve's holder on the lower side of the quarts sleeve **(C)**, perform from **Step 18.** on it.

Figure 10-32: Removing the Cable from the UVIS Sensor

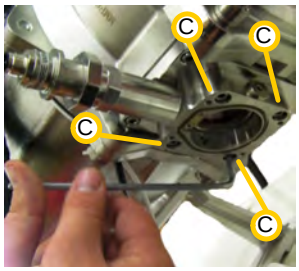


Figure 10-33: Removing the Quartz Sleeve's Holder on the Lower Side

Removing the Quartz Sleeve

Wear protective gloves. To protect the quartz sleeve, do not touch the internal surface with your bare hands.

Step 22. On upper side of the quartz sleeve, place a gloved palm of your hand over the quartz sleeve and gently push in. The quartz sleeve shifts inward. The o-ring on the side where you pushed slips out (if not, take it off).

Step 23. On the lower side of the quartz sleeve, the sleeve sticks out slightly. **To prevent any damage to the quartz sleeve, make sure you pull the quartz sleeve straight out.** Carefully pull out the quartz sleeve **(D)** and remove the O-ring **(E).**

Figure 10-34: Quartz Sleeve and O-ring (Generic View)

Cleaning the Quartz Sleeve

Step 24. If there is a deposit buildup on the sleeve's exterior surface, clean it with a mild, household cleaning solution.

Step 25. Clean the exterior and interior surfaces with alcohol. If the sleeve cannot be cleaned to a pristine condition, replace it with a new one.

Step 26. Ensure that there is no residual moisture or high humidity on the inner surface of the quartz sleeve. If any visible moisture exists, dry it before inserting it in place.

Installing the Quartz Sleeve

Step 27. Mount a new O-ring onto the quartz sleeve (as shown in Figure 10-34 above) and insert the sleeve into the Unit from the lower side until it reaches the other side. Make sure it is centered as you insert it.

Step 28. Add the second new O-ring on the other side of the quartz sleeve.

Step 29. Install the quartz sleeve holder on the lower side and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**

Step 30. Install the quartz sleeve holder on the upper side.

Step 31. Using a #4 Allen wrench, close the screws in diagonal pattern and tighten them.

Step 32. To reinstall the lamp, See ***Replacing a UV Lamp*** on page **123.**

Step 33. Reinstall the lamps' connector assemblies pushing each straight in and using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure them to the Unit.

Step 34. Connect the UVIS sensors' cables.

Step 35. On the corresponding Connection Box to where the sleeves/lamps are located, turn the switch to the **On** position.

For Systems with DPM - Reinstalling the DPM Frame

Step 36. On the top, pull the DPM rods up all the way.

Step 37. Pick up the DPM frame over the rod and put it in place. Fasten the four bolts (two are show **(C)** and there are another two on the other side.

Step 38. Using an applicator stick, dab the bolt that is to fasten the DPM rod to the DPM housing with Loctite glue.

Step 39. Using a #3 Allen wrench, on the DPM, fasten the bolt **(W)** to hold the rod of the Wiper assembly in place.

Step 40. Using the #2.5 Allen wrench, on both sides of the DPM, place covers and fasten its screws. Each side has 10 screws. The top cover has 6 screws.

Step 41.

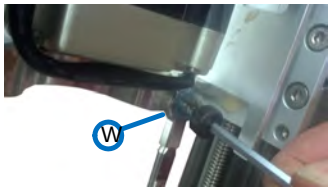


Figure 10-35: Installing the Plate and Wiper Assembly

Reintroducing the Water

Step 42. Carefully open the Inlet and Outlet valves (Be sure to avoid water hammer). Follow the directions in the section ***Filling unit with water*** on page **99**.

Reinstating the Electricity

Step 43. Open the 48VDC Module door and set the circuit breaker to the **On** position.

Step 44. On the corresponding Connection Box to where the sleeves/lamps are located, turn the switch to the **On** position.

Powering Up the System

Step 45. On the **Control Module**, tap . The button turns green. Wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Step 46. In the **Control Module**, Cleaner Settings screen, tap **Home** . The Maintenance mode is deactivated and the **Home** button becomes the **Maintenance** button again. The DPM Cleaning mechanism (inside) moves to its position at the top.

Verifying System Performance

Step 47. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 48. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 49. Run the system with the caps of the lamp off as described in the section, 10.6.4. Replacing the Lamp, see from **Step 17.** to the end of the procedure, starting on page **123**.

WARNING!

While lamp caps are off, be sure to follow safety requirement:

- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

Step 50. Check the main screen of the **Control Module** and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range and record is using the Table in Appendix C. Checking the System Parameters on page **184**.

If the values are out of range, notify the executive maintenance engineer.

10.9 Checking / Replacing Sensors

From time to time, sensors may need to be checked (most frequently for condensation). During the Calibration process, the UVIS sensor also requires to be removed and temporarily replaced with a Reference Master sensor during. At times a sensor needs to be replaced. There are two types of UVIS sensors. The sensor itself is the same in both types. The difference is the sensor casing and how it is fastened to the Atlantium Unit. The two types are:

- **Adjustable UVIS sensor** - The Sensor's position is adjustable and its casing is fastened with nuts. See below.
- **Fixed UVIS sensor** - The sensor's casing screws into a fixed position. This type of sensor may contain a pinhole. See ***Replacing/Checking a UVT Analyzer or Fixed UVIS Sensor*** on page **144**.

10.9.1 Checking an Adjustable UVIS Sensor

Performed as needed

Before removing the UVIS sensor to check it for moisture, do one of the following:

If you do not have a caliper measurement tool: You must mark the placement of the sensor so that you can replace it in the correct position.

If you have a caliper: You must measure the position of the sensor. (See on page **122.**) or by using a marker to mark the position before removal.

Prepare the following:

- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

➲ To check an adjustable sensor (using a marker):

Step 1. In the **Control Module**, note the value for UVIS for the lamp on which the sensor to be replaced. For details on finding the **Control Module** screen with this information, see the section, ***System Tuning*** on page **146**.

Step 2. Mark the placement of the sensor so that you can replace it in the correct position. Use the thin marker to mark the position before removal.

Mark the sensor in two places, a line along the nut and across the nut and Perpendicular to the first line. as shown in **(1)**.

Step 3. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

Step 4. When you return the sensor to its place, make sure the thin lines line up to the nut and align together.

Step 5. In the **Control Module**, check that the value for UVIS for the lamp on which the sensor was replaced is identical to the value before the sensor was removed. For details on finding the **Control Module** screen with this information, see the section, ***System Tuning*** on page **146**.

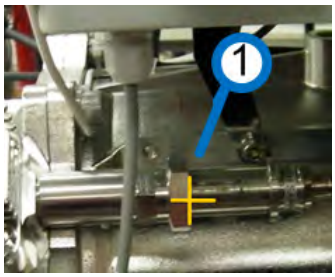


Figure 10-36: Marking the Sensor

➲ To check an adjustable sensor (using a caliper):

Step 1. Place the measurement tool on the UVIS sensor as show in Figure 10-37 below. Use the screw to adjust the placement of the caliper measurement tool.

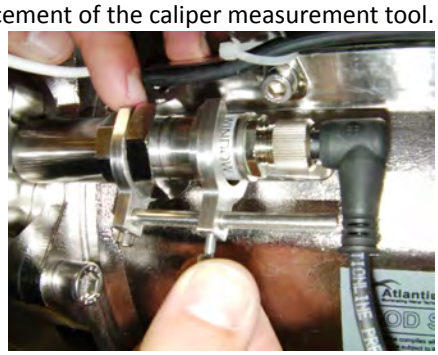


Figure 10-37: Measuring the UVIS sensor

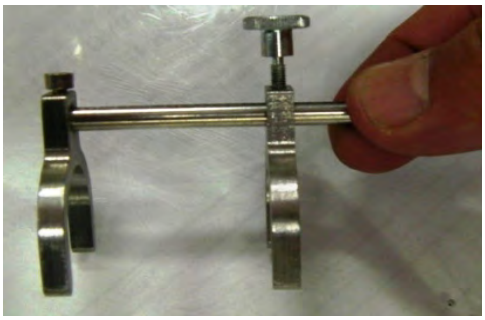


Figure 10-38: Caliper Measurement Tool

Step 2. Carefully remove the caliper, taking care not to accidentally change its position, and set it aside.

Step 3. Remove the sensor according to the relevant steps in ***Replacing an Adjustable UVIS Sensor*** on page **142**.

Step 4. If you are in the midst of a Calibration process for the UVIS sensor, replace the original UVIS sensor with the Reference sensor.

Step 5. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

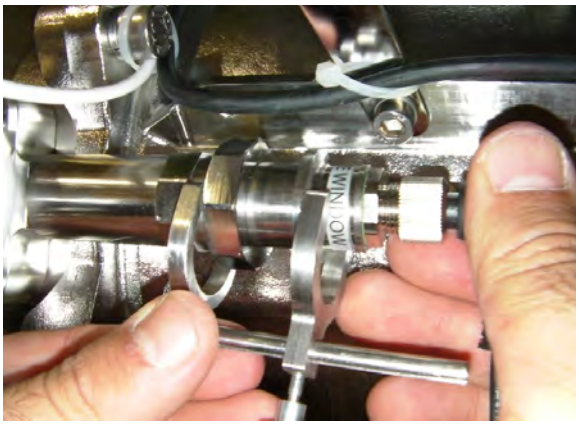


Figure 10-39: Adjusting the Reference sensor

Step 6. When you insert a Reference Master sensor or return a sensor to its place, using the set caliper (calibrated to the original UVIS sensor's position) adjust the sensor so that it is set to the position of the original. Refer to Figure 10-39 to the left.

10.9.2 Replacing an Adjustable UVIS Sensor

Performed as needed

Before removing the UVIS sensor when calibrating it against the Reference Master sensor or checking for moisture, you must mark the placement of the sensor so that you can replace it in the correct position. This is accomplished using the caliper measurement tool (See on page **122.**) or by using a marker to mark the position before removal. Replacing the UVIS sensor involve the loosening of the nut holding it in place.

Prepare the following:

- Replacement UVIS sensor or appropriate type
- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

➲ To replace an adjustable UVIS sensor:

Step 1. To disconnect the harness connector (shown in view **A** in Figure **10-41**), unscrew the sensor's threaded locking nut.

Step 2. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view **B** below. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 3. Holding the left wrench **#24** steady, twist the right wrench **#30** to loosen the nut.

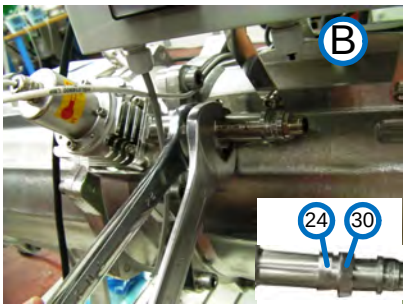


Figure 10-40: Replacing the Sensors (Generic View 2)

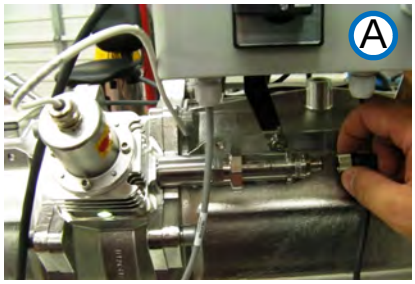


Figure 10-41: Replacing the Sensors (Generic View 1)

Step 4. Carefully pull the sensor out of the housing **(C)**. The Large nut comes off together with the sensor.

Step 5. To enable the system to track relevant data about the sensor, make a note of the serial number of the replacement sensor (for later use).

Step 6. Verify that the sensor lens **(1)** is clean. If needed, using a lens cloth, clean it. (See below.)

Be sure that the lens is completely dry before it is returned to its place.

Step 7. When replacing a sensor, remove the nut **(2)** and the washer **(3)** shown in Figure 10-43 below.

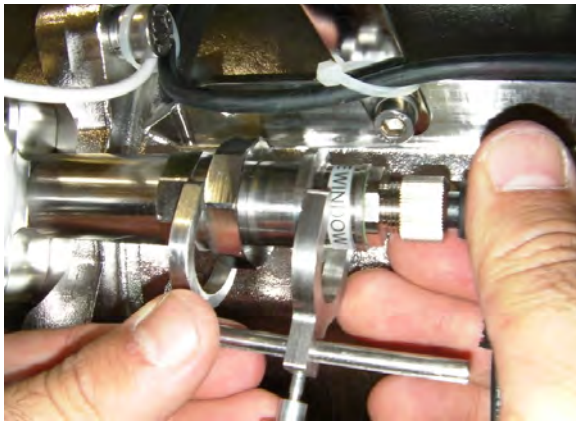


Figure 10-42: Replacing the Sensors (Generic View 3)

Step 8. Position the nut **(2)** on the sensor as shown above. Then position the washer **(3)** on the sensor as shown above.

Step 9. Insert the sensor into the housing.

- If you are replacing the sensor, positioning it about 2/3 the way in. Fasten the nut onto the housing.
- If you are returning a sensor that was removed for checking or calibration, see ***To check a sensor or remove it for calibration*** on page **141**.

Step 10. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view **B** above. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 11. Holding the left wrench **#24** steady, twist the right wrench **#30** to tighten the nut.

Step 12. Reconnect the harness's threaded cap to the new sensor.

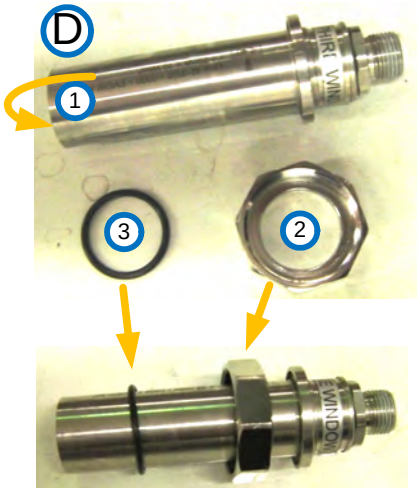


Figure 10-43: Sensor, washer and nut

Step 13. For a replacement sensor, in the **Control Module**, perform **System Tuning**, See the section, ***System Tuning*** on page **146**.

10.9.3 Replacing / Checking a UVT Analyzer or Fixed UVIS Sensor

Performed as needed

For system that may contain a fixed UVIS sensors, this sensor-replacement procedure is required. If your system contains the adjustable UVIS sensor, see ***Checking an Adjustable UVIS Sensor*** on page **141**.

For some of the sensors, there is an pinhole located on the sensor's window. This pinhole is calibrated by Atlantium for the specific measurement point. Do not switch disks between ports. Before inserting the sensor, be sure to include the pinhole if present.

Prepare the following:

- Replacement sensor

➲ To replace a sensor:

Step 1. To disconnect the harness connector (shown by the arrow in view **A** below), unscrew the sensor's threaded locking nut.

Step 2. Unscrew the sensor's threaded locking nut from the bottom of the sensor housing, as shown in view **B**.

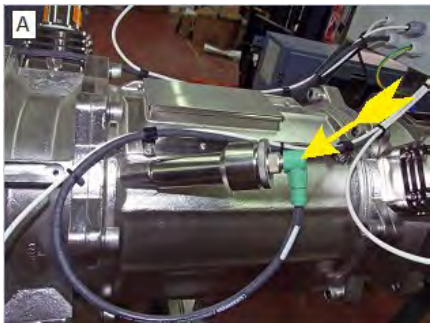


Figure 10-45: Replacing the Sensors 2 (Generic View 1)

Figure 10-44: Replacing the Sensors 2 (Generic View 2)

Step 3. Carefully pull the sensor out of the housing **(C)**.

Step 4. If an pinhole **(E)** is present it slides out.

Step 5. Unpack the new sensor **(D)** and (if it is supplied) the sensor pinhole.

Figure 10-47: Replacing the Sensors 2 (Generic View 3)

Figure 10-46: Replacing the Sensors 2 (Generic View 4)

Step 6. To enables the system to track relevant data about the sensor, make a note of its serial number.

Step 7. Verify that the sensor lens **(F)** is clean. If needed, using a lens cloth, clean it.

Step 8. If an pinhole **(E)** is present, insert it.

Step 9. Insert the sensor into the housing, and screw the threaded cap back onto the housing, securing the sensor.

Step 10. Reconnect the harness's threaded cap to the new sensor.

Step 11. In the **Control Module**, perform **System Tuning**, see the section, ***System Tuning***, below.

10.10 Testing System Lamps

Performed as needed

During normal operations, an authorized facility technician may need to test the lamps that appears on the Tech Monitor screen.

➲ To test the lamps:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **Tech Monitor Tab** . The **Tech Monitor** screen appears.

Figure 10-48: Tech Monitor Tab

Step 3. For each lamp to be tested according to the **Lamp ID** column on the left, make a mark in the checkbox on the right.

Step 4. To adjust the power of the selected lamp(s), using the **-/+** buttons, set the desired power value. Check that the values for the other columns are in normal range.

Step 5. To Turn off the he selected lamp(s), tap **Lamp Off**. The Lamp's **On/Off** icon turns grey. Check that the selected lamp is indeed off. Check that the values for the other columns are in normal range.

Step 6. To Turn on the he selected lamp(s), tap **Lamp On**. The Lamp's **On/Off** icon turns green. Check that the selected lamp is indeed on. Check that the values for the other columns are in normal range.

When leaving the **Tech Monitor tab**, the lamps are returned to the settings of the system's configuration.

10.11 System Tuning

Performed as needed

System tuning is performed on these occasions:

- As the last step in system installation
- As the last step when lamps are changed and after 100 operating hours (as needed)
- Upon advice of an Atlantium Service Representative when the system is reset

➲ To tune the Atlantium system:

Step 1. Just prior to tuning, obtain a water sample from the sampling valve on the inlet side of the Atlantium Unit. Test the water sample in an external spectrophotometer. Write down the results. If a spectrophotometer in not available, record the current UVT value shown on the system's display.

Step 2. In the **Control Module**, on the top of the screen, tap . The button turns grey. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 3. On the Atlantium Unit, locate each UVIS and UVT Intensity Sensor.

Step 4. On each sensor (refer to the figures in the section, ***Checking/Replacing Sensors*** on page **140**):

- **a** Unscrew the locking nut and gently pull out the sensor.
- **b** With a dry and clean lint free cloth, wipe the UVIS sensor's quartz window.
- **c** Reinsert the sensor and replace the locking nut.

Step 5. To turn on the Atlantium system, on the **Control Module** main screen, tap . The button turns green. Wait approximately three minutes for the **Ready Lamps On** signal to transmit.

Step 6. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 7. Tap the **Tuning Tab**. The **Tuning** screen appears.

Figure 10-49: Technician Settings Screen> Tuning Tab (Actual numbers vary according to system configuration)

Step 8. Verify that the lamps are on and enabled (the number of lamps listed in the display is according to the number of lamps in your system configuration).

Step 9. For each lamp to be tuned, according to **Lamp ID** in column **1**, make a mark in the appropriate checkbox in the last column on the right.

Step 10. Under **Status T0 %** , using the **-/+** buttons, set the **Status T0 percentage** up to **100%**.

Step 11. Wait a few minutes to allow the system to stabilize.

Step 12. To verify that the Sensor readings are stable, observe the reading for a few seconds to see that they remain the same numbers. For the UVIS sensor, see the **UVIS mA** column above.For the UVT Intensity sensor, see the **UVTS mA** column above. If the numbers are not stable, unscrew and remove the two caps on each side of the lamp. (See Figure **10-12** on page **131**.)

Step 13. Leave the caps open for 30 minutes to make sure no humidity is present in the lamp housing. Then replace the caps and check the sensor reading again.

When the Lamp's caps are removed, do not look directly into the opening without appropriate eye protection. The UV light emitted via the opening is a danger to the eyes.

Step 14. At the bottom of the screen, for **UVT To**, enter the current UVT value obtained from the spectrophotometer that you wrote down in **Step 18.** above (or the recorded current UVT value).

Step 15. To record the reason for tuning the system, At the bottom of the screen, for **Tuning Reason**, from the dropdown list, select the reason.

The options are:

- **System Integration**
- **Lamp changed**
- **Maintenance**

Step 16. Tap **Start Tuning**. A progress bar appears at the bottom of the screen. The process takes about 6 to 15 minutes.

If the system is not running, or if a lamp is not on, a popup message appears informing you that in order to start a tuning process, the system must be running.

Step 17. Verify that the UVT is the number you set previously (see the **UVTS mA** column) and that the Status on the Monitor tab is the same as the number you set in the **Status T0 %** column.

Step 18. To shutdown the Atlantium system, on the **Control Module** main screen, tap and wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Step 19. On the Atlantium Unit, locate each UVIS and UVT Intensity Sensor.

Step 20. On each sensor (refer to the figures in the section, ***To replace an adjustable UVIS sensor:*** on page **142**):

- **a** Unscrew the locking nut and gently pull out the sensor.
- **b** With a dry and clean lint free cloth, wipe the UVIS sensor's quartz window.
- **c** Reinsert the sensor and replace the locking nut.

Step 21. To turn on the Atlantium system, on the **Control Module** main screen, tap and wait approximately three minutes for the **Ready Lamps On** signal to transmit.

10.12 Replacing / Editing the System Components in the Control Module

The following system components may be replaced from time to time:

- Lamps
- UVIS
- UVTS
- Ballast (Module)s
- Cleaner
- Reference UVIS

If they are replaced, the new component needs to be recorded in the **Control Module**. If a record contains an error, it can be edited.

➲ To record a replacement system component:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **System Parts tab**. The **System Parts** screen appears.

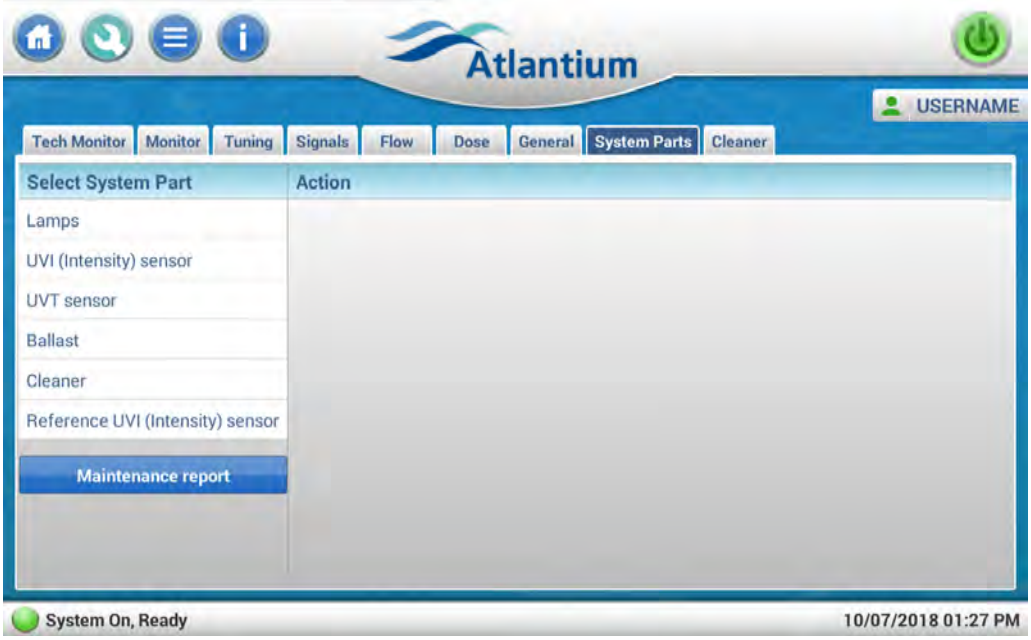


Figure 10-50: System Parts Screen - Lamps Selected

Step 3. Tap the desired system component. The options are:

- Lamps
- UVIS
- UVTS
- Ballasts
- Cleaner
- Reference UVIS

Under **Action**, the **Edit** and **Replace** buttons appear.

Step 4. Tap **Replace**. The field of the record appear.

Figure 10-51: System Parts- Lamps Replace Screen

The fields vary depending of the selected component.

Step 5. For **Unit**, if more than one Atlantium Unit is in the system configuration, select the Unit number.

Step 6. For **Choose Card**, from the dropdown list, select the corresponding Connection Box card (corresponding to the lamp associated with it) to which the component is connected.

Step 7. For **Serial Number**, enter the component's serial number.

Step 8. For **Replace Reason**, select the reason the component is being replaced. Option are:

- Expired
- Malfunction
- Other

Step 9. For **Age**, using the **-** button, reset the number of hours for the component's operating time to **0**.

Step 10. Tap **Done**. The changes are saved in the database and the name of the logged in Operator appears in the Operator field.

10.12.1 Editing a System Parts Record

If you find you made an error in the record of the details of the system part, you can edit the record.

➲ To edit the system part record:

Step 1. Navigate to the System Parts screen as in the procedure above.

Step 2. Tap the desired system component.

Under **Action**, the **Edit** and **Replace** buttons appear.

Step 3. Tap **Edit**. The field of the record appear.

Figure 10-52: System Parts- Lamps Edit Screen

Step 4. Modify the fields as needed (see the procedure above).

Step 5. For Edit Reason, from the dropdown list, select the reason for editing. The options are:

- No Serial
- Serial Mistake
- Other

Step 6. Tap **Done**. The changes are saved in the database and the name of the logged in Operator appears in the Operator field.

10.13 Anti-Corrosion Maintenance

The Atlantium Unit has been coated with the OKS 2101 – Protective Film for Metals to protect the Unit against corrosion. This protective film must be reapplied yearly to make sure the Atlantium Unit is constantly protected from the corrosion that can be affected in a seawater environment.

Prepare the following:

- OKS 2101 Spray
- Safety glasses with side-shields conforming to EN166
- Protective gloves that satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it
- Respiratory protection mask with filter type A-P

WARNING!

Refer to the OKS 2101 Safety Data Sheet according to Regulation (EC) No. 1907/2006 - GB.

Follow these precautions for safe handling

Do not use in areas without adequate ventilation.

Do not breathe vapors or spray mist.

In case of insufficient ventilation, wear suitable respiratory equipment.

Avoid contact with skin and eyes.

Keep away from fire, sparks and heated surfaces.

Persons with a history of skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.

Smoking, eating and drinking should be prohibited in the application area.

Wash hands and face before breaks and immediately after handling the product.

Do not get in eyes or mouth or on skin.

Do not get on skin or clothing.

Do not ingest.

Do not use sparking tools.

These safety instructions also apply to empty packaging which may still contain product residues.

Pressurized container: protect from sunlight and do not expose to temperatures exceeding 50 °C. Do not pierce or burn, even after use.

➲ To apply the protective film:

Step 1. Wear the protective glasses, gloves and filter mask as specified above.

Step 2. For optimum effect, clean the surfaces. Best way is to clean mechanically first and then with OKS 2610/ OKS 2611 universal cleaner. The surfaces must be bright metal and dry. Stir OKS 2100 thoroughly before use.

Step 3. Shake the OKS 2101 can.

Step 4. Spray on a thin protective film in an evenly manner to the prepared surfaces Move the attached cables aside as you spay. Make sure to cover the entire body, including under the cables. Avoid local excesses. The optimal layer thickness is 50µm.

Step 5. At an environmental temperature of 20°C, allow the Unit to dry for 30 minutes. In colder environments, allow additional time.

10.14 DPM Maintenance

The DPM Cleaner require the following maintenance procedures as needed:

- Cleaning the DPM's Opto-Coupler (See below)
- Replacing the DPM Belt (See page 153)
- Replacing the DPM Dynamic Seal, Base PO-Ring, & Wiper Assembly Wipers (See page 154)

10.14.1 Cleaning the DPM's Opto-Couplers' Sensors

Performed as needed

An Opto-Coupler's connector in the DPM mechanism may malfunction due to dust collected, covering its sensor.

Prepare the following:

- Cotton swab
- Alcohol
- #2.5 Allen wrench

To clean the Opto-Couplers' Sensors:

Step 6. Dip a cotton swab in alcohol and wipe the Opto-Coupler's sensor **(A)** clean.

Step 7. At the top of the **Control Module** screen, tap . The **Monitor Tab** is displayed.

Step 8. Tap the **Cleaner tab**. The **Cleaner** screen appears.

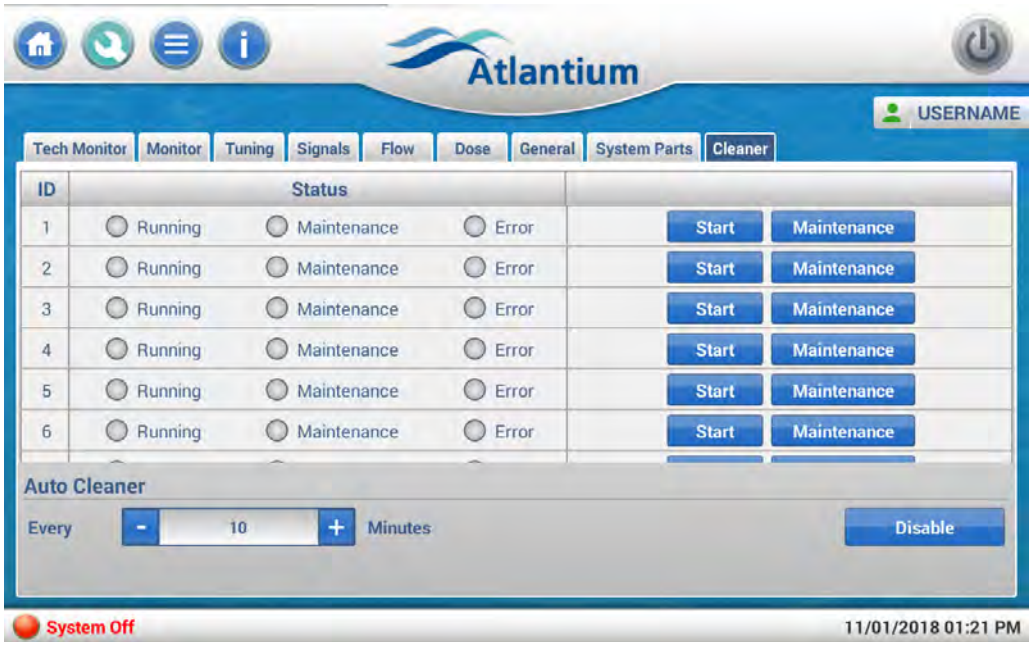


Figure 10-53: Cleaner Screen - Maintenance Mode

For the desired lamp, on its line, tap **Maintenance**. White the wiper assembly moves to the **Maintenance** position, the **Start** button become the **Stop** button and The is **Maintenance** greyed. When the wiper assembly moves to the **Maintenance** position, the **Maintenance LED** displays green and the button becomes the **Home** button.

Step 9. Dip a cotton swab in alcohol and wipe the Opto-Coupler's sensor **(B)** clean.

Step 10. In the **Control Module**, Cleaner Settings screen, tap **Home** . The Maintenance mode is deactivated and the **Home** button becomes the **Maintenance** button again. The DPM Cleaning mechanism (inside) moves to its position at the top.

Step 11. To perform the maintenance process on another lamp, when the process is finished, **Control Module** carry out the steps again again.

Step 12. Reposition the DPM cover and using the #2.5 Allen wrench, refasten the eight screws to secure it.

10.14.2 Replacing the DPM Belt

Performed as needed

If the belt inside the DPM becomes loose or torn, follow the procedure in this section.

Prepare the following:

- Replacement Belt
- #2.5 Allen wrench
- #4 Allen wrench

To replace the DPM belt:

Step 1. Open the 48VDC Module door and set the circuit breaker **(A)** to the **Off** position.

Step 2. On the Connection Box, set the switch to the **Off** position.

Step 3. Using the #2.5 Allen wrench, remove the DPM covers from the left side and the top.

Step 4. To release the DPM engine components, using the #4 Allen wrench loosen the screws **(B)** at the top.

Step 5. Carefully detach the belt **(C)** from one of the gears and then from the other, pulling it free.

Figure 10-54: 48VDC Module Circuit Breaker

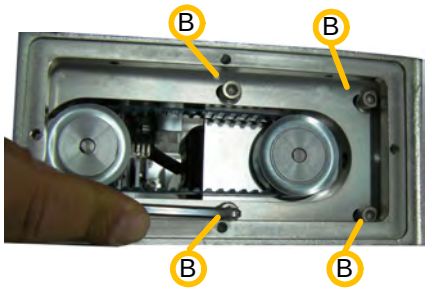


Figure 10-55: DPM Mechanism Top Screws

Figure 10-56: Removing the Belt

Step 6. Carefully insert the replacement belt **(D)** over one of the gears **(E)** and then the other.

Step 7. With on hand, hold the DPM mechanism stead against the side **(F)** and, using the #4 Allen wrench, tighten the four screws on top **(G)**.

Step 8. On the Connection Box, set the switch to the **On** position.

Figure 10-57: Replacement Belt

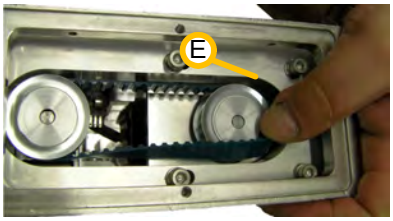


Figure 10-58: Inserting the Replacement Belt

Step 9. Reposition the DPM cover on the top and using the #2.5 Allen wrench, refasten the four screws to secure it.

Step 10. Reposition the DPM cover on the side and using the #2.5 Allen wrench, refasten the eight screws to secure it.

Step 11. Open the 48VDC Module door and set the circuit breaker **(A** see above**)** to the **On** position.

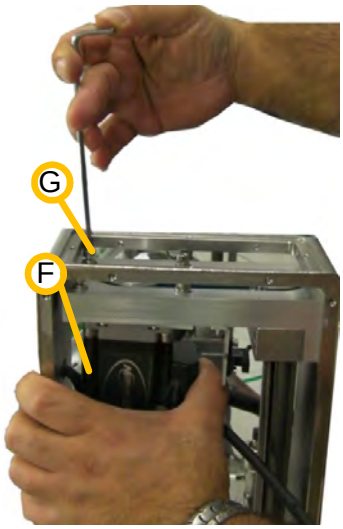


Figure 10-59: Tightening the screws

10.14.3 Replacing the DPM Dynamic Seals, Base O-Ring, & Wiper Assembly Wipers

The DPM's dynamic seals, base o-ring, and wiper assembly wipers wear out. They need to be replaced yearly. In locations where water contains excessive minerals, these may require more frequent replacement. Consult with your Atlantium representative. Follow the procedure in the section.

■ Two Replacement Dynamic Seals	■ #2.5 Allen wrench
■ Replacement wiper assembly wiper	■ #3 Allen wrench
■ Replacement O-rings	■ #4 Allen wrench
■ Protective gloves	■ Small Phillips screwdriver
■ Loctite® Threadlocker Blue 242® glue and applicator stick	■ Small flat screwdriver
	■ #6 Box wrench
■ Appropriate cleaning cloth	■ Adjustable Wrench
■ Alcohol	

While carrying the steps of this procedure, replace all of the O-rings on any component that you open, such as the lamp housings and holders, are not damaged.

➲ To replace the DPM's dynamic seal, base o-ring, and wiper assembly:

Replacing the DPM's dynamic seal, base o-ring, and wiper assembly wipers can be carried out once the quartz sleeves have been removed. Refer to the section, ***Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings*** on page **134**.

Step 1. Perform the steps for removing the lamps and sleeves in the section, ***Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings*** on page **134**,

Removing the DPM Plate and Wiper Assembly

Step 2. Using the #2.5 Allen wrench, unfasten the screws **(A)** of the plate **(B)**.

Step 3. Wedge a flat screwdriver under the plate **(B)** and push it straight out, together with the wiper assembly **(C)** as well.

Step 4. Replace all exposed o-rings.

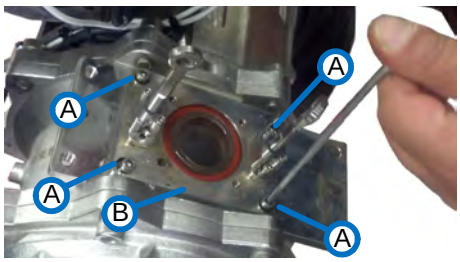


Figure 10-60: Remove the Screws of the Plate

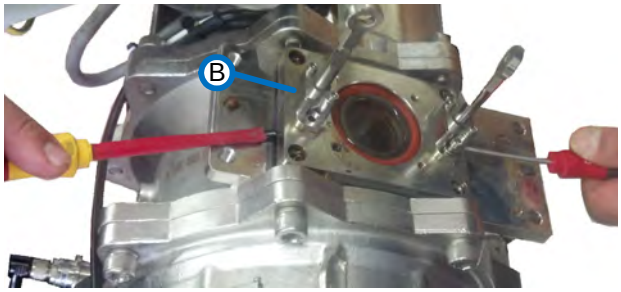


Figure 10-61: Remove the Plate

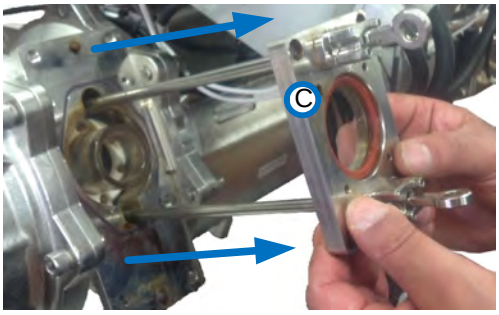


Figure 10-62: Remove the Plate and Wiper Assembly

Removing and Installing the Quartz Sleeve/Replacing the O-Rings

Step 5. In the section, 10.8. Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings, perform the steps from section, ***Removing the Quartz Sleeve*** on page **138**, starting with **Step 22.** until **Step 32.**

Removingthe Wiper Assebly Rods

Step 6. Grip **(E)** the top part of the Wiper assembly's rod **(F)** and hold it steady. At the same time, using an Allen wrench, unfasten the bolt **(D)** of holding the rod to the Wiper assembly.

Step 7. Repeat for the second rod.

Step 8. Remove the wiper assembly **(G)**.

Step 9. Pull the rod out of the guide seal **(F)** and plate **(C)** from the top.

Step 10. Repeat for the second rod.

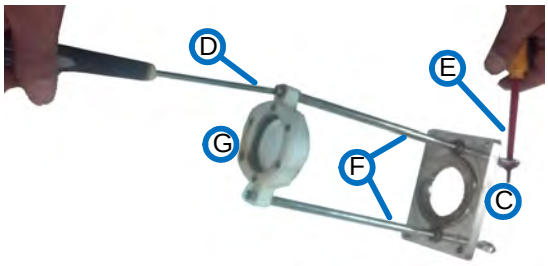


Figure 10-63: Removing the Rods' Screws

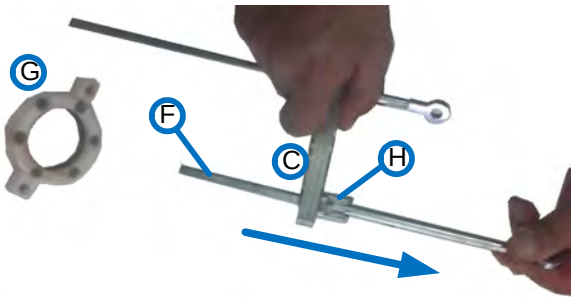


Figure 10-64: Removing the Rods

Replacngthe Dynamic Seals

Step 11. On the guide seal, attached to the DPM base, unfasten the screws **(I)**.

Step 12. Wedge a flat screwdriver under the cover **(J)** of the guide seal and nudge it off. The dynamic seal **(K)** is revealed.

Step 13. Using a tool to help, remove the dynamic seal **(K)**.

Installing the New Guide Seals

Step 14. Insert the new dynamic seal **(K)** as shown below and positioned with the top side **(K1)** facing up and the bottom side **(K2)** with the grooves facing down.

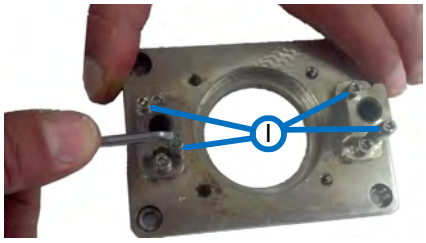


Figure 10-65: Removing the Rods' Screws

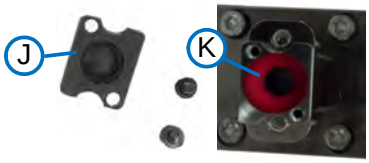


Figure 10-66: Dynamic Seal Revealed

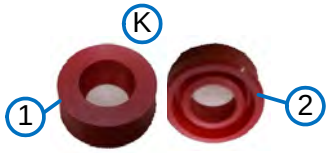


Figure 10-67: Dynamic Seal - Top and Bottom View

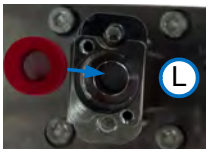


Figure 10-68: Inserting the Dynamic Seal

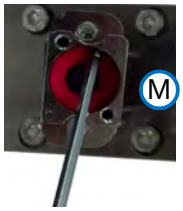


Figure 10-69: Pressing In the Dynamic Sealw

Step 15. Return the cover **(J)** and refasten the screws **(I)**.

Replacing the wiper

Step 16. Using an small Phillips screwdriver, unscrew the six screws on the wiper assembly **(G)**. Set the screws and the nuts in a safe place.

Step 17. Remove the wiper assembly cover **(P)** and then remove the wiper itself.

Step 18. Note the position of the ridge **(N)** on the replacement wiper and insert it -ridge inward **(N)**-into the wiper assembly case.

Figure 10-70: Opening the Wiper Assembly

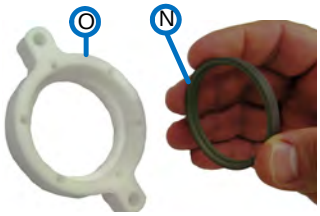


Figure 10-71: The DPM Wiper

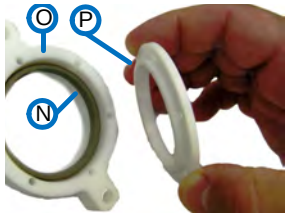


Figure 10-72: The Wiper Assembly Cover

Step 19. Using an applicator stick, dab each of the screws with Loctite glue **(Q)**.

Step 20. Put the cover **(P)** back on and, using a small Phillips screwdriver on one side and a small Box wrench on the other, fasten the six screws and nuts **(R)** back on the wiper assembly bottom **(O)**.

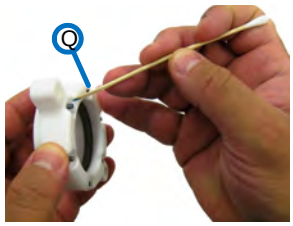


Figure 10-73: Applying Glue

Step 21. With an appropriate cleaning cloth and Alcohol, thoroughly clean the wiper area in the wiper assembly **(S)**.

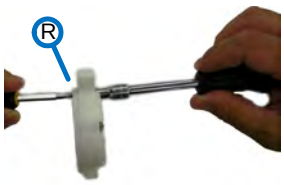


Figure 10-74: Applying Loctite

Figure 10-75: Cleaning the Wiper

Assembling the DPM Plate and Wiper Assembly on the Rod

Step 22. Insert a rod **(H)** into a guide seal **(F)** and plate **(A)** from the top.

Step 23. Repeat for the second rod.

Step 24. Insert the rod **(H)** into the wiper assembly **(G)**.

Step 25. Using an applicator stick, dab the screw **(D)** that is to fasten the wiper assembly to the rod with Loctite glue.

Step 26. Repeat for the second rod.

Step 27. Grip **(E)** the top part of the Wiper assembly's rod **(H)** and hold it steady. At the same time, using an Allen wrench, fasten the central bolt **(D)** of the wiper assembly.

Step 28. Repeat for the second rod.

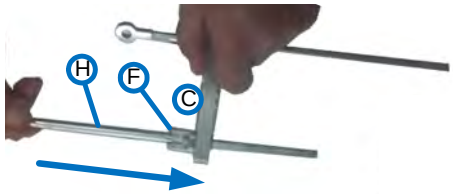


Figure 10-76: Inserting the Rod into the GuideSeal/ Plate

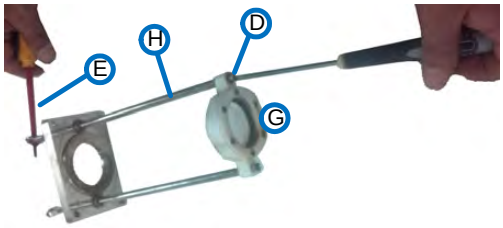


Figure 10-77: Fastening the Central Bolt of the Wiper Assembly.

Installing the DPM Plate and Wiper Assembly

Step 29. Push the wiper assembly **(T)** in, followed by the plate. Be sure to push them straight in.

Step 30. Using a #4 Allen wrench, fasten the plate **(U)** in place tightening each of the four Allen screws and washers in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) that secure the sleeve's holding ring to the Unit. Tighten one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the sleeve (2 turns).

Step 31. Continue to alternate until all the screws are tight.

Step 32. Insert a new o-ring **(V)**.

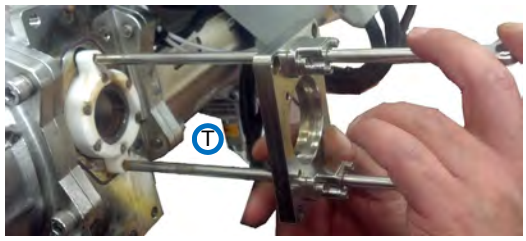


Figure 10-78: Installing the Plate and Wiper Assembly

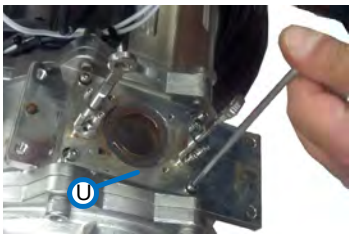


Figure 10-79: Fastening the Plate

Figure 10-80: Inserting the O-ring

Return to the section, 10.8. Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings, and perform the steps for reattaching the DPM frame until the end of the procedure from **Step 36.** on page **139**.

10.15 Advance Maintenance

The advanced maintenance procedures require opening the Connection Box. **Local regulations may require a certified electrician to perform any of these procedures.**

This section contains:

- ***Replacing the Temperature Sensor*** on page 159
- ***Replacing the Temperature Converter Card*** on page 160

10.15.1 Replacing the Temperature Sensor

In the event that the Temperature sensor requires replacement, this section details the procedure.

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature sensor
- #5 Allen wrench
- #3 Flat screwdriver
- #2 Phillips screwdriver

➲ To replace the Temperature sensor:

Step 1. Turn the Mains circuit breaker to the **Off** position.

Step 2. Using the appropriate Allen wrench, open the Connection Box cover.

Step 3. Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.

Step 4. Locate and disconnect the Temperature Sensor connector **(A)**.

Step 5. Detach the wire from the terminal connector and remove the wire from the Connection Box via the gland.

Step 6. Using a #2 Phillips screwdriver, detach the temperature sensor **(C)** from the Atlantium Unit.

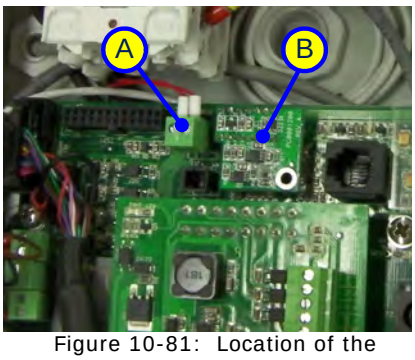


Figure 10-81: Location of the Temperature Sensor Connector (Generic)

Step 7. Attach the replacement Temperature sensor **(C)** to the Atlantium Unit.

Step 8. Thread the wire through the gland and attach it to the terminal connector.

Step 9. Insert the connector **(A)** to its socket on the terminal block.

Step 10. Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.

Step 11. Turn the Mains circuit breaker to the **On** position.

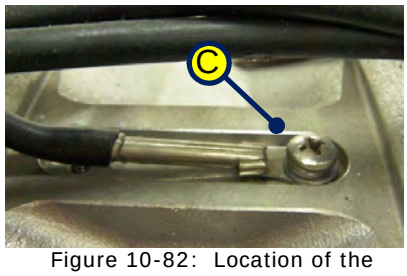


Figure 10-82: Location of the Temperature Sensor on the Atlantium Unit

10.15.2 Replacing the Temperature Converter Card

Performed as needed

In the event that the Temperature sensor card requires replacement, this section details the procedure.

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.
Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature Converter card
- #5 Allen wrench
- #2 Phillips screwdriver

➲ To replace the Temperature sensor card:

Step 1. Turn the Mains circuit breaker to the **Off** position.

Step 2. Using the appropriate Allen wrench, on the first Connection Box, open the Connection Box cover.

Step 3. Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.

Step 4. Locate and carefully pull off the Temperature Sensor card **(C)**.

Step 5. Aligning the pins of the replacement Temperature Sensor card **(C)** with the connector (on the board) on the Connection Box card, carefully insert the small card into the connector of the Connection Box card.

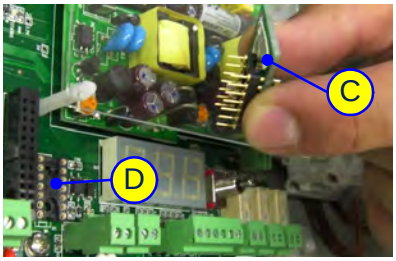


Figure 10-83: Replacing the Temperature Sensor Card (Generic)

Step 6. Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.

Step 7. Turn the Mains circuit breaker to the **On** position.

11 Troubleshooting

11.1 Viewing system errors

The sections headers in this chapter are comprised of message headers that appear as error messages on the Event log screen of the **Control Module.**

Event log screen

Note the Event ID of the relevant event/error.

11.2 Troubleshooting issues

The dose is above the maximum dose on page **165**
The UV dose is too low on page **165**
The temperature is too high on page **166**
Temperature is critically high! on page **166**
Temperature sensor fault on page **171**
The rate of the water flow is too low / too high on page **167**
UVT is too low! on page **168**
UV Lamp ignition failed! on page **168**
Control Module power failure on page **170**
The UV Dose is below the set level on page **171**
Ballast Modulecommunication error on page **173**
The UV Lamp unexpectedly shut down on page **172**
CRC error on page **172**
Malfunction 45 Ballast faults on page **172**
Cleaner fault on page **174**
Tuning process aborted by system on page **174**

Each Event provides the following additional information:

- System power
- System dose
- System flow
- System status
- System UVT
- System Temperature

Events (ID 14, 43, and 45) related to the Ballast Module version 3 contain this additional information:

- Lamp Card (Connection Box Card version)
- State
- Status
- Voltage - For Ballast Module version C, provides values that serves as error codes (See below)
- UVT

Pay attention to the following safety notes when reviewing the troubleshooting issues:

The solution to the problem might involve exposure to UV light.

The system generates ultraviolet (UV) light within the Atlantium Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working.

Do not look directly into the lamp enclosure during system operation, or while examining or servicing the system's internal components when the system is operating, or when a lamp's caps are open.

The solution to the problem might involve working in a high-voltage environment.

Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, **turn the** BM**main circuit breaker to the Off position.**

Place a sign on the relevant switch and **Control Module** alerting others **NOT** to touch the switch and touch screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

The solution to the problem might involve working in an enclosure with moving parts.

Before working on the DPM, replacing parts, or during any other maintenance procedure, turn off all power supplied to the DPM mechanism.

The solution to the problem might involve working with hot surface.

Do not touch the UV lamp with your bare hands. Lamps can reach a temperature of 1,000°C under operating conditions.

Wait at least 10 minutes until the lamp is cooled down before replacing it.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

11.3 The dose is above the maximum dose

Event Index	Control Module displays this warning message
1	The dose is above the maximum defined in the system settings

	Possible Cause	Solution
1	The current flow rate is too low. An upstream process has obstructed the flow rate.	Check the upstream piping and the components installed there, such as the pumps, filters.
2	The **Max Dose** parameter is set too low.	In the **Control Module**, raise the **Max Dose** setting. See ***Configuring Dose Settings*** on page **89.**
3	The system is set for a constant power that is too high given the current water flow.	In the **Control Module**, change the **Operation Mode** from **Power** to **Dose** setting. See ***Configuring Dose Settings*** on page **89.**

11.4 The UV dose is too low

Event Index	Control Module displays this warning message
2	Dose too low! Below minimum!

	Possible Cause	Solution
1	The Water flow rate is too high.	Check the water flow on the Control Module's main screen. If it is higher than normal, decrease the rate of the water flow through the system. See ***Configuring the Flow Settings*** on page **74.**
2	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See ***DPMUltrasonic Cleaning*** on page **110** and***Cleaning In Place (CIP)*** on page **101**. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.
3	UV lamp efficiency has dropped.	Check the UVIS sensors and clean them if necessary. Check the lamp's efficiency reading. If it is low, replace the lamp. See ***Lamp & UVIS Assembly*** on page **107.**
4	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See ***Cleaning In Place (CIP)*** on page **101**. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.
5	The UVT sensor has malfunctioned.	Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See ***Replacing/Checking a UVT Analyzer or UVIS Sensor*** on page **119.**
6	The system is operating on **Power mode** instead of **Dose mode**.	If the system settings are in Power mode, in the **Control Module**, access the Settings screen and change the **Operation Mode** from **Power** to **Dose** setting. Or, change the Power mode settings to allow higher power to respond to operating conditions. See ***Configuring the Operation Mode*** on page **72.**

11.5 The temperature is too high

Event Index	Control Module displays this warning message
4	The lamp's temperature getting too high - check water flow

	Possible Cause	Solution
1	The Max temperature value is too low.	Check the system's **Max Temperature** settings. See ***Configuring the General & Connection Settings*** on page 91.
2	The water flow has significantly decreased.	Check the flow reading in the **Control Module** to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.
3	There are air bubbles or the system is not fully filled with water.	Verify this by looking through the Atlantium Unit's view port.
4	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing a designated air release valve at the highest point on the inlet side of the system.

11.6 Temperature is critically high!

Event Index	Control Module displays this warning message
5	The lamp's temperature is too high and the unit will shut down

	Possible Cause	Solution	
1.	If this message appeared right after a **BM Communication Error** message (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/329°F, then most likely the communication between the **Control Module** and the Connection box is lost.	Follow the solutions offered in ***CRC error*** on page 172.	
2.	The temperature reading dropped to 0°C / 32°F due to a short-circuit in the Temperature sensor or Connection Box Card		Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page 121. If the problem still persists replace the Connection Box card. Contact your Atlantium Service representative.
3.	The Max temperature value is too low.	Check the system's **Max Temperature** settings. See ***Configuring the Technician Settings*** on page 77.	
4.	There is no water flow.	Check the flow reading in the **Control Module**'s main screen to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.	
5.	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.	

	Possible Cause	Solution	
6.	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing designated air release valve at the highest point on the inlet side of the system.	
7.	The Temperature sensor is faulty.		Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page 121.
8.	The Temperature Converter card is faulty.		If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page 160.

11.1 Temperature sensor fault

Event Index	Control Module displays this warning message
6	One of the temperature sensors has malfunctioned or the communication with the Connection Box card was lost.

	Possible Cause	Solution
1	If this message appeared right after a **BM Communication Error** message (and both messages refer to the same lamp) and the temperature display on the monitor is 0°C/32°F, then most likely the communication between the **Control Module** and the Connection box is lost.	Follow the solutions offered in ***CRC error*** on page 172.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuit in the Temperature sensor, or Connection Box Card	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page 121. If the problem still persists replace the Connection Box card. Contact your Atlantium Service representative.

11.2 The rate of the water flow is too low / too high

Event Index	Control Module displays this warning message
10	The water flow rate is higher than the maximum flow rate defined in the system settings. Check the water flow.

	Possible Cause	Solution
1	The water flow rate into the system is too high.	Decrease the rate of water flow from the facility to the Atlantium Unit.
2	**Maximum allowable flow rate** parameter is too low.	Increase the setting of the **Maximum allowable flow rate** parameter to the maximum allowable flow rate.

11.3 UVT is too low!

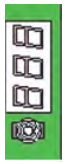
Event Index	Control Module displays this warning message
11	The UVT is lower than the minimum UVT defined in the system settings.

	Possible Cause	Solution	
1	**Minimum allowable UVT** parameter is too high.	Decrease the setting of the **Minimum allowable UVT** parameter to the minimum allowable UVT.	
2	The UVT sensor's lens is fouled.		Pull out the UVT sensor and see if the sensor's lens is foul or damaged. Clean the lens with a dry lens cloth. If it is damaged, replace the sensor with a new one and perform System Tuning according.
3	Deposit has collected in the UV chamber or lamp's quartz sleeve.	Look through the Atlantium Unit's view port. If the color of the light is in a shade of green or yellow, it probably indicate there is deposit present. Perform CIP. ***Cleaning In Place (CIP)*** on page 101	
4	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.	
5	You did not install a designated air release valve at the highest point on the inlet side of the system.	Install air release valve at the highest point on the inlet side of the system.	
6	There is a leak inside the UV chamber		Look through the Atlantium Unit's view port and check if there is condensation on the inner side of the view port and/or if there is water collected between the quartz tube sleeve and the stainless steel underneath. if you see this, contact Atlantium support for further instructions.
7	The UVT sensor has malfunctioned.		Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See ***Replacing/Checking a UVT Analyzer or UVIS Sensor*** on page 119.

11.4 UV Lamp ignition failed!

Event Index	Control Module displays this warning message
14	A UV lamp has failed to ignite. Check if it needs to be replaced, or if there is a faulty connection or ballast malfunction.

For Ballast Module version C, check the Voltage value error code. See on page 164.

	Possible Cause	Solution	
1	The system is set to the wrong model.	In the **Control Module**, access **Technician screen #2**. Verify that the system is set to the correct model, number of lamps and branches. See ***Configuring the Technician Settings*** on page **77.**	
2	One or more connectors are burnt out.		Turn off the system's main power source On each Connection Box, turn the switch to the Off position. Disconnect and thoroughly inspect the Ballast connectors. IF replacement is needed, contact Atlantium Technical Support.
3	The power supply inside the Ballast Module is faulty.	 	In a system with more than one lamp, if possible, switch between the Ballast Module connectors in order to verify the Ballast is faulty, if so replace the Ballast Module. In a system with only one lamp, replace the Ballast Modulewith a new one. See ***Connecting the Cables to the Electrical Cabinet*** on page **61**.
4	A lamp is faulty/burned out.	 	In a system with more than one lamp, switch between the lamps in order to verify the lamp is faulty, if so replace the lamp. In a system with only one lamp, replace the lamp with a new one. See **Inserting the new lamp** on page **111.**
5	The ID number of the Connection Box card does not match ID number of the Ballast Module.		On each Connection Box, open the top cover and locate the 3-digit display (as shown on the right). Check that the set number is identical to the number marked on corresponding Ballast Module. If not identical, use the switch to change it to the correct number. 
6	One or more of the fuses in the Connection Box is burned out.		Replace the burned out fuse in the Connection Box. Contact Atlantium support for further instructions.
7	The Connection Box card is faulty.		On the Connection Box, open the top cover and check that the 3-digit display and LEDs are lit. If they are dark, replace the Connection Box card. Contact Atlantium support for further instructions.

11.5 Deposit has collected on surfaces

Event Index	Control Module displays this warning message
30	The UVT is lower than the UVT Min value defined in the system settings. Clean the system deposit using either the Cleaner or CIP.

	Possible Cause	Solution
1.	The UVT Min value defined in the system settings may be wrong.	Check that the UVT Min value defined in the system settings is within the correct range.
2.	CIP needs to be performed.	Change the CIP frequency in the CIP Plans. See ***Cleaning the Module (CIP)*** on page 113
3.	The CIP is not performed frequently enough.	Change the CIP frequency in the CIP Plans.

11.1 Control Module power failure

Event Index	Control Module displays this warning message
39	The system turned off due to Control Module power failure.

	Possible Cause	Solution
1	No power is feeding into the Connection Box.	Open the Connection boxes and verify that all Connection Boxes have power. See Connection Box Connections Diagram provided.
2	A wire is loose.	With a small flat screwdriver go over all the communication screws in the **Control Module** and Connections Box s and make sure all wires are located properly in their place and are well tightened. ***Control Module Connections*** on page 56
3	A jumper is missing.	A communication jumper might need to be installed, contact Atlantium Support for further instruction.
4	Ground is not connected to the **Control Module**'s power inlet.	A certified electrician is required to verify that the **Control Module** is properly grounded. ***Control Module Connections*** on page 56
5	There is a Hardware problem in the **Control Module's** I/O (ISO) or in the Connection Box.	Contact Atlantium support for further instructions.

11.2 The UV Dose is below the set level

Event Index	Control Module displays this warning message
41	The UV Dose is below the Set For dose value as defined in the system settings

	Possible Cause	Solution
1	The lamp's life has expired.	Check the Lamp's hours displayed on the **Control Module's** main screen and replace the lamp if needed.
2	The water flow rate is too high.	Decrease the water flow rate. See ***Configuring the Flow Settings*** on page **74.**
3	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See***Cleaning In Place (CIP)*** on page **101**. Try to improve the water condition; check on the filter status and upstream chemical processes.
4	UV lamp efficiency has dropped due to condensation.	Check the UVIS sensors and clean them if necessary. See See ***Replacing/Checking a UVT Analyzer or UVIS Sensor*** on page **119.**
5	The system is operating on **Power mode** instead of **Dose mode**.	Change the system settings to allow higher power to respond to operating conditions.See ***Configuring the General & Connection Settings*** on page **91.** In the **Control Module**, access the Settings screen and change the **Operation Mode** from **Power** to **Dose** setting.

11.3 Temperature sensor fault

Event Index	Control Module displays this warning message
42	All temperature sensors in specified system branch have malfunctioned or Connection Box cards communication is lost. The system is shutting down.

	Possible Cause	Solution
1	If this message appeared right after a **BM Communication Error** message (and both messages refer to the same lamp) and the temperature display on the monitor is 0°C/32°F, then most likely the communication between the **Control Module** and the Connection box is lost.	Follow the solutions offered in ***CRC error*** on page **172**.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuit in the Temperature sensor or Connection Box Card	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page **121.** If the problem still persists replace the Connection Box card. Contact your Atlantium Service representative.

11.4 The UV Lamp unexpectedly shut down

Event Index	Control Module displays this warning message
43	A lamp shutdown unexpectedly.

For Ballast Module version C, check the Voltage value error code. See on page 164.

	Possible Cause	Solution
1	A short power failure (up to 15 seconds) to the Atlantium Unit only, while the **Control Module** still had power.	Turn the lamps back on using the On/Off button on the **Control Module**'s main screen.

This situation can happen only in installations were the **Control Module** and the Atlantium Unit have separate power sources.

11.5 CRC error

Event Index	Control Module displays this warning message
44	Cyclic Redundancy Check error. Check if the parameters and counters have been reset to factory defaults. If so, contact Atlantium Technical Support.

	Possible Cause	Solution
1	While writing parameters into the **Control Module**'s flash storage, the system encountered a problem due to an unexpected power failure or software upgrade.	Reenter all system parameters from scratch. Contact Atlantium support for further instructions. See ***Configuring the General & Connection Settings*** on page 91.

11.6 Malfunction 45 Ballast faults

Event Index	Control Module displays this warning message
45	A ballast or lamp has malfunctioned. Check for faulty connection, ballast or lamp.

For Ballast Module version C, check the Voltage value error code. See on page 164.

11.7 Ballast Modulecommunication error

Event Index	Control Module displays this warning message
46	There is lost communication between one of the Connection Boxes and the **Control Module**.
69	There is lost communication between one of the Connection Boxes and the **Control Module**. The system is shutting down

	Possible Cause	Solution
1	No power is feeding into the BM.	Verify that BM and all Connection Boxes have power.
2	A wire is loose.	With a small flat screwdriver go over all the communication screws in the **Control Module** and the BM and make sure all wires are located properly in their place and are well tightened. ***Control Module Connections*** on page **56** and See ***System Messages*** on page **181.**.
3	A jumper is missing.	A communication jumper might need to be installed, contact Atlantium Support for further instruction.
34	Ground is not connected to the **Control Module**'s power inlet.	A certified electrician is required to verify that the **Control Module** is properly grounded. ***Control Module Connections*** on page **56**
45	There is a Hardware problem in the **Control Module's** I/O (ISO) or in the Connection Box.	Contact Atlantium support for further instructions.

11.8 Cleaner fault

Event Index	Control Module displays this warning message
70	The Cleaner mechanism is faulty.

	Possible Cause	Solution
1	The belt inside the DPM is loose or torn.	Open the DPM and check the belt. If loose, preposition it. If torn, replace it.
2	One or more of the Opto-coupler's connectors in the DPM mechanism is malfunctioning due to dust collected, covering its sensor. One or more of the Opto-coupler's connectors in the DPM mechanism is loose or disconnected	Open the DPM and using a cotton swab, clean the Opto-coupler. See ***Cleaning the DPM's Opto-Couplers' Sensors*** on page **153.** Open the DPM and check the Opto-coupler's connectors. If needed, reconnect it properly. If the problem persist contact Atlantium support for further instructions.
3	The DPM engine connector is loose or disconnected. Or, the engine is malfunctioning.	Open the DPM and check the engine's connector. If needed, reconnect it properly. If the problem persist contact Atlantium support for further instructions.

11.9 Tuning process aborted by system

Event Index	Control Module displays this warning message
100	Tuning process aborted by system

	Possible Cause	Solution
1	Communications fault	Check the communications cables and cable connections between the **Control Module Module** and other equipment. Restart the **Control Module Module**. If the problem persists, contact Atlantium Technical Support.

Appendix A Modbus Communication Protocol

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks.

Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

For information on how to configure the communication properties and make the cable connections, see ***Modbus Communication Connections*** on page **59**.

ATTENTION!

Modbus registers are meant to be used for specific commands in a command algorithm. Do not write to any register if you do not intend to use it.

A.1 Modbus Registry Map

The table on the following pages provides the information you need to help you configure your network to properly communicate with the Atlantium system.

All of the registers length are Unsigned 16 bits, even of the Coils Bit

Function code 3 - Holding registers

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read / Write	Status
194	System Temperature	: 100			°C|°F	R	OK
204	Dose	: 1		Actual dose	mJ/cm²	R	OK
205	System On/Off		Coil Bit	0 - System OFF, 1 - System ON		R/W	OK
227	UVT	: 100		Actual UVT	%	R	OK
228	Power	: 1		Actual power	%	R	OK
229	Flow	: 1		Actual flow value - Display according to format set on HMI (m³/h or gal/min)	m³/h	R	OK
0	Min Dose	:10		Up to 6553	mJ/cm²	R	OK
1	Max Dose	:1			mJ/cm²	R	OK
2	Set For	:10		Up to 6553	mJ/cm²	R	OK
3	System Unit format		Coil Bit	0 - Imperial, 1 - Metric		R	OK

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read / Write	Status
4	Flow in FS mode	: 1		Min. Val = 1 , Max.Val = 30000 - Display according to format set on HMI (m^3/h or gal/min)	m^3/h	R/W	OK
5	Min Flow	: 100		Min. Val = 1 , Max.Val = 300 - Display according to format set on HMI (m^3/h or gal/min)	m^3/h	R/W	OK
6	Max Flow	: 1		Min. Val = 1 , Max.Val = 30000 - Display according to format set on HMI (m^3/h or gal/min)	m^3/h	R/W	OK
7	Operation Mode		Coil Bit	0: Power Mode, 1: Dose Mode		R/W	OK
8	System Lamps status	: 1		Average of all lamps status.	%	R	OK
32	Set Power	: 1		Set the power for power mode	%	R/W	OK
Lamp1							
35	UVIS	: 100			4-20 mA	R	OK
36	UVTS A	: 100			4-20 mA	R	OK
37	UVTS B	: 100			4-20 mA	R	OK
38	Lamp Current	: 100			Amper	R	OK
39	Lamp Voltage	: 100			Voltage	R	OK
40	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK
42	Lamp UVT A	: 100			%	R	OK
43	Lamp Status	: 100			%	R	OK
44	Lamp UVT B	: 100			%	R	OK
46	Lamp Age	: 1			Hour	R	OK
47	Lamp Ignitions	: 1				R	OK
51	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
52	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
53	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Lamp 2							
54	UVIS	: 100			4-20 mA	R	OK
55	UVTS A	: 100			4-20 mA	R	OK
56	UVTS B	: 100			4-20 mA	R	OK
57	Lamp Current	: 100			Amper	R	OK
58	Lamp Voltage	: 100			Voltage	R	OK

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read / Write	Status
59	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK
61	Lamp UVT A	: 100			%	R	OK
62	Lamp Status	: 100			%	R	OK
63	Lamp UVT B	: 100			%	R	OK
65	Lamp Age	: 1			Hour	R	OK
66	Lamp Ignitions	: 1				R	OK
70	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
71	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
72	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Lamp 3							
73	UVIS	: 100			4-20 mA	R	OK
74	UVTS A	: 100			4-20 mA	R	OK
75	UVTS B	: 100			4-20 mA	R	OK
76	Lamp Current	: 100			Amper	R	OK
77	Lamp Voltage	: 100			Voltage	R	OK
78	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK
80	Lamp UVT A	: 100			%	R	OK
81	Lamp Status	: 100			%	R	OK
82	Lamp UVT B	: 100			%	R	OK
84	Lamp Age	: 1			Hour	R	OK
85	Lamp Ignitions	: 1				R	OK
89	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
90	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
91	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Lamp 4							
92	UVIS	: 100			4-20 mA	R	OK
93	UVTS A	: 100			4-20 mA	R	OK
94	UVTS B	: 100			4-20 mA	R	OK
95	Lamp Current	: 100			Amper	R	OK
96	Lamp Voltage	: 100			Voltage	R	OK
97	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read / Write	Status
99	Lamp UVT A	: 100			%	R	OK
100	Lamp Status	: 100			%	R	OK
101	Lamp UVT B	: 100			%	R	OK
103	Lamp Age	: 1			Hour	R	OK
104	Lamp Ignitions	: 1				R	OK
108	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
109	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
110	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Lamp 5							
111	UVIS	: 100			4-20 mA	R	OK
112	UVTS A	: 100			4-20 mA	R	OK
113	UVTS B	: 100			4-20 mA	R	OK
114	Lamp Current	: 100			Amper	R	OK
115	Lamp Voltage	: 100			Voltage	R	OK
116	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK
118	Lamp UVT A	: 100			%	R	OK
119	Lamp Status	: 100			%	R	OK
120	Lamp UVT B	: 100			%	R	OK
122	Lamp Age	: 1			Hour	R	OK
123	Lamp Ignitions	: 1				R	OK
127	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
128	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
129	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Lamp 6							
130	UVIS	: 100			4-20 mA	R	OK
131	UVTS A	: 100			4-20 mA	R	OK
132	UVTS B	: 100			4-20 mA	R	OK
133	Lamp Current	: 100			Amper	R	OK
134	Lamp Voltage	: 100			Voltage	R	OK
135	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK
137	Lamp UVT A	: 100			%	R	OK

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read / Write	Status
138	Lamp Status	: 100			%	R	OK
139	Lamp UVT B	: 100			%	R	OK
141	Lamp Age	: 1			Hour	R	OK
142	Lamp Ignitions	: 1				R	OK
146	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
147	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
148	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Lamp 7							
149	UVIS	: 100			4-20 mA	R	OK
150	UVTS A	: 100			4-20 mA	R	OK
151	UVTS B	: 100			4-20 mA	R	OK
152	Lamp Current	: 100			Amper	R	OK
153	Lamp Voltage	: 100			Voltage	R	OK
154	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK
156	Lamp UVT A	: 100			%	R	OK
157	Lamp Status	: 100			%	R	OK
158	Lamp UVT B	: 100			%	R	OK
160	Lamp Age	: 1			Hour	R	OK
161	Lamp Ignitions	: 1				R	OK
165	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
166	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
167	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Lamp 8							
168	UVIS	: 100			4-20 mA	R	OK
169	UVTS A	: 100			4-20 mA	R	OK
170	UVTS B	: 100			4-20 mA	R	OK
171	Lamp Current	: 100			Amper	R	OK
172	Lamp Voltage	: 100			Voltage	R	OK
173	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R	OK
175	Lamp UVT A	: 100			%	R	OK
176	Lamp Status	: 100			%	R	OK

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read / Write	Status
177	Lamp UVT B	: 100			%	R	OK
179	Lamp Age	: 1			Hour	R	OK
180	Lamp Ignitions	: 1				R	OK
184	Lamp Enable		Coil Bit	0 - DIsable, 1 - Enable		R/W	OK
185	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W	OK
186	Lamp state			7 -> lamp off, 6 -> Lamp Ignition, 4-> Lamp On		R	OK
Valve							
200	Bypass Valve		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
201	Inlet Valve		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
202	Outlet Valve/ Ready		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
	Alarms						
203	General Alarm		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
206	Over dose		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
207	Min Dose #2		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
208	High Temp #4		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
209	Over Temp #5		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
210	Shut Down Alarm		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
211	Low UVT		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
212	Lost Comm		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
213	Under flow		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
214	High Flow #10		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok
224	Under dose set #41		Coil Bit	1 - Valve ON , 0 - Valve OFF		R	Ok

Appendix B System Messages

The table below details the message that can appear on the **Control Module**'s screen or in the Event logs. A number of parameters are User Defined and include setting a value and upon reaching, exceeding or falling below it, an Action is set to be triggered. The Action is also configurable. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution** - The caution alerts you to an anomaly based on the user preferences.
- **Warning** - One or more warning messages are issued and it triggers a pop-up window message and a General Alarm (if configured, see ***Configuring the General Alarm*** on page **61**).
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm (if configured, see ***Configuring the General Alarm*** on page **61**).

The other parameters are system defined. The type column in the table below details the type of parameter.

Table B-1: **Control Module**'s Messages

Key	Type	Event Log	Line Message / Popup Title	Popup Message
1	The value is User Defined	Over Max. Dose	The dose is above the maximum dose	The dose is above the maximum defined in the system settings.
2	The value is User Defined (The default is None)	Dose Too Low	The UV dose is too low	Dose too low! Below minimum!
4	Warning - The default temperature threshold is 5°C (9°F) below what is defined for Over Temp	Temp. Too High	Temperature is Too High	The lamp's temperature getting too high -- check water flow.
5	Shutdown - - The default temperature threshold is User Defined	Over Temp	Temperature is Critically High!	The lamp's temperature is too high and the unit will shut down.
6	Shutdown - Hardware malfunction	Temperature Sensor Fault	Temperature Sensor Fault	One of the temperature sensors has malfunctioned or the communication with the Connection Box card was lost.
9	Shutdown ---The Minimum Flow is User Defined	Low Flow	The rate of the water flow is too low	The water flow rate is lower than the minimum flow rate. Check the water flow.
10	The value is User Defined	High Flow	The rate of the water flow is too high	The water flow rate is higher than the maximum flow rate allowed. Check the water flow.
11	The value is User Defined	Transmission Low	UVT is too low!	UVT is below the required value
13	Shutdown - DPM malfunction	Cleaner Fault	Cleaner Fault	Cleaner Mechanism Fault
14	Warning	Lamp Ignition	UV Lamp Ignition Failed!	The UV lamp has failed to ignite due to an electronic fault. The UV lamp may be burnt out or a wire may be disconnected.

Table B-1: **Control Module**'s Messages

Key	Type	Event Log	Line Message / Popup Title	Popup Message
16	Appears in the Event Log only - pops up from additional info from Event Log	User On		The Control Module screen or an external signal turned on the system.
17	Appears in the Event Log only - pops up from additional info from Event Log	User Off		The Control Module screen or an external signal shut down the system.
18	Appears in the Event Log only - pops up from additional info from Event Log	Auto On		System power was restored or flow raise above minimum. The lamps reignite.
19	Appears in the Event Log only - pops up from additional info from Event Log	Auto Off		Flow rate below user low flow setting; automatic shutdown.
20	Appears in the Event Log only - pops up from additional info from Event Log	Lamps On		System Ready. All the enabled lamps are on.
24	Caution	Dose On Spec	Dose On Spec	The UV dose being delivered by the system is above the minimum validated value
25	Warning	Dose Off Spec	Dose Off Spec	The UV dose being delivered by the system is below the minimum validated value
26	Caution	UVT On Spec	UVT On Spec	The UVT value is above the minimum validated UVT
27	Warning	UVT Off Spec	UVT Off Spec	The UVT value is below the minimum validated UVT
28	Caution	Flow On Spec	Flow On Spec	The flow rate is below the maximum validated flow
29	Warning	Flow Off Spec	Flow Off Spec	The flow rate is above the maximum validated flow
33	Warning	UVT 4th calibration failed	UVT Analyzer 4th calibration check failed	UVT Analyzer calibration check failed 4 times in a row. System is off-spec
34	The value is User Defined (Default as None)	Validated Min UVT	Validated Min UVT	UVT is below Validated Min UVT
35	The value is User Defined (Default as None)	Validated Max Flow	Validated Max Flow	Flow is above Validated Max Flow
39	Appears in the Event Log only - pops up from additional info from Event Log	Power Failure	Power Failure	The system was turned off due to power failure
41	The value is User Defined	UV Dose	The UV Dose is below the set level	The UV Dose is below the Operate At dose

Table B-1: **Control Module**'s Messages

Key	Type	Event Log	Line Message / Popup Title	Popup Message
42	Warning - Sensor needs to be replaced or there may be a loose wire	Temp. Sensor Fault	Temperature Sensor Fault	Fault on lamp temperature sensor
43	Shutdown	Lamp Shutdown	The UV lamp unexpectedly shutdown	Lamp shutdown
44	Appears in the Event Log only - pops up from additional info from Event Log	CRC Error	CRC Error	System error parameters and counters might be reset to factory defaults. Please check
45	Shutdown	PS Fault	Power supply fault	Ballast fault [value displayed], error code: [number displayed].
46	Warning	Communication Error	Communication Error	Lost communication with the Ballast Module
49	Appears in the Event Log only - pops up from additional info from Event Log	UVIS calib info	UVIS calib info	UVIS calib lamp [value displayed]
50	Appears in the Event Log only - pops up from additional info from Event Log	UVT calib info	UVT calib info	UVT calib lamp [value displayed]
53	Appears in the Event Log only - pops up from additional info from Event Log	Settings Changed	Settings changed	Settings changed [value displayed]
54	Caution	UVIS is calibrated successfully	UVIS is calibrated successfully	UVIS is calibrated successfully
55	Caution	UVIS calibration failed	UVIS Calibration Failed	UVIS calibration failed, Sensor is out of calibration. Correction Factor is implemented
56	Caution	UVT successful calibration check	UVT successful calibration check	UVT Analyzer successful calibration check
57	Caution	UVT check failed	UVT Analyzer check failed	UVT Analyzer check failed; Auto recalibration was done

Appendix C Checking the System Parameters

You can use the checklist below to keep a record of your checks on Atlantium system parameters. **Print this page**.

Table C-1: Atlantium System Parameters Checklist

Date	Time	Lamp ID	Power	Flow Rate	UVT	UV Dose	Lamp Status	OK Y/N	Remarks/Alarms/ Events
			Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____		

Appendix D Glossary of Terms

Term	Definition
Auxiliaries	Valves, flow devices, etc. essential to the workings of the Atlantium system but not directly a part of the Atlantium system.
Ballast	The power supply that translates the electrical power to UV power.
Connection Box	The box that sits on top of each lamp unit and holds the circuit board for the Electrical Cabinet or Ballast Module electrical configurations. Not relevant for Ballast Module/Cable-style.
Control Module	Control Module that controls operations and dose measurement. Comes in two versions: Standard and Premium. The Standard has a monochromatic screen and several key features. The Premium has a color screen can accommodate a full complement of reporting and trending features.
DPM Cleaner	The Deposit Prevention Mechanism for cleaning the quartz sleeves.
External On/Off	This enables an external source (SCADA, PLC, Process) to turn the Atlantium system on and off.
Electrical configuration styles: ***Atlantium provides a choice of electrical configurations.***	
Electrical Cabinet-style (EC)	**EC** - Electrical Cabinet.
Ballast Module	Waterproof case for the ballast which connects via a consolidated plug from the circuit board which sits on top of each lamp unit.
General Alarm	This enables the Atlantium unit to send an alarm to a buzzer, cell phone or other destination.
Inlet/Outlet Valves	Isolating Valves on the Inlet/outlet sides.
Lamp Chamber	The chamber which holds the lamp.
Lifting Support	The lifting element which can also be used to help support the Unit.
Pinhole	This is a small disc that is fitted for each sensor port to ensure that the right amount of light penetrates.
Portable CIP System	The portable unit Atlantium sells that connects to the Unit to clean and disinfect it without disassembling. It includes several kits – a valve kit, an accessories kit and the CIP kit itself.
Quartz Chamber	The **quartz pipe** in which the disinfection occurs.
System Ready Signal	The signal the Atlantium unit provides to tell the production system that the Atlantium System is ready to supply properly disinfected water.
System Tuning	This is process is performed when key components are replaced or changed to synchronize them.

Term	Definition
Triangle Support Brackets	RZ-Series only- brackets that can be used to support and balance a unit.
UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds the real time dose calculation.
UVT Analyzer	The sensor that measures the lamp UV intensity through the water. Together with the UV Intensity sensor and the Control Module, it analyzes, tracks and reports the water quality.